A Life-Cycle Model of Mortgage Debt Relief: Refinancing and Forbearance during the COVID-19 Pandemic

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Abstract

When a crisis leaves mortgaged households unable to fulfil their mortgage obligations, assistance can be provided by either lowering interest rates, which allow households to refinance their mortgage debt; or by providing direct financial support to households. During the COVID-19 pandemic, low mortgage rates were followed by a huge refinancing wave, while forbearance programmes through the CARES Act allowed affected households to temporarily pause their mortgage payments, without any penalties. Using a life-cycle heterogeneous-agent model with discrete choice over renting and mortgage decisions, this paper investigates which households each instrument helped, and how the outcomes differ by macroeconomic conditions and the programme's design. First, I find that while refinancing would have been beneficial for most households, households that could not meet underwriting requirements could stave off foreclosure and smooth out consumption through forbearance programmes. Second, rising home prices did not add to what forbearance achieved so much as substitute for it: the additional equity moved households across the collateral test and onto the refinancing margin, so take-up roughly halved and the cohort left in forbearance was smaller and more distressed than the one a flat-price crisis produces. For owners as a whole the price rise was costly rather than helpful, because maintenance and rent both scale with the price of housing while the capital gain accrues only at death. Estimating households' willingness to pay for forbearance programmes, I find that households value it at \$17,045 and value the rate cut obtained through refinancing at \$27,328.

*Department of Economics, Michigan State University. **Preliminary and incomplete.** This draft is circulated for discussion only; please do not cite, quote or distribute without the author's consent. The prevention counterfactual and the design-margin exercises are summarised in the Introduction but are not reported in full in this draft.

1 Introduction

A household that cannot meet its mortgage payment in a downturn can be helped in two ways. It can be allowed to *restructure* the loan—refinance into a lower rate, perhaps extracting equity in the process—or it can be allowed to *suspend payments* temporarily. The first works through prices and is delivered by the market: a reduction in the treasury rate lowers the mortgage rate, and a borrower who can pass underwriting can refinance their mortgage, lowering future payments and possibly extracting equity. The second works through eligibility and is delivered by fiat: a borrower who meets a hardship condition defers scheduled payments without being re-underwritten. The two instruments aim at the same outcome, the mortgage default of a household whose income has fallen, but they are gated by opposite conditions. Refinancing requires the borrower to demonstrate, at the moment of application, that she can afford the new loan; payment deferral requires her to demonstrate that she cannot afford the old one. This paper asks which households each instrument helps, and how that answer moves as the macroeconomic state and the design of the programme change.

The question matters because the gate on the price instrument is not one condition but several, and they tighten together in a downturn. Refinancing in this model requires the household to clear a payment-to-income ceiling and a loan-to-value limit, to take a balance no smaller than the one it already owes, and to pay a fee levied on the value of the house rather than on the loan being refinanced. A rate cut delivered through refinancing is therefore a transfer conditional on a joint underwriting-and-cost hurdle, and DeFusco and Mondragon (2020) and Chen et al. (2020) show that such conditions bind hardest in recessions. Which component does the work is a quantitative question, and Section 6.6 answers it: in this model the fee restricts refinancing more than eligibility does. If the households that fail them are the households on the margin of default, then the price instrument cannot reach the population it is meant to protect, and the eligibility instrument is not an alternative to it but the only route to that population. If instead the two overlap—if the households that defer are largely households that could have refinanced—then forbearance is substituting for a cheaper channel. Which of these describes a crisis is an incidence question, and it cannot be answered from realized data alone, because no borrower is observed under the instrument she did not use.

The CARES Act granted forbearance on self-certification to any borrower with a federally backed mortgage; and treasury rates fell sharply in 2020, so that the thirty-year mortgage rate fell from 3.7 per cent in January to 2.8 per cent in December, setting off a refinancing wave of roughly fourteen million loans; and a foreclosure moratorium ran alongside both. The loan-level record already shows the two operative instruments reaching different people. In the Fannie Mae Single-Family Loan Performance file, borrowers who later entered forbearance had originated at a mean loan-to-value ratio of 77.5 per cent and a debt-to-income ratio of 38.3 per cent; borrowers who refinanced in 2020–2022 had originated at 64.8 and 32.1. The underwriting profile that predicts needing relief is the profile that predicts being unable to refinance. That is a fact about origination characteristics,

not about counterfactual access, and it is the gap between the two that a structural model has to close.

I build a life-cycle heterogeneous-agent model of housing tenure and mortgage choice in the incomplete-markets tradition of Eichenbaum et al. (2022) and Wong (2016), extended along three lines. A delinquency state, following Low (2022), lets a missed payment persist for a period before it resolves through cure, foreclosure or relief. A forbearance option routes an eligible household into a deferred state in which it chooses, from an annual menu, what fraction of the year's payment to remit, with the remainder carried as a non-interest-bearing balloon due at maturity, the settlement the GSEs deployed; the option is self-certified and granted on request, and what limits it is the hardship condition that the scheduled payment exceed a fifth of current income. Default is non-recourse. The crisis is an unanticipated event grafted onto an estimated aggregate process, and every counterfactual is run on the same panel of households under common random numbers, so that differences across economies are differences in policy. The model is calibrated to the 2019 Survey of Consumer Finances and the Panel Study of Income Dynamics, reproduces the broad signature of the 2020 mortgage market, and is held to a reduced-form test it was not fitted to: the opening regression of forbearance take-up on origination characteristics, re-estimated on simulated data, returns the same sign on payment burden and the same concentration of take-up in the highest payment-burden quintile as the data, though it reverses the sign on leverage (Section 5).

Before the quantitative model, a two-period version states what determines who each instrument can reach. The rate cut operates through a participation condition in which the present value of the payment reduction must cover a fee levied on the house; deferral operates through an eligibility condition on the payment-to-income ratio. Neither condition contains the other's parameters, so the boundary between the two instruments is a vertical line in payment-burden space that a rate cut cannot move. And because the benefit of refinancing accrues on the balance while its cost is levied on the house, the rate cut's reach falls with age on two margins at once as the balance amortises and the horizon shortens. These two propositions organise the quantitative results.

The first result is that the income test is what divides the two instruments' caseloads. Turning the policy levers on one at a time in the calibrated crisis, forbearance alone reaches 19.6 per cent of pre-retirement mortgagors; adding the rate cut more than halves that to 9.4 per cent while 38.7 per cent refinance instead. The two are substitutes in take-up but not in what they prevent: the rate cut leaves crisis default exactly where the no-instrument crisis leaves it, at 0.33 per cent, while forbearance takes it to 0.04 per cent. The households forbearance reaches enter the crisis with about three-fifths the income of those who refinance, 0.62 against 1.03 in model units, and are markedly more levered, 0.81 against 0.58. Those four figures are measured in the arm where both instruments operate and house prices are held flat, which is the comparison that isolates the two caseloads; adding the house-price rise shrinks the forborne cohort further and leaves it poorer still, at 0.44. Relaxing each underwriting restriction in turn confirms the margin. Loosening the down payment, permitting cash-in refinancing or removing the refinancing fee changes who holds a mortgage and raises forbearance take-up; flattening the maturity rule changes what the deferral is worth without

changing who takes it; only removing the payment-to-income ceiling cuts take-up, the take-up rate and the value of the deferral together. Measured in crisis-year cash flow the asymmetry is entirely in reach, not in size: per recipient the two instruments are indistinguishable, about $14,000 each, because a household reached by either makes no mortgage payment at all that year—the refinancer because it originates a new loan, the forborne because it draws $\xi = 0$. What separates them is that the rate cut reaches nearly five times as many households—24.1 per cent of crisis mortgagors against 4.9 per cent. Averaged over every crisis mortgagor rather than only those reached, the rate cut is worth $3,470 per household against the deferral's $688.[1] Both are progressive on the extensive margin alone: what falls with income is the probability of being reached, not the size of the relief to those who are. Neither number is a transfer. The rate cut's is a payment holiday financed by a new and larger loan, against which the interest the lower rate actually saves is about $1,076 a year; the deferral is repaid at maturity. Whether the two populations stay separate or are reached in sequence depends on how long cheap credit lasts: under a one-period cut, a household that defers has forfeited the refinance, while under a sustained and anticipated cut the forbearance cohort refinances out at above the non-applicant rate, and the reversal is confined to borrowers past their mid-forties for the reason the two-period model gives.

The second result is that forbearance changes when its recipients lose the home far more than whether they do. The cohort it reaches is severely selected—36.2 per cent eventually lose the house against 0.06 per cent of mortgagors who never applied, a ratio of nearly six hundred to one against a rate of 4.7 per cent among crisis mortgagors as a whole—but that is a cohort share, not a treatment effect. Solving the identical economy with the option unavailable in every state and following the same households through both worlds, 2.1 per cent of those reached avoid a foreclosure they would otherwise have suffered, 17.2 per cent suffer one later, and a group larger than the prevented, 6.5 per cent, forecloses *earlier* with the option than without it. Lifetime foreclosures across the panel differ by 0.09 per cent, and in the direction of slightly more rather than fewer. Seen from the lender's side the same fact is a cost: model-implied foreclosure credit losses are higher with the option than without at every foreclosure cost considered. Not because there are more foreclosures—over the life cycle there are marginally fewer—but because each one is larger: a household that skipped a year of payments amortised no principal in it, so when it does foreclose it owes slightly more against a house worth the same. And 40.5 per cent of every dollar advanced sits with a household that forecloses anyway. This is loss accounting rather than a full lender cost—it discounts nothing and omits servicing advances, prepayment and the guarantee structure. It is also specific to the deployed interest-free deferral, under which the balance never grows and the balloon is written off at foreclosure: a settlement that capitalised the deferred amount instead would add a carrying-cost

[1] Both figures are differences in scheduled payment between policy arms, taken over the households whose payment actually moves, and are measured on the crisis year alone. The per-household averages divide those totals by the same population the totals are measured over: crisis owner-mortgagors carrying a balance and present in all three policy arms, of all ages. The shares quoted earlier in this paragraph are shares of pre-retirement mortgagors, which is Table 7's base. They are not the same object as the deferred dollar advances reported in Appendix G, which are the increment in the deferred balloon over every approved forbearer in a given arm. Per recipient the two measures agree to within a few hundred dollars.

channel this accounting does not contain. The relief is insurance against a liquidity event for a segment the price instrument cannot reach, and not a durable reduction in that segment's default risk.

The third result is about the margin. Holding the instruments fixed and varying the world they arrive in, deepening the output contraction from essentially nothing to twenty-six per cent raises crisis delinquency from 0.9 to 5.9 per cent and default from nothing at all—not one pre-retirement mortgagor defaults in the mildest arm—to 1.4 per cent, while forbearance approvals rise only from 4.0 to 6.0 per cent: the eligibility test selects on a payment burden set largely by the pre-crisis cross-section, so the households a deeper crisis endangers are mostly not new entrants to eligibility. The housing backdrop matters more: replacing 2020's eight per cent appreciation with a ten per cent fall more than doubles approvals, from **[needs: falling-price arm]** to **[needs: falling-price arm]**, while adding **[needs: falling-price arm]** lifetime foreclosures. And the design of the programme matters most of all. Whether the deferred amount capitalises immediately or waits as a non-interest-bearing balloon moves take-up by a factor of eight, from **[needs: regime-4 arm]** to **[needs: regime-4 arm]** households; sweeping the eligibility floor from the deployed fifth of income down to **[needs: PTI sweep]** raises the households reached from **[needs: PTI sweep]** to **[needs: PTI sweep]** while the deferral per recipient falls from **[needs: PTI sweep]** to **[needs: PTI sweep]**, and lifetime foreclosures move from **[needs: PTI sweep]** to **[needs: PTI sweep]**. In each case the lever sets how much is lent and to whom; none of them sets how many homes are kept.

The paper contributes to three literatures. To the structural work on monetary transmission through mortgage refinancing (Eichenbaum et al., 2022; Wong, 2016; Beraja et al., 2019; Di Maggio et al., 2017) and on refinancing frictions in recessions (DeFusco and Mondragon, 2020; Greenwald, 2026; Chen et al., 2020; Agarwal et al., 2023; Fuster et al., 2021), it adds a second, non-price instrument and shows that the households the frictions exclude are the households the second instrument reaches. To the empirical literature on CARES forbearance (Goodman and Zhu, 2024; Cherry et al., 2021; Gerardi et al., 2022; An et al., 2022; Boctor, 2024; Kim et al., 2024), which estimates the effect of relief for the borrowers who used it, it supplies what that design cannot: the counterfactual reach of the instrument a borrower did not use, and the lifetime trajectory of the treated cohort beyond the within-shock window. To the structural literature on mortgage design and foreclosure policy (Guren et al., 2021; Campbell et al., 2021; Mitman, 2016), which asks what contract or policy is optimal, it takes the deployed instruments as given and asks about their incidence; the recession option of Campbell et al. (2021)—interest-only payment with maturity extension—is the closest relative of the forbearance technology studied here. The mechanical ingredients are established: that employment documentation and closing costs can close the refinancing channel (DeFusco and Mondragon, 2020); that the aggregate effect of a rate cut depends on the distribution of refinancing gains (Eichenbaum et al., 2022); that a recession-contingent reduction in required payments stabilises consumption and shifts the timing of default (Campbell et al., 2021); and that negative equity is necessary but not sufficient for default, which usually requires an income trigger as well

(Foote et al., 2008; Ganong and Noel, 2023). The contribution is to put the two instruments in one household problem and measure how much of the observed gap between who refinances and who forbears each underwriting rule accounts for, what relief does to the households it reaches, and how both move at the margin.

Section 2 sets out the institutional setting and the loan-level facts. Section 3 states the two-period model. Section 4 presents the quantitative model and Section 5 its calibration and fit. Section 6 reports which instrument reaches whom, which underwriting restriction gates that reach, and what each instrument is worth to the households it reaches (Section 6.8). The prevention counterfactual and the design-margin exercises summarised above are not written up in full in this draft.

2 The Setting and the Loan-Level Evidence

2.1 Two instruments on opposite sides of an underwriting boundary

The United States responded to the pandemic with three concurrent mortgage-market interventions. The CARES Act of March 2020 granted forbearance to any borrower with a federally backed mortgage who attested to a pandemic-related hardship; no documentation or underwriting was required, and by mid-2020 roughly eight per cent of Fannie Mae loans were in active forbearance. The Federal Reserve cut the policy rate to zero, the thirty-year fixed mortgage rate fell to a historic low, and roughly fourteen million loans were refinanced. A federal foreclosure moratorium ran alongside both through July 2021. The first two instruments sit on opposite sides of an underwriting boundary. A refinance required positive equity, a passing payment-to-income ratio and a credit history clearing post-2010 standards; forbearance required none of these. The moratorium suspended foreclosure administratively, which makes the recorded collapse in 2020–2021 foreclosure largely mechanical, and it is set aside here. The question is which of the two operative instruments reached the borrowers facing default-margin stress.

The boundary was institutional rather than absolute. CARES covered federally backed loans only, and the FHFA permitted borrowers in forbearance who remained current, and borrowers who had exited and made three consecutive payments, to refinance.[2] The deployed settlement for GSE loans let scheduled interest accrue during the spell and then deferred the missed principal and interest into a non-interest-bearing balance due at payoff, sale or maturity, with rate, term and payment unchanged. This is the settlement the quantitative model takes as its baseline (Section 4.3). Segmentation rather than disjointness is therefore the right description of what the data can show.

2.2 Three facts

The evidence comes from a two-per-cent random sample of the Fannie Mae Single-Family Loan Performance file, loans originated 2000–2024, which covers fully amortising, fully documented, conventional fixed-rate conforming mortgages—the segment on which both instruments operated. Appendix A reports the full set of exhibits; three facts carry the argument.

Forbearance users and refinancers have different origination profiles. Table 1 reports mean origination characteristics for loans that eventually entered forbearance, loans that never did, and rate-and-term and cash-out refinances originated during the low-rate window. Future forbearance users originated at a mean LTV of 77.5 per cent and DTI of 38.3, against 70.3 and 33.6 for never-event loans; 2020–2022 rate-and-term refinancers sit at 64.8 and 32.1. The ordering holds vintage by vintage, not only in the pooled means (Figure 13). The profile that makes a borrower likely to need forbearance is the profile that makes the same borrower unable to clear post-2010

[2] Federal Housing Finance Agency, "FHFA Announces Refinance and Home Purchase Eligibility for Borrowers in Forbearance," news release, May 19, 2020.

Table 1: Origination characteristics by eventual outcome and by 2020–2022 refinance type.

	By eventual outcome, all vintages		Refinances originated 2020–2022	
Mean at origination	Forbearance	Never-event	Rate-and-term	Cash-out
Original UPB ($)	251,178	217,252	258,870	228,578
DTI (%)	38.3	33.6	32.1	34.3
Original LTV (%)	77.5	70.3	64.8	62.7
Credit score	725.5	748.4	765.5	746.0
N	24,359	1,095,118	172,571	118,654

Notes. Two-per-cent Fannie Mae sample. "Forbearance" is every loan that ever entered forbearance; "never-event" is every loan that neither entered forbearance nor foreclosed. The refinance columns are loans with purpose R (rate-and-term) or C (cash-out) originated 2020–2022. The public file does not link a terminated loan to the loan that refinanced it, so the two blocks compare origination profiles rather than alternative outcomes for a common population. Standard deviations and the foreclosure group are in Table 12.

underwriting.

Forbearance was drawn from the population foreclosure draws from. On the leverage and payment-stress dimensions, forbearance users look like foreclosed borrowers—LTV of 77.5 against 78.1 per cent, DTI of 38.3 against 39.0—but with higher credit scores, 725.5 against 696.6 (Table 13, in which the foreclosure group is restricted to loans whose foreclosure occurred after 2010, dropping the crisis-era wave; taken over all foreclosures the two means are 79.0 and 692.6, in Table 12). The same selection is visible without any regression: across deciles of estimated income at origination, lifetime foreclosure falls by a factor of roughly eleven, and it peaks at the ninety-per-cent LTV decile (Figure 14). The borrowers exposed to default-margin stress are low-income and high-leverage, and because refinancing into the record-low rate required passing a payment-to-income test, that is the population the price instrument screens out. Linear probability models confirm the pattern at the loan level: origination LTV, DTI and the note rate raise the probability of both serious delinquency and forbearance take-up, and a higher credit score lowers both (Table 14, Appendix A.2). The two populations do differ in their vintage timelines: ever-foreclosure loans are overwhelmingly the pre-2010 book, ever-forbearance loans a post-2015 cohort (Figure 15), so part of the raw 2020–2021 decline in foreclosure reflects a better-underwritten stock of loans rather than relief.

Table 2: Foreclosure among seriously delinquent loans, by cohort and forbearance use.

Group	N	Foreclosed	Foreclosure rate (%)	Prepaid (%)
A: First 90+ delinquency 2016–2019	4,166	709	17.0	46.7
B: COVID-onset, entered forbearance	13,766	133	1.0	41.2
C: COVID-onset, no forbearance	3,531	214	6.1	25.3

Notes. Loans reaching 90+ days delinquent, two-per-cent sample. The foreclosure rate is the share of the group that foreclosed over its observation window; the three windows differ in length (median months observed 37.5, 13 and 35), so Figure 1 compares the groups at fixed horizons. All pairwise differences in the foreclosure rate are significant at $p < 0.001$.

Among the seriously delinquent, forbearance replaced foreclosure as the resolution—and the comparison group did not share in it. Table 2 partitions loans that reached ninety-day delinquency into three groups: those whose first serious delinquency occurred in 2016–2019, with no access to CARES (A); COVID-onset delinquencies that used forbearance (B); and COVID-onset delinquencies that did not (C). Group C is the within-COVID comparison group, which holds the macro environment fixed but is not a control in any causal sense: entry into B is selected on exactly the hardship that predicts foreclosure, and Section 2.3 shows how much of the raw gap that selection and the recording of delinquency during a spell account for. If the moratorium and house-price appreciation alone drove the decline, C should look like B; it looks like A. Group C foreclosed at 6.1 per cent against Group B's 1.0, and at thirty-six months after onset the cumulative incidence in C had overtaken the pre-COVID cohort's while B remained near one per cent (Figure 1). Two further features of the table matter for what follows. Group B *prepaid* at 41.2 per cent, which in 2021–2022 is overwhelmingly refinancing: a large share of the forbearance cohort refinanced out once the spell ended. And among the loans that did eventually foreclose, forbearance shifted the terminal event about two years to the right (Figure 16), the same postponement the model's same-household counterfactual produces.

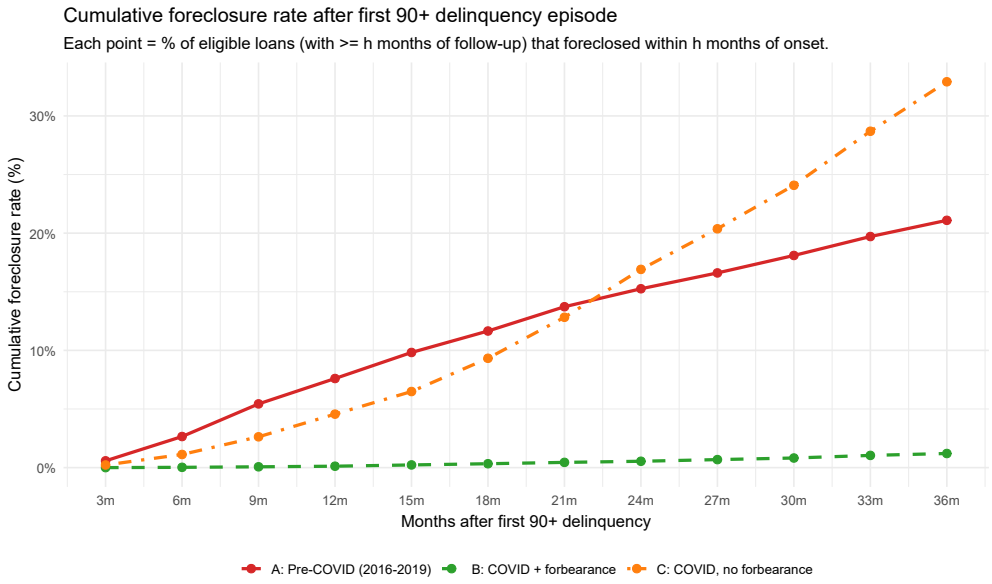


Figure 1: Cumulative foreclosure by months since first 90+ delinquency, for the three groups of Table 2. The COVID-without-forbearance group (C) overtakes the pre-COVID group (A) about two years after onset; the COVID-with-forbearance group (B) stays near one per cent.

2.3 What the association is, and is not

A Cox model of time to foreclosure among these borrowers, controlling for origination LTV, DTI, credit score, note rate, balance and purpose, puts the forbearance hazard at 0.117 of the no-forbearance hazard. Correcting the design for immortal time, by switching the exposure on at

the observed entry month, and for competing risks, by treating prepayment as a competing event rather than independent censoring, moves the ratio to 0.126 (Table 18), and that landmark specification is the corrected estimate the section rests on. Because proportionality is rejected for the forbearance term, the hazard ratio is not a single summary of the whole follow-up, and the cumulative incidence functions of Table 19—which put thirty-six-month foreclosure at 1.5 per cent among borrowers who entered forbearance within a year against 17.2 per cent among those who did not—show how the contrast accumulates over the horizon. Those two figures are descriptive only. They classify on forbearance within twelve months while running the clock from the first 90+ flag, so a loan must survive a year without foreclosing to be eligible for the treated arm: the comparison retains the immortal-time problem that the landmark specification removes, and the gap it reports is correspondingly wider.

Most of that association is definitional. The ninety-day threshold that defines the risk set counts missed payments, and a borrower in forbearance is exercising a contractual right to miss them: entry into forbearance precedes the ninety-day flag for 92 per cent of the treated loans that reach it. Rebuilding the risk set on a delinquency clock that does not accrue during forbearance (Appendix A.4) removes roughly two-thirds of the protective association. What survives is a reduction in thirty-six-month foreclosure of 1.3 to 1.8 percentage points, estimated on the only frame with adequate common support, and a positive association over the whole pre-determined book, because take-up concentrated among the far more distressed. The loan-level data support a bracket, not a point, and none of it is causal: borrowers opt in, servicers differ in how readily they grant relief, and distress severity drives both take-up and foreclosure. What the data cannot answer at all is the counterfactual the paper is about—whether the rate cut would have reached these same borrowers—because no borrower is observed under both instruments. That is what the model is for.

3 What Each Instrument Can Reach: A Two-Period Model

Two inequalities determine, in principle, which households a rate cut reaches and which a deferral reaches, and they organise every cross-sectional pattern the quantitative model produces.

A household enters the crisis with liquid assets x, income y, a mortgage balance b at a locked rate R with n periods remaining, and a house worth Ph. Its scheduled payment is the annuity

$$\overbrace{M(b,R,n)}^{\text{scheduled payment}} = b \cdot \overbrace{k(R,n)}^{\text{annuity factor}}, \qquad k(R,n) = \frac{R(1+R)^n}{(1+R)^n - 1}, \tag{1}$$

and it chooses among four actions. *Pay* leaves consumption $y + x - M$. *Refinance* replaces the loan at the crisis rate $R' < R$ at a cost $\Phi = \phi_r Ph + F$, proportional in the house rather than the balance. *Defer* pays a fraction ξ of the scheduled payment and carries the remainder as a non-interest-bearing balloon due at maturity, so the balance itself is unchanged and the household owes an additional $(1-\xi)M$ at settlement; eligibility is restricted by the payment-burden floor. *Default* surrenders the house, and the household enters the next period as a renter. Two features of the environment do the work. The term is set by the household's age, $n = \min(30, T - a)$, not by the loan's vintage, so a refinance does not reset it. And default is non-recourse with no equity recovery: the walking household forfeits $\max(0, Ph - b)$ rather than realising it.

Three boundaries. Deferral requires a payment burden above a floor, $M \geq \tau^{\text{pti}} y$ with $\tau^{\text{pti}} = 0.20$. Writing $\ell = b/(Ph)$ for leverage, the condition is

$$\overbrace{\ell}^{\text{leverage } b/Ph} \geq \overbrace{\frac{\tau^{\text{pti}}\, y}{k(R,n)\, Ph}}^{\text{floor translated into leverage}}, \tag{2}$$

a ray through the origin in (y, ℓ) space and the vertical line $M/y = \tau^{\text{pti}}$ in payment-burden space. A household refinances when the present value of the payment reduction covers the fee,

$$\underbrace{\big[k(R,n) - k(R',n)\big]\, b}_{\text{annual saving}} \cdot A(R',n) \;\geq\; \underbrace{\phi_r Ph + F}_{\Phi}, \qquad A(R',n) = \frac{1 - (1+R')^{-n}}{R'}. \tag{3}$$

With no equity recovery the household defaults when the value of renting exceeds that of paying, which for given y defines a leverage threshold $\bar{\ell}(y)$ that falls as liquidity tightens. Because downsizing is not permitted, a distressed owner with positive equity cannot monetise it; the choice is to pay, defer or forfeit.

Proposition 1 (The instruments are separated by the eligibility floor)**.** *A rate cut operates through* (3)*, which does not involve* τ^{pti}*; deferral operates through* (2)*, which does not involve* Φ*. The boundary between them in payment-burden space is vertical at* $M/y = \tau^{pti}$ *and does not move with* R'*.*

Figure 22 draws this boundary in the quantitative model, across the five arms of the decomposition, which turn the rate cut and the deferral on in turn.

Proposition 2 (The rate cut's reach falls with age on two margins)**.** *Holding Ph fixed, the left-hand side of* (3) *is increasing in both n and b while the right-hand side is independent of both. Over the life cycle b amortises throughout and $n = \min(30, T - a)$ falls once the remaining horizon binds, so the benefit base shrinks on two margins while the fee base does not shrink at all. The two margins engage in sequence rather than together: with $T = 84$ the cap binds and $n = 30$ at every age up to 54, so before that age the decline is amortisation alone and the horizon margin only adds to it thereafter.*

Proposition 2 is the sharper of the two, because the fee is levied on the house and the benefit accrues on the balance. Table 3 evaluates (3) at the quantitative model's own median leverage by age band: the ratio of present-valued saving to fee crosses one between the 35–44 and 45–54 bands, which is where the simulated refinance share halves. Because n is still at its cap of 30 in all three of those bands, that crossing is produced by amortisation of the balance and not by the shortening horizon, which only begins to operate in the 55–64 band. The analytical condition and the simulated share are independent calculations, reported side by side rather than fitted to one

Table 3: The refinance participation condition at the model's median leverage, by age.

Age	Median ℓ	Balance	n	PV saving	PV / Φ	Refinance share
25–34	0.749	\$187,250	30	\$30,297	3.19	59.4%
35–44	0.558	\$139,500	30	\$22,571	2.38	55.4%
45–54	0.183	\$45,750	30	\$7,402	0.78	29.6%
55–64	0.023	\$5,750	24	\$778	0.08	18.4%
65–74	0.016	\$4,000	14	\$343	0.04	10.5%
75–84	0.007	\$1,750	4	\$52	0.01	4.1%

Notes. Condition (3) at $Ph = \$250{,}000$, $R = 4.95\%$, $R' = 3.69\%$, $\Phi = 0.03\,Ph + \$2{,}000 = \$9{,}500$. Annuity factors are evaluated exactly, not rounded. Median leverage is measured on pre-crisis mortgagor-years in the baseline of Section 6. The final column is the share of crisis mortgagors in each band who refinance when only the rate cut is available.

Two implications carry into the quantitative sections. The instruments are complements in a specific sense: the rate cut reaches low-burden, high-balance, long-horizon households, and deferral is the only instrument available to the rest. And the complementarity has an age dimension that owes nothing to the eligibility floor and everything to the contract's arithmetic, which is why it survives changes in τ^{pti}. The two-period model is silent on two things the quantitative model measures: how much mass sits in each region (Section 6.2 and Figure 22), and what each instrument is worth to the households it reaches (Section 6.8).

4 The Quantitative Model

The model is a life-cycle, heterogeneous-agent, incomplete-markets model of housing tenure and mortgage choice, solved by backward induction and simulated forward on a panel of households hit by an unanticipated aggregate shock. Its structure follows Eichenbaum et al. (2022), Wong (2016) and Low (2022); the additions are a delinquency state, a forbearance technology, and the underwriting restrictions that separate the two instruments. This section states the environment and the recursive problem; Appendix D gives the estimation of the income and aggregate processes.

4.1 Households, income and the aggregate state

Time is annual. Households enter at age 25, work to 64, retire at 65 and live at most to $T = 84$, surviving from age a to $a+1$ with probability π_a taken from U.S. life tables. At death the household values terminal net worth W through a bequest function $\text{Beq}(W) = B\,(\max\{W, 0\}^{1-\sigma} - 1)/(1-\sigma)$. Flow utility is Cobb–Douglas over nondurable consumption c and housing services h with relative risk aversion σ; a renter receives

$$u^{\text{rent}}(c, h) = \frac{(c^{\alpha} h^{1-\alpha})^{1-\sigma} - 1}{(1-\sigma)}, \tag{4}$$

and an owner receives a premium $\theta^{\text{own}} \geq 1$ on housing services,

$$u^{\text{own}}(c, h) = \frac{(c^{\alpha} (\theta^{\text{own}} h)^{1-\alpha})^{1-\sigma} - 1}{(1-\sigma)}. \tag{5}$$

The -1 in each numerator is an additive constant in every period, so lifetime value is not proportional to the consumption aggregate. That is what rules out the usual homothetic consumption-equivalent calculation and motivates the exact decomposition of Appendix E.

Working-life income is the product of a permanent component and an aggregate-exposure term,

$$\begin{aligned} y(\eta, s, a) &= \overbrace{\chi_a \exp(\eta)}^{\substack{\text{permanent income } \bar{y} \\ \text{(age profile} \times \text{idiosyncratic)}}} \cdot \overbrace{f\big(\bar{y}, Y(s)\big)}^{\text{aggregate exposure}}, \\ f(\bar{y}, Y) &= 1 + \overbrace{(Y-1)}^{\text{output gap}}\ \overbrace{\big(\chi_1 + \chi_2 \exp(\chi_3 \bar{y})\big)}^{\text{loading, falling in } \bar{y}}, \qquad \chi_3 < 0, \end{aligned} \tag{6}$$

where χ_a is a hump-shaped age profile, η follows the autoregression $\eta' = \rho_\eta \eta + \varepsilon'$, and $Y(s)$ is detrended aggregate output in state s. Because $\chi_3 < 0$ a contraction falls more heavily on low permanent income, the pattern Guvenen et al. (2014) document; the loadings are rescaled so that the age-40 cross-section has an aggregate labour-income elasticity of about one. From 65 the household receives a pension equal to a replacement ratio ϱ of its own last working-period income, invariant to the aggregate state; retirees are therefore exempt from the income shock and from the crisis job-loss draw described in Section 5.

The aggregate state $s = (r^{\text{short}}, Y, P^h)$ comprises the short real rate, detrended output and the house-price level, respectively. Its normal component is an eight-state Markov chain estimated from FRED and BEA series (Appendix D.2). I append crisis states reached with probability zero from the normal states, so each crisis is an unanticipated (MIT) event: normal-times value functions and the stationary distribution are unaffected by the crisis states' existence, which are nonetheless solved so the simulator can drop households into them. The mortgage rate in state s follows an estimated rule

$$r^m(s) = \max\Big\{ \overbrace{r_0^m}^{\text{intercept}} + \overbrace{\beta_r r^{\text{short}}(s)}^{\text{short-rate pass-through}} + \overbrace{\beta_g \log Y_d(s)}^{\text{output term}}, \overbrace{\underline{r}^m}^{\text{floor}} \Big\}, \tag{7}$$

through which a short-rate cut transmits with loading $\beta_r \approx 0.55$. The rate is *locked* at origination, $r^{\text{lock}} = r^m(s_o)$, and the only way to capture a lower market rate is to refinance. The baseline crisis sets the output multiplier to 0.90 with the short rate cut to zero and house prices *rising* eight per cent, with forbearance available: an income shock against a strong housing market. Because a crisis state's output is anchored to the highest normal state rather than to the state the economy is in, that multiplier delivers a 4.6 per cent fall in output relative to the pre-crisis state rather than a ten per cent one; equation (29) in Appendix D.2 gives the mapping, and output changes are quoted as realised changes throughout. Evaluated at the crisis state the rule (7) returns 4.24 per cent, a cut of only 53 basis points from the normal-state 4.77 per cent, because the output term offsets part of the short-rate term; the crisis mortgage rate is therefore set at 3.69 per cent, a cut of 108 basis points, close to the one observed in 2020.

4.2 The mortgage contract and underwriting

Mortgages are fixed-rate and fully amortising. A loan of balance b at locked rate r^{lock} with term n carries the annuity payment

$$\overbrace{M(b, r^{\text{lock}}, n)}^{\text{annuity payment}} = b\,\frac{r^{\text{lock}}(1+r^{\text{lock}})^n}{(1+r^{\text{lock}})^n - 1}, \qquad \overbrace{b' = b(1+r^{\text{lock}}) - M(b, r^{\text{lock}}, n)}^{\text{amortised balance}} \tag{8}$$

in a period in which the household pays. The term is tied to remaining life, $n = \min(30, \max(T - a, 1))$, so no loan extends past the terminal age and the term shortens near the end of the horizon. A new loan—at purchase, refinance or trade-up—must satisfy a down-payment and a payment-to-income constraint,

$$\overbrace{b \le (1-\delta) P^h(s) h}^{\text{down-payment (LTV) limit}}, \qquad \overbrace{M\big(b, r^m(s), n\big) \le \text{PTI}_{\max} \cdot y(\eta, s, a)}^{\text{payment-to-income limit at the market rate}}, \tag{9}$$

and a refinancing owner additionally faces the *no-cash-in* restriction

$$\overbrace{b_{\text{new}} \;\geq\; b}^{\text{no cash-in: the new balance cannot be smaller}} \qquad , \tag{10}$$

so an owner at the LTV cap whose income has fallen cannot refinance into a lower payment unless appreciation relaxes the cap—which the 2020 price path does and a 2008 path would not. Because (10) permits $b_{\text{new}} > b$, refinancing is also a route to liquidity, as in Chen et al. (2020), and the same three restrictions govern the rate motive and the liquidity motive alike. Housing adjustment is upsize-only: an owner may trade up but may not sell and re-buy a smaller home, so shedding housing is not an alternative to default.[3] Default is non-recourse: a defaulting owner surrenders the home, is released from the balance, incurs a utility penalty ψ and a T_{lock}-period re-purchase lockout, and re-enters as a renter.

Three features of this contract bind asymmetrically across the life cycle and generate the segmentation the paper measures. The payment-to-income test gates refinancing on *flow* income, so a household whose income falls loses the refinance precisely when it most needs a lower payment. The no-cash-in rule blocks a high-LTV household from refinancing when equity is scarce. And the retirement-tied term raises the required payment on late-career originations. Forbearance requires none of these.

4.3 Delinquency and forbearance

Distress does not resolve instantly. An owner may *become delinquent*, skipping the scheduled payment while the balance accrues at the locked rate and entering a state with delinquency count $d = 1$, which must resolve within a year. A delinquent owner may *cure*, paying the arrears plus a late fee ϕ_{late}; *foreclose*; or, in a crisis state, *apply for forbearance*. An owner in good standing may also apply directly. Applying requires eligibility,

$$e(\Omega) = \overbrace{\mathbf{1}\{b > 0\}}^{\text{balance left to defer}} \cdot \overbrace{\mathbf{1}\big\{M(b, r^{\text{lock}}, n) \geq \tau^{\text{pti}}\, y(\eta, s, a)\big\}}^{\text{hardship: payment at least } \tau^{\text{pti}} \text{ of current income}} \qquad , \tag{11}$$

so that an owner with no balance left to defer, or with a scheduled payment below the hardship threshold τ^{pti} of current income, does not have the application in her choice set. Approval is near-automatic in a CARES state, as the statute intended, but it is neither certain nor confined to the crisis: $p^{\text{forb}}(s) = 0.99$ in a CARES state and 0.02 otherwise. The rejection branch of the application

[3] In the 2020 episode households did not trade down, and the owners who would use such a margin are equity-rich households disjoint from the leveraged, income-stressed households the paper studies. A sale-to-rent margin is an open extension.

value

$$V^{\text{app}} = \overbrace{p^{\text{forb}}(s)}^{\text{approval prob.}} V^{\text{forb}} + \overbrace{\left(1 - p^{\text{forb}}(s)\right)}^{\text{rejection prob.}} \overbrace{\begin{cases} \max\{V^{\text{def}}, V^{\text{bdel}}\} & d = 0 \\ \max\{V^{\text{forecl}}, V^{\text{cure}}\} & d = 1 \end{cases}}^{\text{fallback if rejected}} \tag{12}$$

is therefore live rather than decorative. Two fingerprints of this appear in the results and would be impossible under a degenerate approval rule: a small rejected cohort appears in the baseline, drawn from households whose applications the normal-state probability turns down.

Forbearance intensity is a household choice. An approved forbearer selects, once at application, the fraction ξ of the year's scheduled payment it will remit from the thirteen-point menu

$$\overbrace{\xi}^{\text{paid fraction}} \in \Xi = \{0, \tfrac{1}{12}, \tfrac{2}{12}, \ldots, 1\}, \qquad \overbrace{1-\xi}^{\text{deferred fraction}} = \frac{\text{deferred months}}{12}, \tag{13}$$

one point per deferred month, and commits to it for the year. The household then remits ξM plus maintenance.

In the baseline settlement regime the mortgage balance *does not* capitalise. The deferred amount $(1-\xi)M$ is carried as a separate non-interest-bearing balloon, due at the original maturity and callable earlier on sale, refinance or payoff, while the scheduled balance evolves exactly as it would have: $b' = b$ over the forbearance year. This is the deferment-to-maturity settlement Fannie Mae adopted in LL-2021-07, and it is why the mean change in the balance over the forbearance year is zero in every arm rather than merely small. The balloon is priced in the household's problem, so deferral is not free at the margin: what limits take-up is the obligation the household takes on, not capitalised interest, and both the application and its intensity are its own decisions. Eligibility is a restriction on the choice set rather than a servicer screen, and carries no entry fee.

How the deferred balloon is carried. The balloon is priced without entering Ω as a continuous state. Because forbearance is taken once per reset and the deferred amount is bounded by one year of scheduled payments, the outstanding obligation is reconstructed from the binary indicator f as a single annual payment rather than tracked as a balance. It is deducted from resources at every event that settles the loan—refinancing, selling and rebuying, and paying the mortgage off—and so appears in the budget constraints of equations (21), (22) and (20) as a charge against total resources. It is *not* deducted in the default branch, where the household walks away from the loan and the obligation with it, nor in the terminal bequest of equation (28). Two consequences follow and neither is hidden. The reconstruction charges the obligation once per settlement event, so a household that refinances after a spell and later retires the new loan pays it twice; take-up measured under this pricing is therefore a lower bound on what an exactly-priced balloon would deliver. And because the bequest carries no balloon, a household that defers and dies still owing leaves a slightly larger estate than it should. The two push in opposite directions and neither is large enough to reverse the comparisons the paper draws, but the measure to read is the direction of take-up across

settlement regimes rather than its level.

Who may apply, and what happens to the indicators. Both an owner in good standing and a delinquent owner may apply, but not on the same terms. A delinquent applicant is admitted only in a state that resets the programme, and the reset also clears f for her, so a household that has already used forbearance is not barred from a second spell across a reset. Her arrears are not forgiven and not separately charged: the missed payment has already accrued into the balance, $b(1 + r^{\text{lock}})$, and the forbearance year amortises that larger balance, so no late fee ϕ_{late} is levied on entry. An approved applicant of either kind continues as an owner in good standing, $d' = 0$, with $f' = 1$; the indicator clears on curing from delinquency, on clearing the mortgage, and on any transition into a reset state. A rejected applicant falls back on the choice set her delinquency status leaves her, which is what equation (12) conditions on. Section 6.6 varies the settlement regime. The two regimes carried there—deferral to maturity and a level repayment plan—move take-up by less than a tenth of a percentage point, and both avoid capitalising the deferred amount. Whether settlement is a large lever or a small one therefore remains open: the case that would separate them is capitalised reinstatement, in which the deferred amount accrues interest, and that regime is not among the solves reported here.

4.4 The recursive problem

State variables. A household is summarised by the state vector

$$\Omega = \big(\overbrace{x}^{\substack{\text{liquid}\\ \text{assets}}}, \overbrace{b}^{\substack{\text{mortgage}\\ \text{balance}}}, \overbrace{h}^{\substack{\text{house}\\ \text{size}}}, \overbrace{r^{\text{lock}}}^{\substack{\text{locked}\\ \text{rate}}}, \overbrace{\eta}^{\substack{\text{income}\\ \text{shock}}}, \overbrace{s}^{\substack{\text{aggregate}\\ \text{state}}}, \overbrace{d}^{\substack{\text{delinq.}\\ \text{count}}}, \overbrace{f}^{\substack{\text{forb.}\\ \text{used}}}, \overbrace{\ell}^{\substack{\text{lockout}\\ \text{periods}}}, \overbrace{a}^{\text{age}} \big). \tag{14}$$

Here $x \geq 0$ is liquid saving carried at the risk-free return r^f; $b \geq 0$ is the mortgage balance and r^{lock} the rate locked at its origination (both zero for a non-owner); h is the owned house size; η and s are the idiosyncratic and aggregate states of Section 4.1; $d \in \{0, 1\}$ is the delinquency count; $f \in \{0, 1\}$ indicates that the forbearance option is currently in use and so unavailable; it is *not* a once-per-life marker—it clears when the household cures from delinquency or clears the mortgage, and in a reset state, so a household may forbear more than once over its life; $\ell \in \{0, 1, \ldots, T_{\text{lock}}\}$ is the number of post-default lockout periods still to run; and a is age. Tenure is read off the state rather than carried separately: a household is a *non-owner* when $b = h = 0$, an *owner in good standing* when it owns and $d = 0$, and a *delinquent owner* when it owns and $d = 1$.

The continuation operator. Let Ω' denote next period's state, split into the *endogenous* block $(x', b', h', r^{\text{lock}\prime}, d', f', \ell')$ chosen this period and the *exogenous* block (η', s') drawn from the income and aggregate Markov chains of Section 4.1. Every continuation value is the survival-weighted

expectation

$$\mathbb{E}V' \;\equiv\; \beta\Big[\;\overbrace{\pi_a}^{\text{survive}}\;\overbrace{\mathbb{E}_{\eta',s'|\eta,s}\,V\big(\Omega',a{+}1\big)}^{\text{expected continuation value}}\;+\;\overbrace{(1-\pi_a)}^{\text{die}}\,\overbrace{\mathrm{Beq}\big(W'\big)}^{\text{bequest}}\;\Big],\qquad \overbrace{W' = x' + P^h(s)\,h' - b'}^{\text{terminal net worth}}, \tag{15}$$

with discount factor β and age-a survival probability π_a. The tenure superscript on V inside (15)—N, O, or D—is the tenure the current choice transitions into. Every budget below is understood to also impose the consumption floor $c \geq c_{\min}$ and the no-borrowing condition $x' \geq 0$, which I leave implicit. New price and cost symbols are ρ^{rent}, the unit rental rate on rental services h^{rent}, which the household chooses subject to a floor $\underline{h}^{\text{rent}}$ and the ceiling $\bar{h}$ that also bounds owned houses; δ_m, the maintenance rate; the purchase fee Φ^{new}, the refinance cost $\Phi^{\text{refi}} = \phi_r P^h(s)h + F_r$ (a proportional plus a fixed component), and the selling cost ϕ^{sell} levied on the value of a home that is sold; the late fee ϕ_{late}; the default utility penalty ψ; and the lockout length T_{lock}. The floor $\underline{h}^{\text{rent}}$ binds only where a household could not otherwise afford any rental services; with Cobb–Douglas preferences the interior choice is $h^{\text{rent}} = (1-\alpha)X/\big(\rho^{\text{rent}}P^h(s)\big)$ in disposable resources X, so it is the ceiling rather than the floor that binds over most of the state space.

Non-owner. A non-owner rents or, once the post-default lockout has expired ($\ell = 0$), may buy:

$$V^N(x,\eta,s,\ell,a) = \max\Big\{\;\overbrace{V^{\text{rent}}(x,\eta,s,\ell,a)}^{\text{keep renting}},\;\; \mathbf{1}_{\{\ell=0\}}\,\overbrace{V^{\text{purch}}(x,\eta,s,a)}^{\text{buy a home}}\;\Big\}. \tag{16}$$

A *renter* rents h^{rent} units of housing services and continues as a non-owner with the lockout counting down, $\ell' = \max\{\ell - 1, 0\}$:

$$\begin{aligned} V^{\text{rent}} \;=\; & \max_{c,\,x',\,h^{\text{rent}}} \;\; u^{\text{rent}}(c,h^{\text{rent}}) + \mathbb{E}V'^{N} \\ \text{s.t.} \quad c + x' \;=\; & \overbrace{x(1+r^f)}^{\text{cash on hand}} + \overbrace{y(\eta,s,a)}^{\text{income}} - \overbrace{\rho^{\text{rent}}P^h(s)\,h^{\text{rent}}}^{\text{rent}}, \\ & \underline{h}^{\text{rent}} \leq h^{\text{rent}} \leq \bar{h}. \end{aligned} \tag{17}$$

A *buyer* chooses a house h' and a loan b' at the current market rate, $r^{\text{lock}\prime} = r^m(s)$, pays the down payment and the purchase fee out of cash on hand, and enters next period as an owner in good

standing ($d' = 0$, $f' = f$):

$$
\begin{aligned}
V^{\text{purch}} = \max_{h', b', c, x'} \; & u^{\text{own}}(c, h') + \mathbb{E}V'^{O} \\
\text{s.t.} \quad c + x' = \; & \overbrace{x(1+r^f) + y(\eta, s, a)}^{\text{resources}} - \overbrace{\left(P^h(s)h' - b'\right)}^{\text{down payment}} - \overbrace{\Phi^{\text{new}} P^h(s)h'}^{\text{purchase fee}}, \\
& \overbrace{b' \le (1-\delta)\, P^h(s)\, h'}^{\text{down-payment (LTV) limit}}, \qquad \overbrace{M\big(b', r^m(s), n\big) \le \text{PTI}_{\max} \cdot y(\eta, s, a)}^{\text{payment-to-income limit}}, \\
& n = \min\{30, \max(T - a, 1)\}.
\end{aligned}
\tag{18}
$$

Owner in good standing. An owner with $d = 0$ chooses among paying, refinancing, trading up, applying for forbearance (only while the option is free, $f = 0$, and the eligibility condition (11) holds), becoming delinquent, and defaulting:

$$
V^O(x, b, h, r^{\text{lock}}, \eta, s, f, a) = \max\Big(\left\{ V^{\text{pay}},\; V^{\text{refi}},\; V^{\text{up}},\; V^{\text{bdel}},\; V^{\text{def}} \right\} \cup \left\{ V^{\text{app}} \;:\; f = 0,\; e(\Omega) = 1 \right\} \Big).
\tag{19}
$$

Pay remits the scheduled payment and maintenance, amortises the balance, and keeps the locked rate:

$$
\begin{aligned}
V^{\text{pay}} = \max_{c, x', p} \; & u^{\text{own}}(c, h) + \mathbb{E}V'^{O} \\
\text{s.t.} \quad c + x' = \; & \overbrace{x(1+r^f) + y}^{\text{resources}} - \overbrace{M(b, r^{\text{lock}}, n)}^{\text{mortgage payment}} - \overbrace{p}^{\text{prepayment}} - \overbrace{\delta_m P^h(s) h}^{\text{maintenance}}, \\
& \overbrace{b' = b(1 + r^{\text{lock}}) - M(b, r^{\text{lock}}, n) - p}^{\text{amortised balance, net of prepayment}}, \quad 0 \le p \le b, \quad r^{\text{lock}\prime} = r^{\text{lock}},\; d' = 0.
\end{aligned}
\tag{20}
$$

Refinance replaces the loan with a new balance b_{new} at the market rate, pays the refinance cost Φ^{refi}, and resets the locked rate; it must clear the LTV, payment-to-income, and no-cash-in tests:

$$
\begin{aligned}
V^{\text{refi}} = \max_{b_{\text{new}}, c, x'} \; & u^{\text{own}}(c, h) + \mathbb{E}V'^{O} \\
\text{s.t.} \quad c + x' = \; & \overbrace{x(1+r^f) + y}^{\text{resources}} + \overbrace{(b_{\text{new}} - b)}^{\text{net cash from refi}} - \overbrace{\delta_m P^h(s) h}^{\text{maintenance}} - \overbrace{\Phi^{\text{refi}}}^{\text{refi cost}}, \\
& \overbrace{b_{\text{new}} \le (1-\delta) P^h(s) h}^{\text{LTV}}, \qquad \overbrace{b_{\text{new}} \ge b}^{\text{no cash-in}}, \\
& \overbrace{M\big(b_{\text{new}}, r^m(s), n\big) \le \text{PTI}_{\max} \cdot y}^{\text{payment-to-income, at the market rate}}, \quad r^{\text{lock}\prime} = r^m(s),\; d' = 0.
\end{aligned}
\tag{21}
$$

The origination year carries no payment, and what that does. The budget constraint above deducts the refinancing cost and maintenance but no scheduled payment: a household that refinances makes no mortgage payment in the year it originates, and begins paying on the new loan

in the year after. A household that simply pays, by contrast, remits the full annuity. Measured on the baseline crisis this is worth a median of \$10,437 to each of the 6,162 refinancers, and \$61.3 million across them.

Two things follow. The comparison the paper draws between the instruments is unaffected, because an approved forbearer choosing $\xi = 0$ also remits nothing that year: both instruments deliver a full year of deferred payment in the crisis, and the per-recipient figures are therefore like for like. What is affected is the *level* of refinancing's crisis-year cash flow set against the data, where the interval between the retired loan's last payment and the new loan's first is of the order of a month rather than a year. An annual model cannot represent a month, so this is a corner the time aggregation forces rather than a modelling choice made on its merits, and the crisis-year cash-flow figures should be read as an upper bound on what refinancing delivers in the year itself. The lifetime interest saving, which is computed from the rate reduction over the remaining term rather than from the skipped year, does not inherit the bias.

Trade-up sells the current home, realising its equity, and buys a strictly larger home $h' > h$ under the same origination constraints:

$$
\begin{aligned}
V^{\mathrm{up}} = \max_{h'>h,\, b_{\mathrm{new}},\, c,\, x'} \; & u^{\mathrm{own}}(c,h') + \mathbb{E}V'^{O} \\
\text{s.t.} \quad c + x' = \overbrace{x(1+r^f)+y}^{\text{resources}} + & \overbrace{\big((1-\phi^{\mathrm{sell}})P^h(s)h - b\big)}^{\text{equity released on sale, net of the selling cost}} \\
& - \overbrace{\big(P^h(s)h' - b_{\mathrm{new}}\big)}^{\text{down payment on new home}} - \overbrace{\Phi^{\mathrm{new}}P^h(s)h'}^{\text{purchase fee}}, \\
& \overbrace{b_{\mathrm{new}} \le (1-\delta)P^h(s)h'}^{\text{LTV}}, \quad \overbrace{M\big(b_{\mathrm{new}}, r^m(s), n\big) \le \mathrm{PTI}_{\max}\cdot y}^{\text{payment-to-income}}, \; r^{\mathrm{lock}\prime} = r^m(s).
\end{aligned}
\tag{22}
$$

Become delinquent skips the payment—remitting only maintenance while the balance accrues at the locked rate—and moves to the delinquent state:

$$
\begin{aligned}
V^{\mathrm{bdel}} = \max_{c,\,x'} \; & u^{\mathrm{own}}(c,h) + \mathbb{E}V'^{D} \\
\text{s.t.} \quad c + x' = \overbrace{x(1+r^f)+y}^{\text{resources}} - \overbrace{\delta_m P^h(s)h}^{\text{maintenance}}, & \quad \overbrace{b' = b(1+r^{\mathrm{lock}})}^{\text{balance accrues, no payment}}, \; d' = 1.
\end{aligned}
\tag{23}
$$

Default surrenders the home and, because default is non-recourse, walks away from the balance b; it bears the utility penalty ψ and the lockout $\ell' = T_{\mathrm{lock}}$, and continues as a non-owner:

$$
\begin{aligned}
V^{\mathrm{def}} = \max_{c,\,x',\,h^{\mathrm{rent}}} \; & u^{\mathrm{rent}}(c,h^{\mathrm{rent}}) - \overbrace{\psi}^{\text{default penalty}} + \mathbb{E}V'^{N}\big|_{\ell'=T_{\mathrm{lock}}} \\
\text{s.t.} \quad c + x' = \overbrace{x(1+r^f)+y}^{\text{resources}} - & \overbrace{\rho^{\mathrm{rent}}P^h(s)h^{\mathrm{rent}}}^{\text{rent}} \qquad \text{(balance } b \text{ released, non-recourse).}
\end{aligned}
\tag{24}
$$

Delinquent owner. An owner with $d = 1$ cures, forecloses, or—in a CARES/reset state, if the option is free and the eligibility condition (11) holds—applies for forbearance:

$$V^D(x, b, h, r^{\text{lock}}, \eta, s, d, f, a) = \max\Big(\{V^{\text{cure}},\ V^{\text{forecl}}\} \ \cup\ \{V^{\text{app}}\ :\ f = 0,\ e(\Omega) = 1\} \Big). \tag{25}$$

Cure pays the arrears plus a late fee ϕ_{late} on the missed payment and returns to good standing ($d' = 0$):

$$\begin{aligned} V^{\text{cure}} &= \max_{c,\,x'}\ u^{\text{own}}(c, h) + \mathbb{E}V'^{O} \\ \text{s.t.}\quad c + x' &= \overbrace{x(1 + r^f) + y}^{\text{resources}} - \overbrace{(1 + \phi_{\text{late}})\, M(b, r^{\text{lock}}, n)}^{\text{arrears + late fee}} - \overbrace{\delta_m P^h(s) h}^{\text{maintenance}}. \end{aligned} \tag{26}$$

Foreclose, V^{forecl}, is the delinquent-state counterpart of default (24): surrender, non-recourse release, penalty ψ, lockout $\ell' = T_{\text{lock}}$, continue as a non-owner. The *application* value is the approval-weighted average (12). *Approved forbearance* defers the fraction $1-\xi$ of the payment, with ξ chosen from the menu Ξ of equation (13), so the household remits only ξM plus maintenance; the balance does not capitalise ($b' = b$), the deferred amount $(1 - \xi)M$ becomes a non-interest-bearing balloon due at maturity, and the option is marked in use ($f' = 1$) until it resets:

$$\begin{aligned} V^{\text{forb}} &= \max_{c,\,x'}\ u^{\text{own}}(c, h) + \mathbb{E}V'^{O} \\ \text{s.t.}\quad c + x' &= \overbrace{x(1 + r^f) + y}^{\text{resources}} - \overbrace{\xi\, M(b, r^{\text{lock}}, n)}^{\text{partial (paid) payment}} - \overbrace{\delta_m P^h(s) h}^{\text{maintenance}}, \\ &\qquad \overbrace{b' = b}^{\text{balance unchanged}}, \quad \overbrace{(1 - \xi)\, M(b, r^{\text{lock}}, n)}^{\text{added to the balloon}}, \quad f' = 1. \end{aligned} \tag{27}$$

Terminal condition and solution. At the terminal age the value equals the bequest of net worth,

$$V(\Omega, T) = \text{Beq}\big(W\big), \qquad W = \overbrace{x}^{\text{liquid saving}} + \overbrace{\max\big\{0,\ P^h(s)\, h - b\big\}}^{\text{housing equity, non-recourse at death}}. \tag{28}$$

The problem is solved by backward induction on a discretised state space with 25 asset and 25 mortgage nodes placed where the refinancing and forbearance margins bind, and simulated on a panel of 30,000 households under common random numbers; each counterfactual flips one lever relative to a baseline, re-solving when the lever changes the household's problem and reusing the baseline solve when it changes only the realised path.

5 Calibration, Identification and Fit

5.1 Data and parameters

Four public datasets discipline the model. The 2019 Survey of Consumer Finances supplies the steady-state moments—the homeownership rate, wealth-to-income ratios, and the distributions of LTV and payment-to-income among owners. The Panel Study of Income Dynamics identifies the idiosyncratic income process and the age-earnings profile, following Guvenen et al. (2021). The Fannie Mae loan-level file supplies the crisis-year moments and the reduced-form facts of Section 2. FRED and BEA series supply the aggregate state process and the mortgage-rate rule (7).

Table 4: Model parameters.

Parameter	Description	Value	Source
Demographics and preferences			
T, a^{ret}	Horizon (ages 25–84), retirement age	60 periods ($T = 84$), 65	Set
β, σ, α	Discount factor, risk aversion, consumption share	0.94, 2.0, 0.85	Standard
r^f	Return on liquid saving (constant across aggregate states)	0.02	Set
B	Bequest weight	20.0	SCF old-age assets
θ^{own}	Ownership premium	2.0	SCF homeownership rate
ψ	Default disutility	20.0	Fannie Mae lifetime foreclosure
T_{lock}, ϱ	Post-default lockout, pension replacement	5 yrs, 1.0	Set
Income process			
ρ_η, σ_η, n_η	Persistence, innovation s.d., nodes	0.961, 0.131, 11	PSID
(χ_1, χ_2, χ_3)	Output-exposure loadings	(0.807, 11.11, −3.31)	Estimated
Housing and mortgage market			
δ	Minimum down payment	0.20	Landvoigt et al. (2015)
δ_m, ϕ^{sell}	Maintenance rate, selling cost	0.03, 0.09	Set
Φ^{new}	Purchase fee ($\times P^h h$)	0.03	Turnover frequency
Φ^{refi}	Refinance cost ($\times P^h h$ + fixed)	0.03 + \$2k	2020 refinancing share
$\underline{h}^{\text{rent}}$, $\bar{h}$	Rental floor, size ceiling	0.10, 2.0	Set
h	Owned house-size grid	10 nodes on [1.0, 2.0]	Set
ρ^{rent}	Unit rental rate ($\times P^h(s) h^{\text{rent}}$)	0.10	Set
$c_{\min}$	Consumption floor	0.05 ($\approx$ \$3k)	Set
$\text{PTI}_{\max}$	Maximum payment-to-income at origination	0.60	Set (see text)
λ, n	Recourse (0 = non-recourse), term	0, $\min(30, T-a)$	Set
Mortgage rate, crisis, forbearance			
$(r_0^m, \beta_r, \beta_g)$	Rate rule (7)	(.0385, .552, −.051)	FRED/BEA
$\underline{r}^m$	Floor on the rate rule	0.005	Set
$r^{m,\text{crisis}}$	Crisis mortgage rate, overriding the rule	0.0369	2020 rate cut
$p^{\text{forb}}(s)$	Forbearance approval, crisis / normal state	0.99, 0.02	CARES
τ^{pti}	Eligibility floor on M/y	0.20	Set (see text)
u^{crisis}, ρ^{u}	Crisis job-loss incidence, replacement rate	0.10, 0.35	CPS/BLS
ϕ_{late}	Late fee on cure	0.05	Loewenstein et al. (2025)
N_A, N_M, n_{agg}	Asset, mortgage nodes; normal agg. states	25, 25, 8	Set

Notes. The baseline crisis sets the output multiplier to 0.90, which is a 4.6 per cent fall in output relative to the pre-crisis state once the anchoring of equation (29) is applied; the short rate is cut to zero, house prices rise eight per cent, and a job-loss draw reaches ten per cent of working-age mortgaged owners at a 35 per cent replacement rate for the drawn spell. Retirees are exempt from both the aggregate shock and the draw. One model unit is approximately \$60,000.

Table 4 lists the parameters. The model is calibrated rather than estimated. The minimum down payment is the pre-boom standard of Landvoigt et al. (2015). The payment-to-income ceiling at origination is set at 0.60, deliberately above any observed underwriting standard—Greenwald (2026) sets 36 per cent to match the pre-boom underwriting standard and contrasts it with 45 per cent over

2014–2016—so that the test binds only where income has collapsed and the segmentation the model produces cannot be an artefact of a tight cap excluding the forbearance cohort by construction; Section 6.6 reports what removing it does. The eligibility floor $\tau^{\text{pti}} = 0.20$ is set rather than fitted, and it is set at the mean of the object it gates. In the 2019 SCF the mean mortgage payment-to-income ratio among mortgaged owners lies between 0.166 and 0.226 in every age bin from 25 to 84, and rises with leverage from 0.126 below an LTV of 0.10 to 0.232 at an LTV of 0.80–0.90; the model reproduces both profiles (Figure 3). A floor at 0.20 therefore sits at about the mean payment burden of a mortgagor carrying meaningful leverage, and above it for one carrying little. It is not, however, a typical borrower's payment: the burden distribution is right-skewed, so its mean lies well above its median, and only 30.7 per cent of pre-retirement SCF mortgagors carry a burden above 0.20, against 41.2 per cent in the model. Those two shares describe the payment a household actually makes; the forbearance rule is tested against the scheduled amortising payment alone, which is weakly smaller, and under that payment the floor admits about a fifth of pre-retirement mortgagors at the crisis. Section 6.7 reports it. It also sits below the 31 per cent affordability standard HAMP applied (Agarwal et al., 2017), which is the right direction given that CARES forbearance was self-certified hardship rather than an underwritten test.

What the floor stands for. The floor is a reduced form, and it is worth saying what for. The CARES Act set no quantitative threshold: it required a borrower with a federally backed mortgage to attest to a pandemic-related hardship, and servicers were directed to grant the request. What rationed relief in practice was not a payment-burden test but the frictions around one—whether a borrower knew the option existed, what it was believed to cost later, and how readily a particular servicer granted it, which Kim et al. (2024) show explains a substantial part of who was actually reached. None of those is in the model. The floor stands in for all of them, and it is the only thing doing so: the application itself carries no fee and no stigma here, so if the floor were removed the option would be free and universal among mortgagors, which is neither the statute nor the data. Two consequences should be read with that in mind. The floor has no claim to be the right reduced form, only a transparent one, which is why Section 6.7 reports reach as a property of this threshold rather than as a schedule over thresholds. And because the test is on payment burden rather than on hardship, it admits households the crisis did not touch and excludes households it did: Appendix G measures three of the first for every one of the second. A rule conditioned on a realised income fall would target better, and would be a different model.

The origination figures of Section 2 are not the comparison: at 33.6 per cent for never-event loans those are back-end debt-to-income, covering all monthly debt service, whereas τ^{pti} and the survey series above are the scheduled mortgage payment alone. Where the floor sits therefore governs how large the programme can be, and Section 6.7 returns to it. The discount factor, consumption share, ownership premium, bequest weight and default disutility match the 2019 SCF homeownership rate, working- and late-life wealth-to-income ratios, and the Fannie Mae lifetime foreclosure rate.

The crisis is a job-loss shock as well as an output shock: ten per cent of working-age mortgaged

owners are drawn into a hardship spell at a 35 per cent replacement rate, and the draw operates in every arm of every experiment, including the no-policy arm. The single-lever decompositions below therefore isolate the effect of each *instrument* against a common shock, not the effect of the shock's components against one another. In the simulated panel, 9.03 per cent of working-age mortgaged owners fall below half their own prior income at the crisis, against none of the retired.

5.2 What is targeted, and what is a test

Table 5 maps each parameter to the moment it was set against, so that the fit reported below can be read for what it is. The crisis refinancing share is a calibration input and is not reported anywhere as evidence of fit. The forbearance take-up rate is not an input: the eligibility floor is set, take-up is an outcome, and it is reported as a test the model can fail.

Table 5: Parameter-to-target mapping.

Parameter	Target moment	Source
(a) Assigned externally		
β, σ, α; π_a	—	standard; SSA life tables
(ρ_η, σ_η); (χ_1, χ_2, χ_3)	—	PSID; estimated (Appendix D)
δ; ϕ_{late}; $(r_0^m, \beta_r, \beta_g)$	—	Landvoigt et al. (2015); Loewenstein et al. (2025); OLS
δ_m, $\text{PTI}_{\max}$, T, a^{ret}, ϱ, λ, n, T_{lock}, τ^{pti}	—	set by design
(b) Calibrated to a stated target (inputs, not tests)		
$\theta^{\text{own}} = 2.0$	aggregate homeownership rate	2019 SCF
$B = 20.0$	old-age asset holdings	2019 SCF
$\psi = 20.0$	lifetime foreclosure rate	Fannie Mae
$\Phi^{\text{new}} = 0.03$	purchase and turnover frequency	industry estimates
$\Phi^{\text{refi}} = 0.03 + \$2\text{k}$	crisis refinancing share	Fannie Mae, 2020
$p^{\text{forb}} = 0.99$ in the crisis	approval on request	CARES Act
(c) Untargeted (validation)		
—	life-cycle income, asset, ownership, LTV profiles	2019 SCF, 2018 PSID
—	crisis take-up rate; composition of crisis decisions	Fannie Mae, 2020
—	forbearance duration distribution	Sánchez and Wilkinson (2022)

Notes. The aggregate homeownership *rate* is targeted; its *profile* over the life cycle is not. The near-zero 2020–21 foreclosure rate is excluded from panel (c): the moratorium made it administratively determined, so reproducing it would test the moratorium's presence rather than the model's behaviour. What forbearance does to foreclosure is instead addressed against an explicit no-forbearance counterfactual, which the Introduction summarises and which this version does not reproduce in full.

5.3 Fit

Figure 2 overlays the model's pre-crisis life-cycle profiles on the SCF and PSID. Four features bear on the results.

The model reproduces the PSID age profile of mean income by construction, to within about six per cent on average and twelve at the worst age, and its median sits between the two surveys' medians at every age. Median total assets track the SCF through working life. Median *liquid* assets do not: the model's median household holds no liquid buffer for most of its life, where the data's holds tens of thousands of dollars. The gap is partly definitional—the SCF series nets only the primary residence, so it includes retirement accounts, vehicles and business equity—but its direction matters for the exercise. A household with no liquid buffer can absorb a payment shock only through

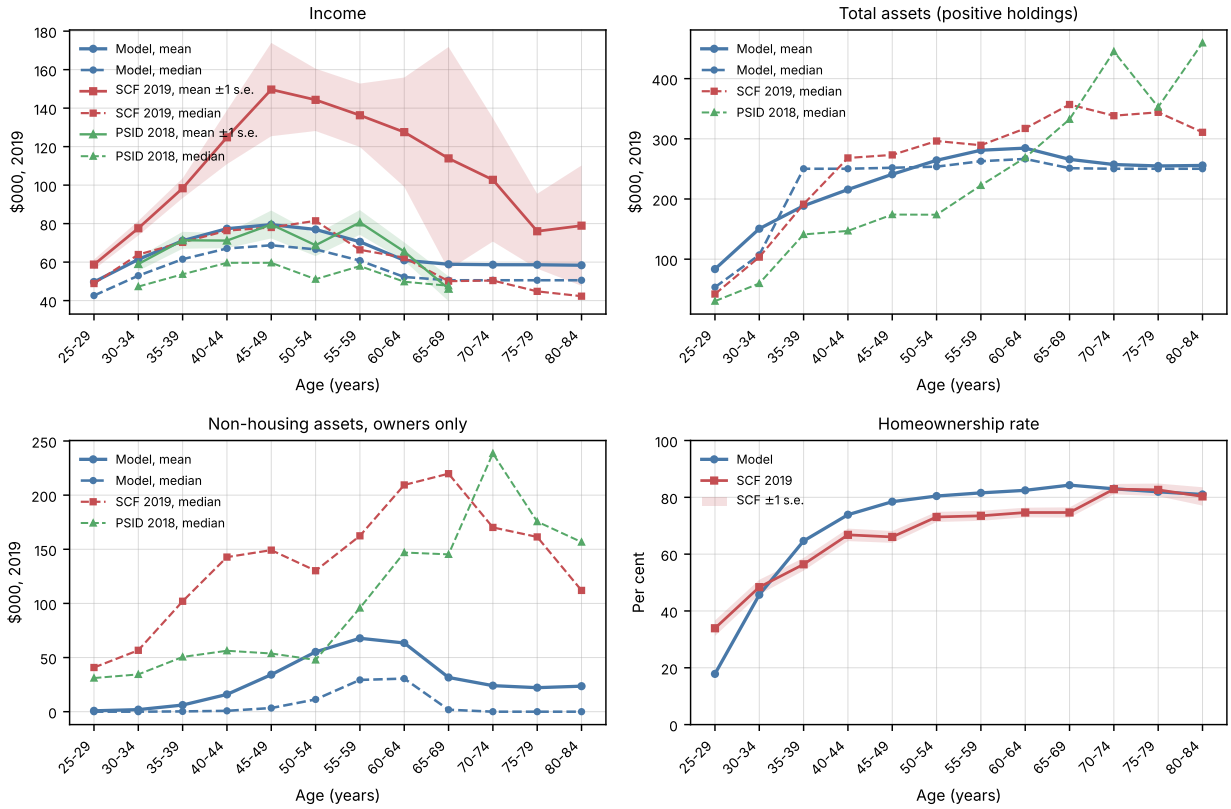


Figure 2: Life-cycle fit: model against the 2019 SCF and the 2018 PSID.

Notes. Solid lines are means, dashed lines medians; the homeownership panel is a rate. Model profiles are averages over all pre-crisis periods of the simulated panel in five-year bins, converted at \$60,000 per model unit; all series in 2019 dollars. The wealth panels report survey medians only, since both surveys carry long right tails the model's single savings vehicle cannot reproduce. Shaded bands are plus and minus one standard error on the data mean where a mean is shown.

the instruments the paper studies, so the model should if anything overstate the demand for relief; that it nonetheless understates observed take-up makes the counterfactuals a lower bound.

At ages 25–29 mean LTV is 0.757 against the SCF's 0.742 and mean payment-to-income 0.200 against 0.226; by 45–49 the payment-burden gap is under a percentage point, and the burden rises with LTV at close to the observed gradient (Figure 3). After the mid-forties the model's mortgagors are bimodal—a spike of nearly-paid-off loans and a body carrying a substantial payment—because model households can only amortise down, with no cash-out, home-equity borrowing or trade-down to re-lever them. Over the pre-retirement window, 41.2 per cent of the model's mortgagors carry a payment burden above 0.20 against 30.7 per cent in the SCF, so the model places somewhat more weight in the upper tail of the burden distribution than the survey does. This is a statement about fit rather than about eligibility: both shares use the payment a household actually makes, and the model's includes voluntary prepayment, whereas the forbearance rule is tested against the scheduled amortising payment alone. On these same households that payment puts 11.6 per cent above the floor.

Appendix G.1 compares the composition of outcomes in the simulated crisis with the corresponding episode in the loan data, classified by the data's own rule.

The aggregate rate is 69.8 per cent against 64.9 in the SCF; the profile is too low at the young end, because households are born with no assets and cannot make a down payment at 25, and too high at the old end for a reason that is mechanical. Housing adjustment is upsize-only, there is no sale-to-rent option for a solvent owner, and the only route out of ownership is default, which costs the utility penalty ψ and forfeits all home equity. Retirees draw a constant pension, so no old-age income decline pushes them out, and the bequest values the house at market less balance with no selling cost deducted, so dying an owner is never penalised relative to selling. Once a household owns, it has almost no way to stop owning. Every arm shares the same panel, so the comparisons between arms are unaffected; the levels should not be read as population forecasts.

The loan-level file records income at origination and no borrower age, so the model is conditioned the way the data can be: every simulated mortgage is tagged with its origination characteristics, and the crisis-period forbearance cohort is compared with all mortgagors on each side separately. The model's forbearance takers differ from all loans by 13.3 percentage points of origination payment burden against 4.6 in the data, and by -0.7 points of origination LTV against 7.0: the model selects on payment burden and on nothing else, because origination leverage is pinned at the down-payment cap for nearly every simulated loan and house size is close to degenerate. Re-estimating the opening linear probability model on simulated data gives the same sign on payment burden—between three and ten times steeper in the model—and the opposite sign on leverage; adding age at origination, which the data cannot, drives the leverage coefficient to zero, so what presents as a leverage gradient in the data is at least partly a life-cycle gradient.

The 2020 mortgage market. The crisis-year moments are computed over owner-mortgagors of all ages, matching the loan-level data; every result from Section 6 on is reported for households

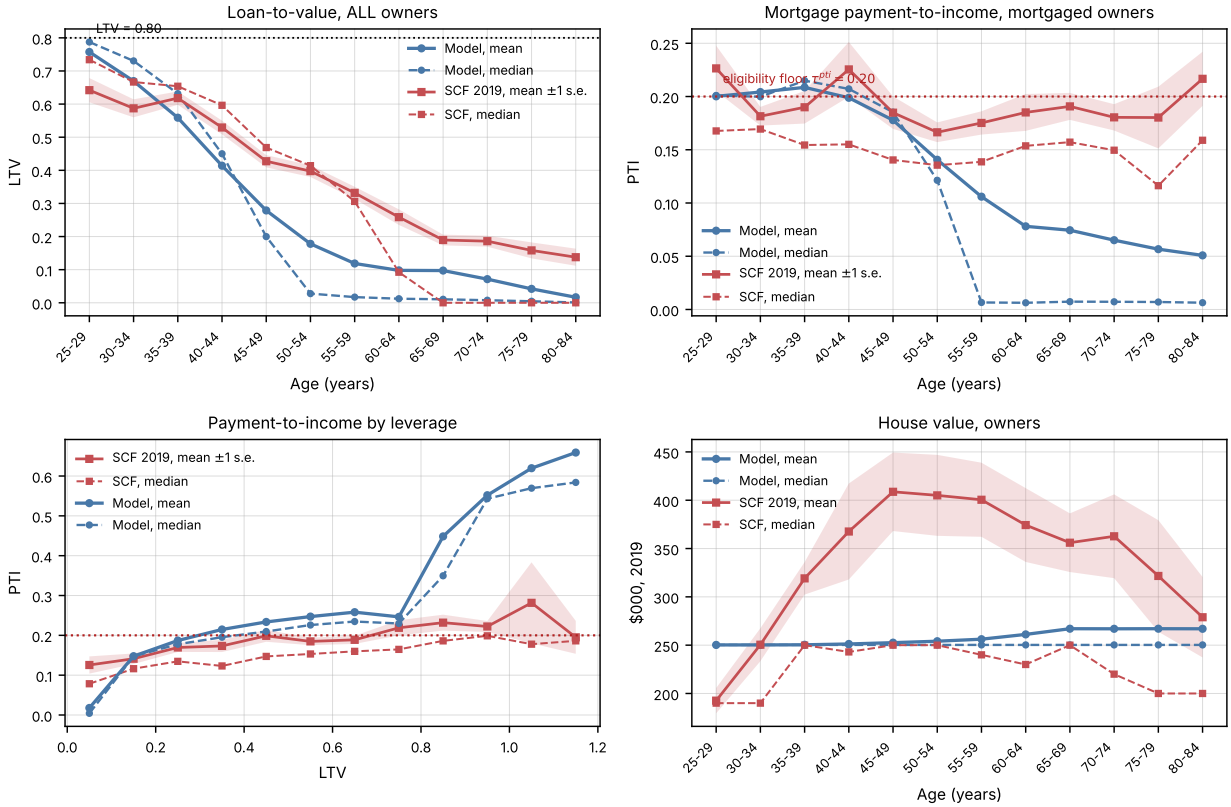


Figure 3: Leverage and payment burden: model against the 2019 SCF.

Notes. Pre-crisis periods of the simulated panel. Solid lines are means, dashed medians; shaded bands are plus and minus one standard error on the SCF mean. Definitions follow the survey exporters: LTV is mortgage balance over house value and payment-to-income is the scheduled mortgage payment over income, restricted to $0 < \text{PTI} < 5$. The dotted lines mark the down-payment cap $1 - \delta = 0.80$ and the eligibility floor $\tau^{\text{pti}} = 0.20$. **The leverage panel is drawn over *all* owners**, including those who own outright at an LTV of zero, which is why its SCF mean at ages 25–29 sits below the 0.742 quoted in the text: that figure is the mortgagor-only sample, matched to the model's mortgagors. The payment-to-income panels are mortgagors on both sides. In the payment-to-income-by-leverage panel the horizontal axis runs past the origination cap $1 - \delta = 0.80$. Nothing originates above that cap: the mass beyond it is delinquent loans, whose balance accrues unpaid interest at the locked rate while no payment is remitted (equation (23)). Because the pre-crisis panel holds the house price fixed at normal state 1, accrual is the only route above 0.80 here; in Figure 19, where the aggregate state moves freely, house-price declines are a second route.

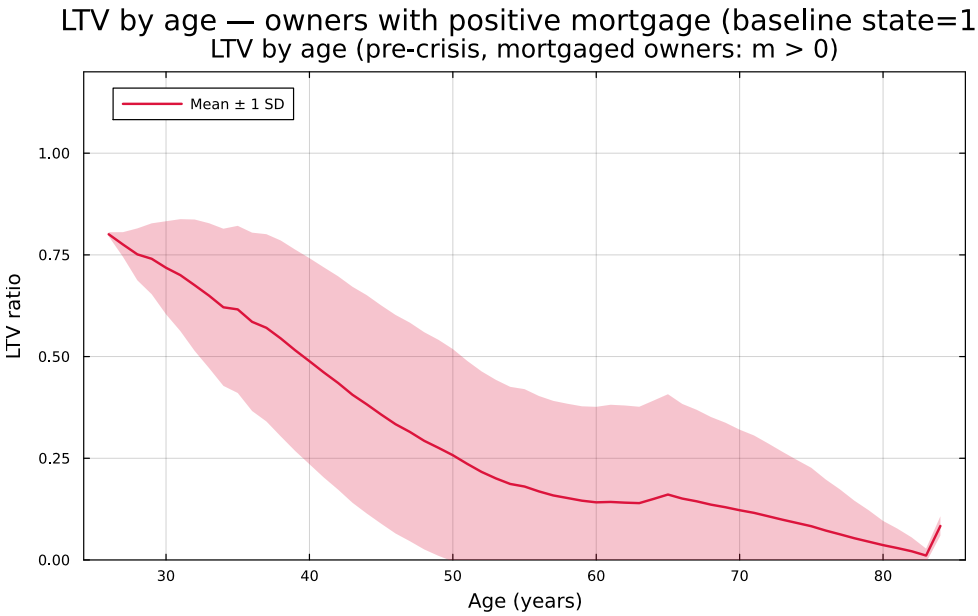


Figure 4: Model LTV by age among mortgaged owners, pre-crisis.

Notes. Model only; the survey comparison is in Figure 3. Pre-crisis periods of the simulated panel. The spread after the mid-forties is the bimodality discussed in the text: model households amortise down with no cash-out, home-equity borrowing or trade-down to re-lever them.

below 65 at the crisis, because CARES relief was an income-loss instrument and the model's retirees, whose pensions are invariant to the shock, hold mortgages and take forbearance for balance-sheet reasons the programme was not aimed at.

Table 6: Crisis-year moments: model against the loan-level data.

Crisis-year moment	Model	Data	Source
Refinancing share of owner-mortgagors	29.88%	26.8%	Fannie Mae loan-level, 2020
Forbearance take-up	4.44%	5.4%	loans entering forbearance, 2020
Crisis-period foreclosure	8 households	floor	not used as validation

Notes. Owner-mortgagors of all ages at the crisis, matching the loan-level frame; the base is the six mortgagor decisions listed in the text, 20,662 households. The refinancing share is a calibration input, not a test. The crisis-period foreclosure count is at the model's floor and is not used as validation, because the foreclosure moratorium made the data's near-zero rate administratively determined.

The refinancing share is targeted and the model overshoots it by about three percentage points. Forbearance take-up is not targeted, and the model undershoots the observed rate by about a percentage point—close enough that it should not be read as a calibrated match, since nothing was fitted to it. The model does not reproduce the distribution of spell *lengths*: committing to ξ at the outset leaves a household no way to end a spell early when income recovers, so 100.0 per cent of spells run the full year against roughly half resolving within three months in the servicing data. A variant that replaces the once-a-year commitment with a month-by-month re-employment hazard does generate a non-degenerate distribution, with a mean of about ten months (Section G.2, Figure 27); it is reported there rather than here because it changes two other parameters at the

same time. The results that follow are therefore about the incidence of relief, not its duration.

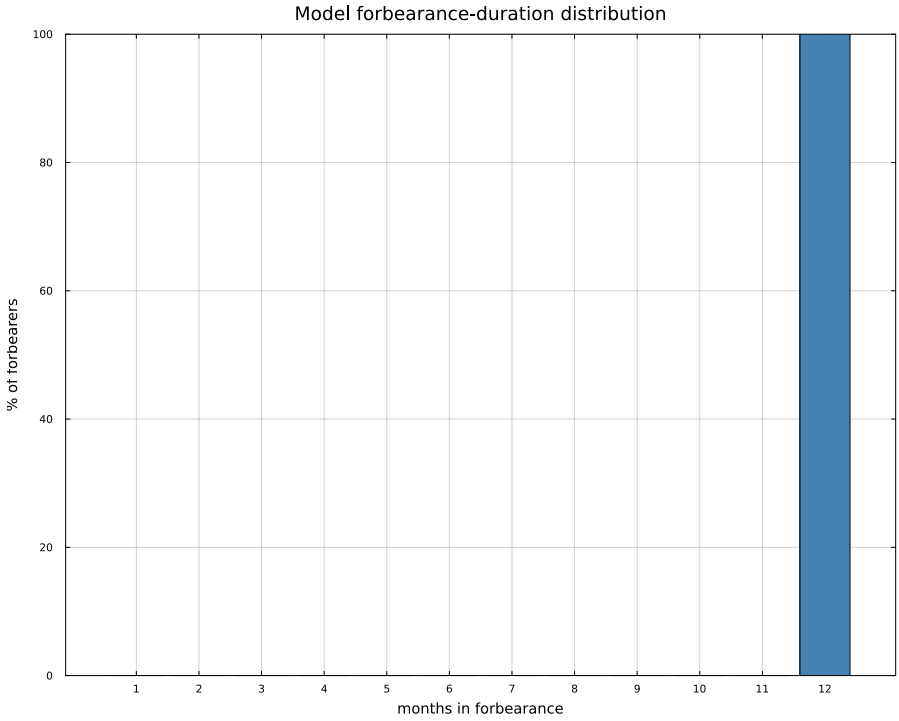


Figure 5: Distribution of forbearance spell lengths in the simulated crisis.

Notes. Model, crisis-year forbearance spells, all ages. Every one of the 917 spells runs the full twelve months; no shorter bin is populated. The servicing data of Sánchez and Wilkinson (2022) has roughly half of spells resolving within three months. Committing to ξ at entry gives a household no way to end a spell early when income recovers, which is the mechanism behind the gap.

6 Which Instrument Reaches Whom

6.1 Design

The solved model is simulated as a panel of 30,000 households, burned in to the stationary age distribution so that the cross-section entering the crisis is the one the calibrated economy generates. The crisis is grafted on at period 861, with the periods either side the last pre-crisis and the first post-crisis year. All experiments share the same seeded draws, so that a difference between arms is the policy rather than the noise. Outcomes are reported as shares of the pre-retirement mortgagor population, defined throughout as households carrying a positive mortgage balance. Outright owners are excluded: they appear in the model's generic keep-the-house branch alongside paying borrowers, but no mortgage instrument can reach them, and including them would understate every share by about a seventh.

Two baselines are carried, because the duration of cheap credit changes what the instruments do to one another. **Baseline A** is a single crisis period: output contracts, the mortgage rate is cut, forbearance becomes available, and the economy returns to normal the following year. **Baseline B** is the same crisis followed by four periods in which output has recovered but the mortgage rate remains at its crisis level. The path is anticipated, so households price the whole chain, and forbearance does not return in the low-rate periods. The two share every parameter, the grid, the panel and the random numbers.

6.2 What each instrument does

The crisis bundles four changes: output falls, the mortgage rate is cut, house prices rise, and forbearance becomes available. Because the solved chain carries each combination as a separate aggregate state reached with probability zero from normal times, any subset can be realised without re-solving. The resulting decomposition is in Table 7 and Figure 6.

Table 7: Crisis-period outcomes under each realised state, shares of pre-retirement mortgagors.

Realised state	Refinance	Forbearance	Default	Delinquent	Pay
Output alone	1.98	0.21	0.33	12.37	81.83
Rate cut alone	38.88	0.08	0.33	5.20	52.17
Forbearance alone	0.58	19.62	0.04	3.38	75.22
Output + rate + forbearance	38.70	9.41	0.04	1.43	49.20
All four (Baseline A)	45.65	4.48	0.05	1.54	47.48

Notes. Per cent of pre-retirement mortgagors in the crisis period, where a mortgagor is a household carrying a positive balance; 13,112 households, against 15,075 if outright owners were included. All five rows read the same solve and the same random draws and differ only in the realised aggregate state. Rows do not sum to one hundred because purchases, sales and deaths are omitted. The fourth row is the full crisis without the house-price rise.

The two relief margins are substitutes, and the substitution is quantitatively large. Where forbearance is the only instrument available, 19.6 per cent of mortgagors take it and almost none refinance. Introducing the rate cut alongside it halves take-up to 9.4 per cent while refinancing

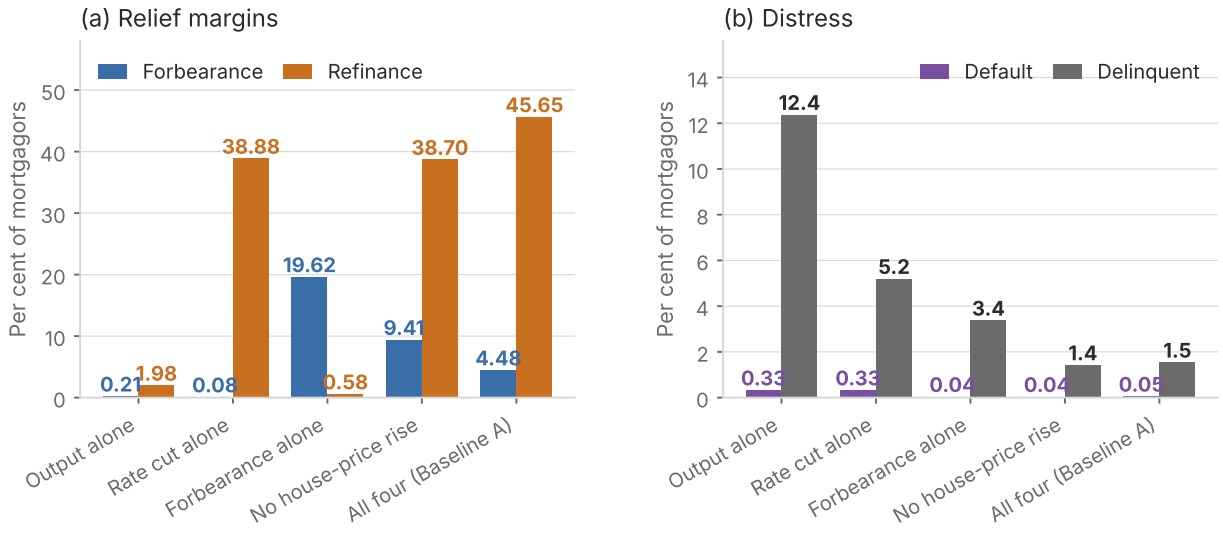


Figure 6: Crisis-period outcomes by realised state. Panel (a): the two relief margins. Panel (b): distress. Shares of pre-retirement mortgagors.

rises to thirty-nine per cent of the population: the price instrument absorbs half the forbearance caseload without any change in eligibility.

A second substitution operates through collateral values, and it is of comparable size. The fourth and fifth rows of Table 7 differ only in the eight per cent house-price rise, and take-up falls from 9.4 to 4.5 per cent across them while refinancing rises further, to 45.7 per cent. Higher house prices relax the leverage test that gates refinancing, so households who would otherwise have deferred payments refinance instead. Since the 2020 episode combined a rate cut with rapidly appreciating housing, both channels were operating, and the model implies that forbearance did considerably less work than a comparison with a flat-price counterfactual would suggest.

The counterfactual in the first row is the relevant one for assessing the programme. With output falling and no instrument available, default is 0.33 per cent of mortgagors and delinquency 12.4 per cent, against 0.05 and 1.5 per cent in the full crisis. The same output shock therefore produces roughly six times the default rate and eight times the delinquency rate once relief is withdrawn, which bounds the programme's contribution from below: the comparison attributes to relief everything that the rate cut, the house-price path and forbearance accomplish jointly.

6.3 Consumption

Figure 7 reports consumption. Mean consumption before the crisis is 0.8876 model units in every arm, identical to four decimal places, which is the MIT construction behaving as intended: states reached with probability zero cannot enter the normal-times value functions. Crisis consumption orders the arms as the distress shares do, from 0.851 under output alone to 0.874 in the full crisis, and the recovery is close to complete within three years in all of them. What the instruments

purchase is therefore the depth of the trough rather than the subsequent level.

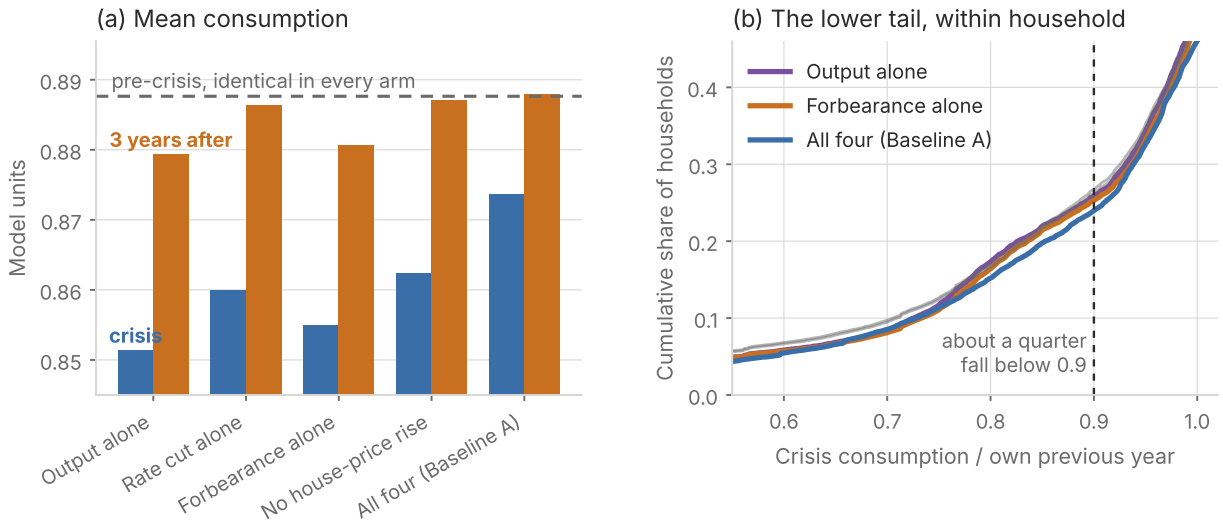


Figure 7: Consumption. Panel (a): mean consumption in the crisis year and three years after, against a pre-crisis level identical across arms. Panel (b): the lower tail of each household's crisis consumption relative to its own consumption the previous year.

The dispersion of consumption is better measured within household than across. The cross-sectional coefficient of variation moves from 0.674 before the crisis to between 0.665 and 0.690 during it, a range too narrow to separate the arms, because it is dominated by permanent-income differences the crisis does not disturb. Panel (b) therefore plots the distribution of each household's crisis consumption relative to its own the previous year, restricted to the lower tail where the instruments operate. At the tenth percentile that ratio is 0.705 under the rate cut alone, 0.726 under output alone and 0.728 in the full crisis; the share of households consuming below ninety per cent of their previous year ranges from 0.240 to 0.267. Relief compresses the lower tail, but the compression is modest, and the arms are close enough that this margin should not be asked to carry an argument on its own.

6.4 Who is reached

Table 8 reports mean characteristics of the approved under each realised state. The composition of the forborne cohort changes with the instruments available, and the direction of that change is informative about what the other instruments do.

As more instruments become available the forborne cohort contracts and its composition shifts. Where forbearance operates alone, the approved have mean income 0.623; in the full crisis the cohort is a fifth of that size and its mean income is 0.438, some thirty per cent lower, with mean liquid assets three times higher and mean age five years greater. What the price instrument removes is the higher-income portion of the caseload, the households able to satisfy a payment-to-income test once the rate falls. The residual that forbearance serves in the full crisis is poorer and older than

Table 8: Characteristics of the forborne cohort, by realised state.

Realised state	Share (%)	Income	Liquid assets	Balance	LTV	Age
Output alone	0.21	0.643	0.025	2.920	0.701	40.1
Rate cut alone	0.08	0.746	0.058	3.442	0.826	35.3
Forbearance alone	19.62	0.623	0.102	2.995	0.719	42.7
Output + rate + forbearance	9.41	0.622	0.187	3.357	0.806	41.5
All four (Baseline A)	4.48	0.438	0.324	3.393	0.754	47.3

Notes. Means over approved pre-retirement households in the crisis period; the share column uses the positive-balance base of Table 7. Every approved household carries a balance by construction, so only the share column moves with that definition, not the means. Income, liquid assets and balance are in model units of roughly \$60,000. The first two rows rest on a few dozen households and are reported for completeness rather than comparison.

the cohort it would have served on its own, so the programme becomes more targeted as it becomes smaller, though not by design.

Appendix F maps the same five arms over payment burden against leverage and against cash held, by age band, so the boundary the eligibility floor draws can be seen directly rather than inferred from counts. Appendix G.2 asks the narrower question of whether relief reaches the households whose income collapses in the crisis.

6.5 Crisis severity

The severity ladder varies the output multiplier alone, holding the rate cut, the house-price path and forbearance availability at their baseline values. Six realised states are simulated off a single solve.

One feature of the construction governs how the ladder should be read. A crisis state's output is anchored to the highest normal state and then scaled, rather than to the state the economy occupies, so a multiplier κ^g delivers a realised change of $1.0298\,\kappa^g/0.9711 - 1$ relative to the pre-crisis level. The baseline's $\kappa^g = 0.90$ is a 4.6 per cent contraction rather than a ten per cent one, and the mildest arm is a 0.7 per cent expansion. The ladder therefore spans approximately no change to a 25.8 per cent contraction; panel (a) of Figure 9 plots both quantities, and realised changes are used throughout.

Table 9 and Figure 8 report the ladder. All three margins rise with severity, but at very different rates, and it is their ratio that carries the economics. Across the whole ladder forbearance take-up rises by half, from 4.0 to 6.0 per cent, and delinquency sevenfold, from 0.9 to 5.9; default rises from nothing at all—not a single pre-retirement mortgagor defaults in the mildest arm—to 1.4 per cent. Where the ratio can be formed, the programme delivers 84 approvals for every household that defaults at the baseline severity and 4.3 at the deepest; both margins are shares of the same base, so the ratio is also percentage points of approvals per percentage point of default. A shallow contraction is absorbed almost entirely on the forbearance margin; a deep one is not, because deferring a year of payments does not assist a household whose income has fallen by a quarter and whose equity has been exhausted. The pattern is monotone with one exception, at $\kappa^g = 0.75$, where

Table 9: The severity ladder: one solve, six realised crisis states.

κ^g	Output (%)	Forb. (%)	Delinq. (%)	Default (%)	Refi. (%)	Consumption	Utility
0.95	+0.74	4.009	0.85	0.000	45.48	0.8893	−0.6305
0.90	−4.56	4.477	1.54	0.053	45.65	0.8736	−0.7361
0.85	−9.86	4.624	2.54	0.099	46.51	0.8605	−0.8920
0.80	−15.16	4.982	3.77	0.237	47.33	0.8448	−1.1427
0.75	−20.47	4.951	4.38	0.853	49.15	0.8261	−1.4994
0.70	−25.77	6.037	5.89	1.415	49.98	0.8057	−1.8781

Notes. Built from the six severity panels, on the *same* base as Tables 7 and 8: pre-retirement households carrying a positive mortgage balance, between 12,936 and 13,119 of them depending on the arm. The $\kappa^g = 0.90$ row is the baseline crisis and reproduces the baseline arm decision for decision—5,985 refinances, 587 approvals, 202 delinquencies and 7 defaults on a base of 13,112—which is the internal check on the fan-out.

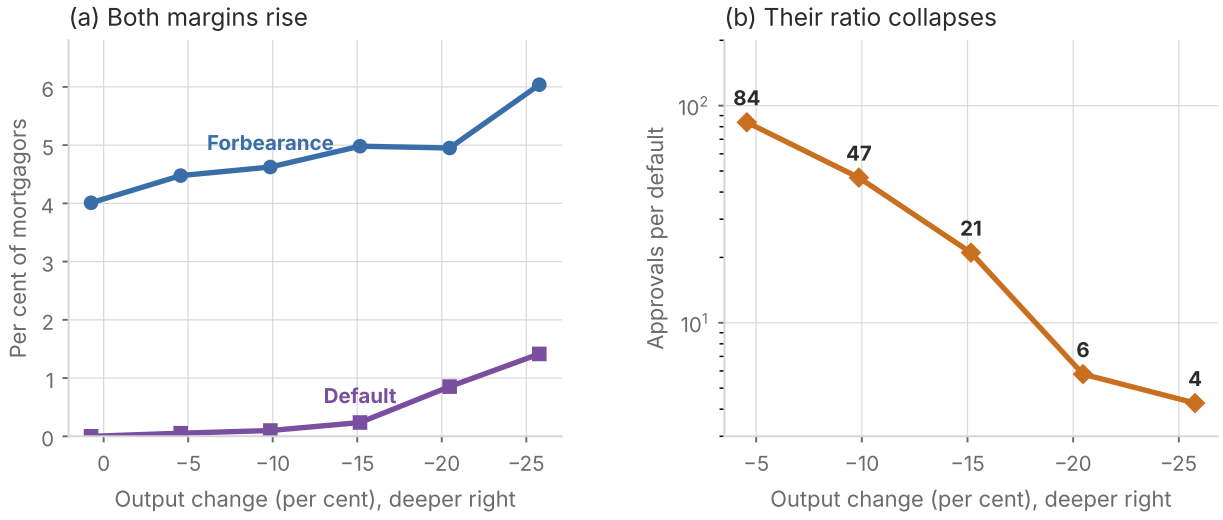


Figure 8: The severity ladder. Panel (a): forbearance and default as crises deepen. Panel (b): the ratio of approvals to default, on a logarithmic scale.

take-up falls against $\kappa^g = 0.80$ by 0.031 points. That is not sampling noise: the arms share common random numbers. It reflects discrete transitions across the asset and mortgage grids together with movement in the denominator, which runs between 12,936 and 13,119 positive-balance mortgagors across the ladder as refinancing, payoff and default choices change who enters the crisis carrying a balance. It is not interpreted here.

The aggregate multiplier conceals substantial heterogeneity in who bears the contraction. The model's income loadings decline with permanent income, following the pattern documented by Guvenen et al. (2014), so a given fall in aggregate output is a much larger shock at the bottom of the distribution than at the top. Measured in every arm rather than projected from one, crisis income at the baseline multiplier is 0.93 of its previous value in the lowest permanent-income quintile and 0.93 in the highest, and the gap opens sharply as the crisis deepens: at $\kappa^g = 0.70$ the lowest quintile keeps 0.45 of its previous income against 0.80 for the highest. The ratio is taken against the same household's own previous year and so carries normal-times income growth within it, which is why the baseline reading is flat: at $\kappa^g = 0.95$, a realised expansion, the lowest fifth posts a gain. The gradient the exposure function implies is visible once the contraction is large enough to dominate that growth. Panel (b) of Figure 9 shows the same point over the life cycle: the pre-crisis income profile peaks near age 47 and the crisis profiles fall below it in proportion to severity, while incomes after 65 are essentially unchanged because pensions are invariant to the aggregate state.

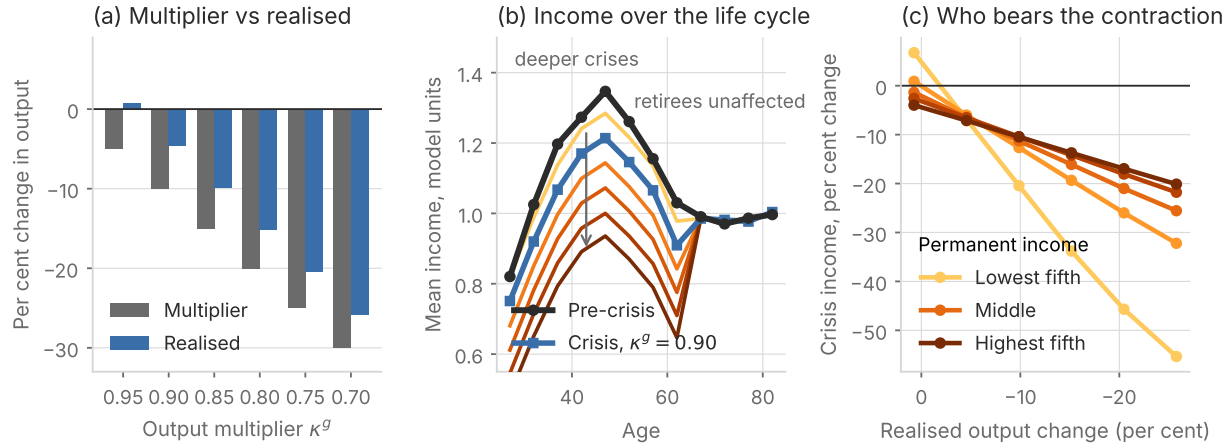


Figure 9: Output and income under each severity specification, measured in every arm. Panel (a): the multiplier against the realised output change. Panel (b): mean income over the life cycle, pre-crisis and in the crisis under each multiplier. Panel (c): the crisis-year change in income by quintile of permanent income, where permanent income is the household's own mean income over all pre-crisis periods. All three panels are measured on the six severity panels directly; nothing is extrapolated from a single arm.

6.6 Which underwriting restriction gates reach

Each row of Table 10 is a separate backward induction in which one underwriting rule is relaxed and the crisis held at the baseline. Figure 10 plots the take-up column.

Table 10: Underwriting relaxations and repayment regimes, at the baseline crisis.

Specification	Forb. (%)	Default (%)	Refi. (%)	Ownership (%)	Utility
Baseline A (regime 3)	3.057	0.027	20.58	69.82	−0.6314
Repayment plan (regime 5)	3.143	0.027	20.52	69.82	−0.6315
Forbearance unavailable	0.000	0.157	21.21	69.82	−0.6315
Payment-to-income cap removed	3.797	0.007	26.64	71.32	−0.6192
Down payment 20 → 5 per cent	4.690	0.063	27.19	75.47	−0.6476
Cash-in refinancing allowed	3.270	0.020	21.45	69.12	−0.6870
Refinancing costless	2.227	0.030	73.75	79.42	−0.5823

Notes. From the arms' moments files, whose percentages are shares of the full 30,000-household panel, renters and non-owners included, and not of mortgagors. Levels are therefore *not* comparable with Tables 7, 8 or 9, all of which use the pre-retirement positive-balance base; comparisons down this table are unaffected, since every row shares the denominator. Each row is its own solve at the baseline crisis, so the pre-crisis distribution entering the crisis differs across rows as well as the crisis-period choice set. Mean utility is the panel average and is used for within-table ordering, not as a welfare measure in levels.

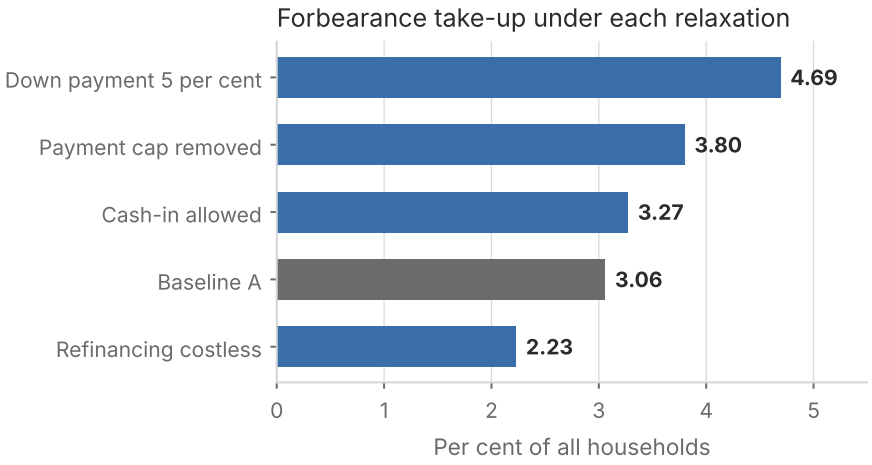


Figure 10: Forbearance take-up under each underwriting relaxation, at the baseline crisis. Shares are of the full 30,000-household panel, as in Table 10, not of mortgagors, and are therefore not comparable with the mortgagor-base take-up rates of Figure 6.

Withdrawing the forbearance option raises default from 0.027 to 0.157 per cent while refinancing rises only from 20.6 to 21.2: households that lose the option do not, for the most part, find another. At the opposite extreme, setting the refinancing fee and its fixed component to zero raises refinancing to 73.8 per cent and lowers forbearance take-up to 2.2 per cent, below the baseline, with ownership at 79.4 per cent and the highest mean utility in the table. This is the substitution of Section 6.2 at its limit, and it locates the binding friction: what restricts refinancing in this model is its cost rather than eligibility.

The collateral and cash-in relaxations do not improve outcomes, and one of them worsens them. Reducing the down payment requirement to five per cent raises ownership to 75.5 per cent and take-up to 4.7 per cent, but default rises and mean utility falls; permitting cash-in refinancing leaves take-up near the baseline and produces the lowest utility in the table. Both allow households to enter the crisis with less equity, and the additional relief they subsequently draw does not compensate. The exception is the payment-to-income ceiling, whose removal lowers default to 0.007 per cent while raising both ownership and utility.

The repayment regime, by contrast, is close to immaterial for reach. Regime 5, under which missed payments do not accrue interest and are spread evenly across the remaining schedule, yields 3.14 per cent take-up against regime 3's 3.06, with default and ownership unchanged and mean utility indistinguishable. Within this model the terms on which deferred payments are repaid are a second-order determinant of who uses the programme, while eligibility and the availability of the price instrument are first-order.

6.7 The eligibility floor

Eligibility for forbearance is governed by a payment-burden floor: a household may apply only if its scheduled payment is at least twenty per cent of its income. The restriction is far from cosmetic. At that threshold it excludes 79.6 per cent of pre-retirement crisis mortgagors. Among those excluded the median scheduled payment-to-income ratio is 0.045; across all pre-retirement crisis mortgagors, excluded and eligible together, it is 0.083. In the arms where the margin can be observed directly, take-up below the floor is zero to two decimal places while take-up above it reaches double figures. Together with the results of Section 6.6, this suggests that the size of the programme is determined primarily by the eligibility margin and by the availability of cheap refinancing, rather than by the approval probability or the repayment terms.

Two features of the environment sit behind the floor and are easy to mistake for separate restrictions. The first is that the floor is a leverage condition in disguise. Written as equation (2) it is a ray through the origin in income–leverage space, so what determines leverage at origination determines who later clears it—and the collateral rule is what determines leverage at origination. Cutting the down-payment requirement from twenty to five per cent in Table 10 raises take-up from 3.057 to 4.690 per cent with the floor itself held at a fifth of income, the largest caseload response in that table apart from the costless-refinancing row, which moves it the other way. The loan-to-value

cap and the eligibility floor are therefore not independent levers: the first sets the distribution the second cuts.

The second is that exclusion by the floor is not cushioned inside the model. As Section 5.3 reports, the median household here holds no liquid buffer through most of working life, where the data's holds tens of thousands of dollars. A household the floor turns away has no stock of liquid assets to run down instead, which is the same observation that makes the model's demand for both instruments an overstatement rather than an understatement. The floor's bite is consequently a property of this model's thin buffers as much as of the threshold itself.

This is reported as a property of the baseline rather than as a schedule, since the floor has not been swept at the present specification. The model can therefore establish that reach is dominated by the eligibility margin, but not what reach would be at any other threshold.

6.8 What the instruments are worth

The preceding subsections measure reach. This one measures value. For each household the realised stream of payoffs from the crisis to the end of its life is recorded under each realised state, and the two are compared by asking what proportional increase in consumption, applied to the household's whole remaining path in the no-instrument arm, would leave it as well off as the arm being valued. Because the arms share one solve and one set of random draws, and because the realised date of death is identical across them, this is a within-household comparison along a common shock path. The measure is exact for this model's preferences rather than the usual homothetic approximation; the derivation, and the reason the approximation cannot be used here, are in Appendix E.

Table 11 and Figure 11 report the result. The headline figure is the aggregate, or utilitarian, equivalent variation: the single factor that equates total realised welfare across the two arms. It is reported in preference to the average of the household-level factors because an individual factor diverges whenever an instrument spares a household damage that scaling its consumption cannot replicate – which is precisely what happens to a household saved from foreclosure, whose loss is partly in the house and in the bequest. Eleven households record a factor above a hundred, eight of them foreclosed in the no-instrument arm, and one alone would move the average by about two fifths. The median and the percentiles describe the household distribution and are robust.

Three features of the table deserve comment. The first is the size of the numbers. The full policy response is worth 0.45 per cent of lifetime consumption in aggregate, and forbearance on its own 0.29 per cent, which among pre-retirement households holding a balance is 1.61 and 0.67 per cent respectively; the corresponding column of Table 11 is all-age rather than pre-retirement and reads 1.42 and 0.65. These are small relative to the reach figures of Section 6.2, and the reason is in the share-unmoved column: in the three arms that hold house prices fixed, between seventy-three and ninety per cent of households end the crisis with a realised path that is unchanged to the last bit. In the four-instrument arm no household is left untouched, because the house-price rise moves maintenance for every owner and rent for every renter, which is why that row reads zero.

Table 11: Consumption-equivalent value of the crisis instruments

	All households					Aggregate, by group		
Arm	Aggregate	Median	p_{10}	p_{90}	Share unmoved	Balance > 0	Renters	Approved for forb.
Forbearance only	0.29	0.00	0.00	0.00	90	0.65	0.00	2.15
Rate cut only	0.43	0.00	0.00	1.86	78	0.87	0.06	–
Forbearance + rate cut	0.68	0.00	0.00	2.41	73	1.43	0.06	3.38
All four instruments	0.45	-0.09	-0.46	2.33	0	1.42	-0.38	3.58
Decomposition, aggregate CEV								
Forbearance alone	0.29							
Rate cut alone	0.43							
Interaction of the two	-0.04							
House-price rise	-0.23							
All four together	0.45							

Notes. Per cent of consumption, against the arm in which output falls and no instrument is switched on. All five arms read the same backward induction and share random draws, so each household is the same person facing a different realised crisis; the measure is a realised-path CEV, not an ex-ante expected-value one. The panel covers every household's entire remaining life, so nothing is right-censored. "Aggregate" is the single scaling factor that equates total realised welfare across the two arms, which is the headline; the mean of the household-level factors is not reported because a factor diverges when the instrument spares a household damage that consumption scaling cannot replicate, which is why the eleven largest belong overwhelmingly to households foreclosed in the reference arm, and one of them alone would move the mean by about two fifths. The median and percentiles describe the household-level distribution and are robust. "Share unmoved" is the per cent of households whose realised path is unchanged by the arm; it is zero in the four-instrument arm, because the house-price rise alters maintenance for every owner and rent for every renter, so the unmoved shares reported in the text are those of the three constant-price arms. "Approved for forb." conditions on the households approved in that arm, and is suppressed where fewer than 50 qualify. A mortgagor is a household with a positive balance; outright owners are excluded throughout. $n = 29403$ households, of which 16876 hold a positive mortgage balance and 8736 rent. The variant that scales bequeathed wealth alongside consumption gives 0.26, 0.38, 0.60, 0.39 respectively.

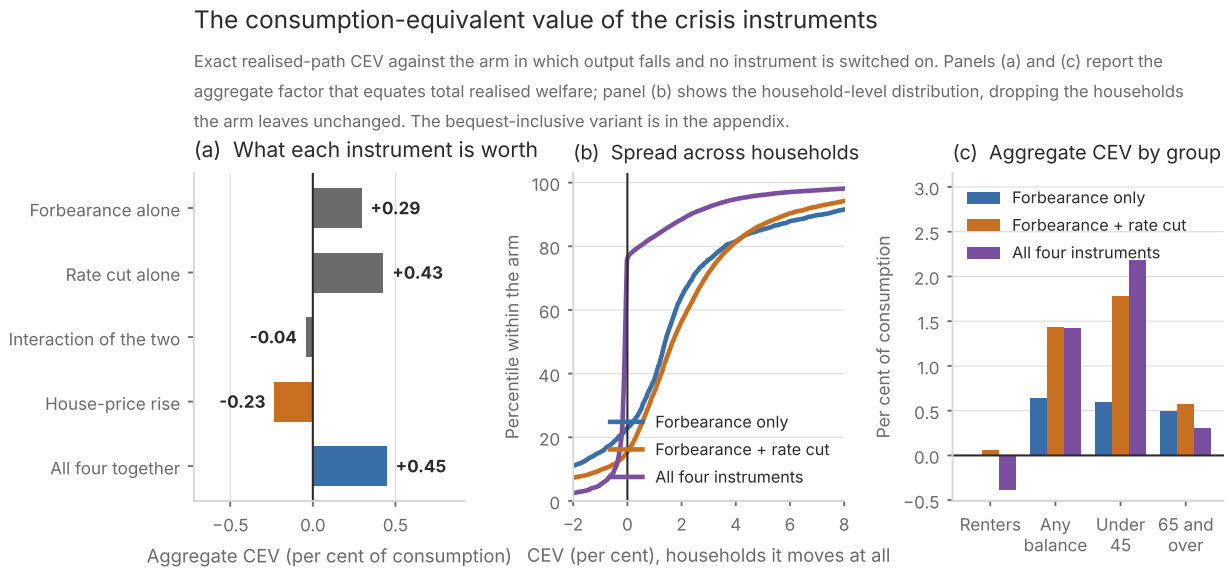


Figure 11: The consumption-equivalent value of the crisis instruments. Panel (a): the aggregate equivalent variation attributable to each instrument. Panel (b): the household distribution, over those households the arm moves at all; the three arms that hold house prices fixed leave between 73 and 90 per cent of households with a realised path that is unchanged, and those are excluded from the panel, while the four-instrument arm leaves none unchanged because the house-price rise moves maintenance and rent for everyone. Panel (c): aggregate equivalent variation by group, where "any balance" is a household carrying a mortgage into the crisis. The variant that scales bequeathed wealth alongside consumption is in Appendix E.

A programme reaching a few per cent of mortgagors cannot be worth much when averaged over everyone, however much it is worth to those it reaches.

The second is that the instruments are not additive, and one of them is negative. Forbearance alone is worth 0.29 and the rate cut alone 0.43, but the two together are worth 0.68 rather than 0.72: they overlap, as the substitution in Section 6.2 implies they must. The house-price rise is worth −0.23. Three channels run in opposite directions. Higher house prices relax the leverage test and move households onto the refinancing margin. They also raise the flow cost of occupying housing in the crisis period, since maintenance enters every owner's budget as $\delta_m P^h(s)h$ and rent enters every renter's as a proportion of the same product, both one-for-one in the state's price. And the offsetting capital gain enters only through the terminal value of equation (28), so it is discounted over the whole remaining horizon. The net sign is negative, and the full crisis is accordingly worth less than the crisis without the house-price rise. Which channel carries the sign is not settled here: the crisis is a single period, so the extra maintenance it charges is too small on its own to account for a lifetime equivalent of this size, and separating the flow cost from the collateral and behavioural responses requires re-solving with maintenance anchored to a price-invariant structure value. Renters, who gain nothing from the collateral channel and pay more for housing in the crisis period, lose 0.38 per cent.

The third is that the value of forbearance is overwhelmingly the value of averting a foreclosure. Among households approved under forbearance alone, those who are foreclosed in the no-instrument

arm gain 4.89 per cent while the rest gain 0.68; in the full crisis the corresponding figures are 7.24 and −0.63. The deferral of a year's payments is worth little on its own. What the programme is worth is concentrated on the minority for whom it is the difference between keeping the house and losing it, and in the full crisis the households it reaches who were never at that margin are made slightly worse off by the accompanying house-price rise.

Appendix E carries three further cuts of the same measure: the equivalent variation grouped by the decision a household actually takes in the crisis rather than by its characteristics, the variant that scales bequeathed wealth alongside consumption, and the conversion of the whole measure into dollars. That last is worth flagging here. The equivalent variation asks how much extra consumption the household would need in the no-instrument world; its mirror image, the willingness to pay, asks how much it would surrender in the instrument world to keep it. The two agree to the fourth decimal in aggregate but not household by household, where only the second is bounded. On the households that use them, the forbearance option is worth \$17,261 and the rate cut \$27,722 per household in present value, against first-year payment reductions an order of magnitude smaller. Those are single pooled figures and condition on nothing, which matters more than it may appear: because a household in its late twenties has an order of magnitude more remaining consumption to scale than one past retirement, the dollar figures and the proportional ones rank age in opposite directions. The rate cut is worth a nearly flat 2.1 to 2.3 per cent of lifetime consumption at every age below 65, and \$43,327 against \$3,054 across the same range. Appendix E stratifies both by age and by balance.

Figure 12 shows where the value sits, and on fixed rather than quantile bins the separation between the two instruments is complete. Forbearance is worth nothing at any age below a loan-to-value of 0.6. Above 0.8 it is worth a great deal, and its value rises steeply with age: nothing to households under 35, then 1.46, 2.40, 3.60 and 7.17 per cent across the successive decades. The rate cut occupies the band immediately below, between 0.4 and 0.8, and is worth *exactly* zero at 0.8 and above in every one of the five age bands. The most levered households cannot satisfy the leverage test however far the price of credit falls, and they are precisely the households that forbearance reaches. The two instruments are therefore not merely substitutes in aggregate; they are close to disjoint in incidence, and the one that reaches the most levered is the one that cannot be delivered through the price of credit.

Two cells qualify that reading. The youngest and most levered households, the 457 under 35 with a loan-to-value above 0.8, are reached by neither instrument: their scheduled payments are small relative to income early in the amortisation schedule, so they fail the eligibility floor of Section 6.7 just as they fail the leverage test. In the full crisis the same cell is worth 2.65 per cent, which is the house-price rise relaxing the leverage test on their behalf. And at the other end of the grid the full crisis is mildly negative, from −0.02 per cent for the youngest households with little leverage to −0.21 for the oldest, the flow cost of more expensive housing falling on households with no debt for the collateral channel to relieve.

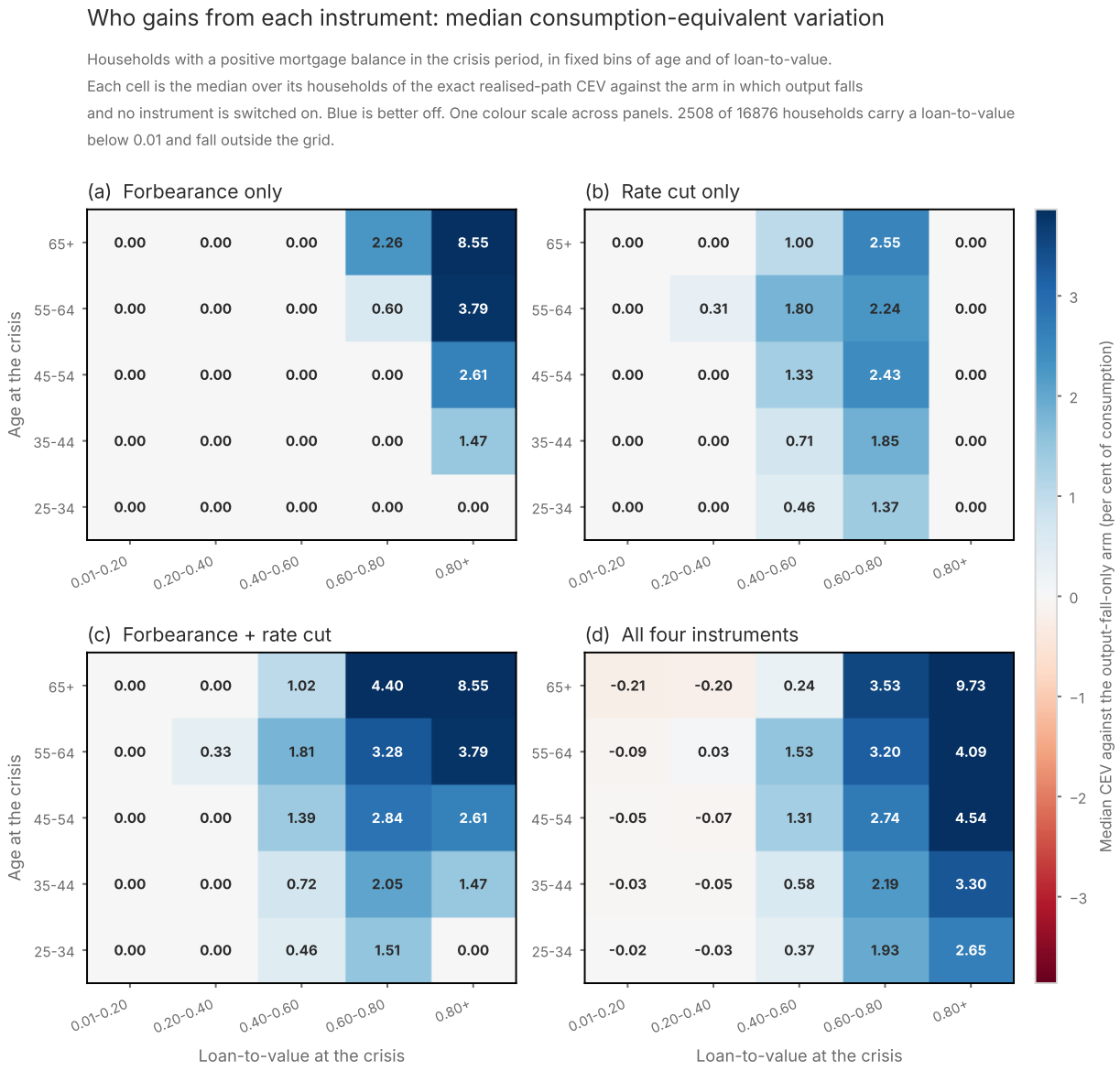


Figure 12: Median consumption-equivalent variation by age and loan-to-value at the crisis, for households with a positive mortgage balance, against the arm in which output falls and no instrument is switched on. Blue is better off. Bins are fixed rather than quantile-based: ten-year age bands to retirement with everything past it in one cell, and loan-to-value in steps of 0.2 from 0.01, the last band open. Households below a loan-to-value of 0.01, 2,508 of 16,876, fall outside the grid. The colour scale is common to the four panels and is clipped at the ninetieth percentile of the absolute cell medians.

7 Conclusion

Two instruments were used at once in 2020 and they were gated by opposite conditions: refinancing asks a borrower to demonstrate that she can afford a new loan, deferral that she cannot afford the old one. The loan-level record shows that the two reached different populations, but it cannot say whether they *would* have reached the same ones, because no borrower is observed under the instrument she did not use. This paper puts both in one household problem to ask that question.

The answer is that the division is almost entirely one of reach rather than of size. Where deferral is the only instrument available it reaches 19.6 per cent of pre-retirement mortgagors; introducing the rate cut alongside it halves take-up to 9.4 per cent while 38.7 per cent refinance instead, with no change in eligibility. Measured in crisis-year cash flow the two are indistinguishable per recipient, about \$14,000 each, and the asymmetry is in how many households each reaches—24.1 per cent of crisis mortgagors against 4.9 per cent, or \$3,470 against \$688 when averaged over everyone. What they prevent differs more sharply than what they pay: the rate cut leaves crisis default where the no-instrument crisis leaves it, at 0.33 per cent, while deferral takes it to 0.04. Relaxing the underwriting restrictions one at a time locates which part of the hurdle binds. Removing the payment-to-income ceiling raises crisis refinancing from 20.6 to 26.6 per cent; setting the refinancing fee to zero raises it to 73.8. In this model the price instrument is restricted by its cost more than by its eligibility rules, while the deferral is restricted by its eligibility floor, which excludes 79.6 per cent of pre-retirement crisis mortgagors. The terms on which deferred payments are later repaid are second-order for reach by comparison, moving take-up from 3.06 to 3.14 per cent.

What the instruments are worth is small in aggregate and concentrated at one margin. The full policy response is worth 0.45 per cent of lifetime consumption averaged over everyone and deferral on its own 0.29 per cent, which among pre-retirement households carrying a balance are 1.61 and 0.67 per cent. The two are not additive—0.29 and 0.43 separately against 0.68 together—which is the same substitution the reach figures show. Almost all of the value sits with the households for whom the instrument is the difference between keeping the house and losing it: among those approved under deferral alone, the ones foreclosed in the no-instrument arm gain 4.89 per cent against 0.68 for the rest. Households value the deferral at \$17,045 and the rate cut at \$27,328 in willingness to pay. The accompanying house-price rise is worth -0.23 per cent to owners as a whole and -0.38 to renters, because maintenance and rent both scale with the price of housing while the capital gain accrues only at the end of life.

Two further results qualify what relief accomplishes. The cohort deferral reaches is severely selected—36.2 per cent of them eventually lose the house, against 0.06 per cent of mortgagors who never apply and 4.7 per cent of crisis mortgagors as a whole—and following the same households through an economy without the option shows that it changes the timing of foreclosure far more than its incidence: 2.1 per cent of those reached avoid a foreclosure they would otherwise have suffered, 17.2 per cent suffer one later, and 6.5 per cent, a larger group than the prevented, foreclose earlier with the option than without it. Lifetime foreclosures across the panel differ by 0.09 per cent

and in the direction of slightly more rather than fewer, and 40.5 per cent of every dollar advanced sits with a household that loses the home anyway. Varying the world the instruments arrive in points the same way: deepening the output contraction raises crisis delinquency from 0.9 to 5.9 per cent and default from nothing to 1.4, while approvals rise only from 4.0 to 6.0 per cent, because the eligibility test selects on a payment burden set largely by the pre-crisis cross-section rather than by the depth of the shock.

Four limits bound all of this, and they are limits of the exercise rather than caveats about precision. The loan-level evidence of Section 2 supports a bracket and not a point—a reduction in thirty-six-month foreclosure of 1.3 to 1.8 percentage points on the one frame with adequate common support, against an increase of 0.34 points over the whole pre-determined book—and none of it is causal, which is why the counterfactual is asked of a model. The welfare measure is a consumption equivalent and therefore overstates the lump sum that would compensate a household, most of all for the foreclosure-averted households on which it already strains. Housing adjustment is upsize-only, so a distressed owner with positive equity cannot monetise it; the eligibility floor τ^{pti} is a reduced form for frictions the statute did not specify, since CARES set no quantitative threshold; and the crisis is a single unanticipated period, which prices none of the option value of relief that might arrive later. The paper is also incomplete in a plainer sense: the prevention counterfactual and the design-margin exercises summarised in the Introduction are not written up in full here, and the eligibility-floor sweep, the falling-price arm and the alternative settlement regime have not been solved at the present specification.

What the exercise does establish is narrower than a ranking of the two instruments and, I think, more useful than one. The underwriting rules that gate the price instrument and the hardship condition that gates the deferral select on different things, so the two reach populations that barely overlap, and the size of the relief conditional on being reached is close to the same. Which households a crisis programme helps is therefore settled almost entirely by the conditions written into its gate, and the levers available to a designer—the floor, the fee, the collateral test, the settlement terms—govern how much is lent and to whom. None of them governs how many homes are kept.

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A Additional Loan-Level Evidence

This appendix supports the facts of Section 2. Throughout, the sample is a two-per-cent random draw from the Fannie Mae Single-Family Loan Performance file, loans originated 2000–2024, which covers fully amortising, fully documented, conventional fixed-rate mortgages and excludes adjustable-rate, government-insured, HARP, interest-only and other non-standard loans. The evidence therefore describes the conventional conforming segment, the population on which both CARES forbearance and the 2020 refinancing wave operated, and the descriptive shares should not be read as market-wide.

A.1 Origination profiles

Table 12: Summary statistics at origination by eventual loan outcome.

	Forbearance loans			Foreclosure loans			Never-event loans		
Variable	N	Mean	SD	N	Mean	SD	N	Mean	SD
Original UPB ($)	24,359	251,178	135,695	13,464	160,537	94,716	1,095,118	217,252	129,079
DTI (%)	24,257	38.3	8.6	13,004	39.0	11.9	1,080,304	33.6	11.1
Original LTV (%)	24,359	77.5	15.7	13,464	79.0	12.1	1,095,118	70.3	17.8
Credit score	24,338	725.5	50.0	13,353	692.6	57.0	1,091,759	748.4	50.8
	Rate-and-term refinance, 2020–22			Cash-out refinance, 2020–22					
Original UPB ($)	172,571	258,870	136,197	118,654	228,578	129,254			
DTI (%)	172,274	32.1	10.1	118,389	34.3	9.8			
Original LTV (%)	172,571	64.8	17.0	118,654	62.7	15.6			
Credit score	172,520	765.5	43.2	118,634	746.0	50.0			

Notes. "Foreclosure loans" comprises all loans reaching foreclosure; the two distressed groups are not mutually exclusive. The refinance rows are the first occurrence of each loan with purpose R or C originated 2020–2022.

Table 13: Pairwise differences in origination characteristics.

Variable	Comparison	Mean A	Mean B	Diff.	p-value
Original LTV (%)	Forbearance vs. Never	77.5	70.3	7.3	< 0.001
	Foreclosure vs. Never	78.1	70.3	7.9	< 0.001
	Forbearance vs. Foreclosure	77.5	78.1	-0.6	< 0.001
DTI (%)	Forbearance vs. Never	38.3	33.5	4.8	< 0.001
	Foreclosure vs. Never	39.0	33.5	5.5	< 0.001
	Forbearance vs. Foreclosure	38.3	39.0	-0.7	< 0.001
Credit score	Forbearance vs. Never	725.5	748.4	-22.9	< 0.001
	Foreclosure vs. Never	696.6	748.4	-51.8	< 0.001
	Forbearance vs. Foreclosure	725.5	696.6	28.8	< 0.001

Notes. Two-sample Welch t-tests of the mean difference. The foreclosure group is restricted to loans whose foreclosure occurred after 2010, across all origination vintages; the restriction drops the crisis-era foreclosure wave rather than the crisis-era origination book. Means here are taken from a generator that rounds to two decimals before reporting, so a never-event figure may differ from Table 1 in the last printed digit.

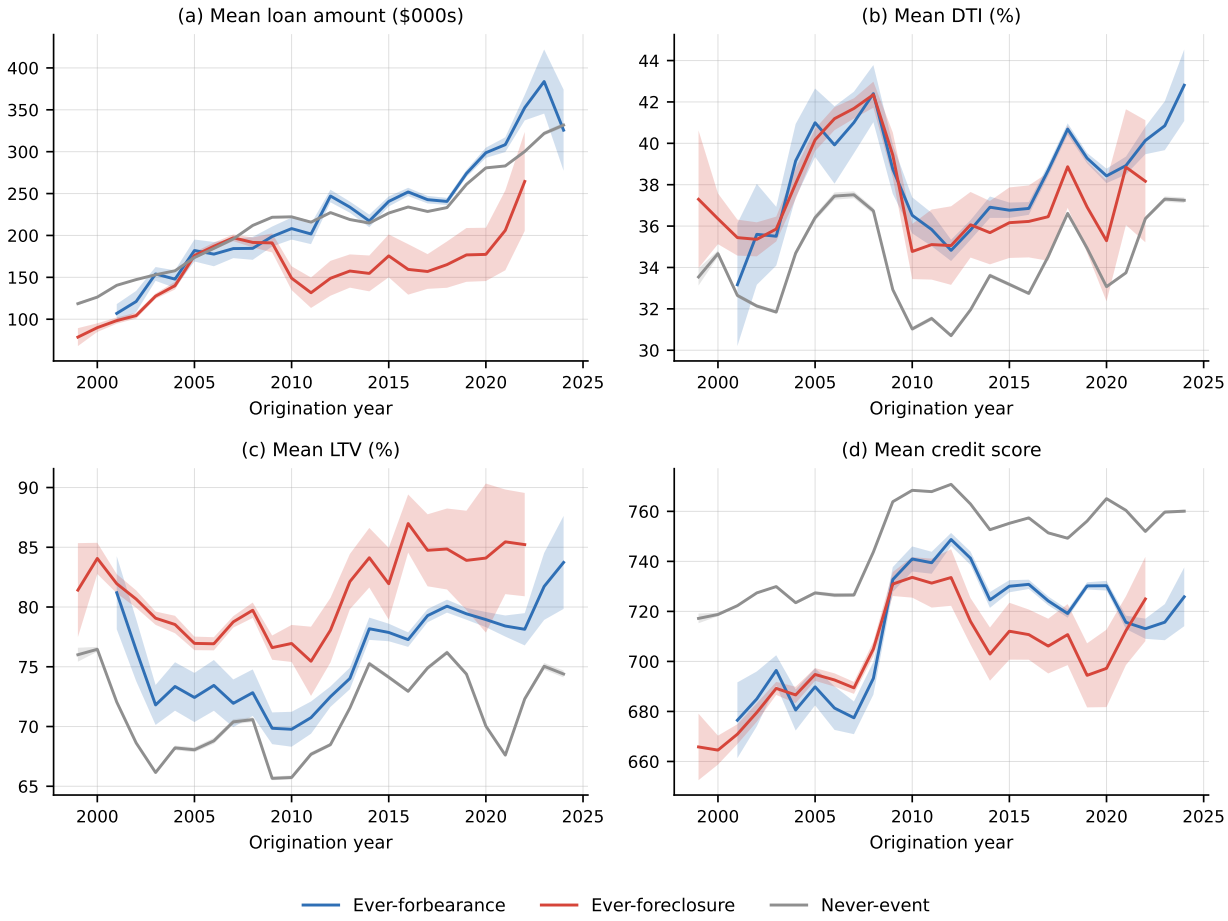


Figure 13: Mean origination characteristics by eventual-outcome group, over origination vintage. Years with fewer than 30 loans in a group are omitted; shaded bands are 95 per cent confidence intervals. The forbearance and foreclosure groups sit above the never-event group in LTV and DTI and below it in credit score at essentially every vintage; ever-foreclosure cohorts thin sharply after 2012 because post-QM originations rarely foreclose.

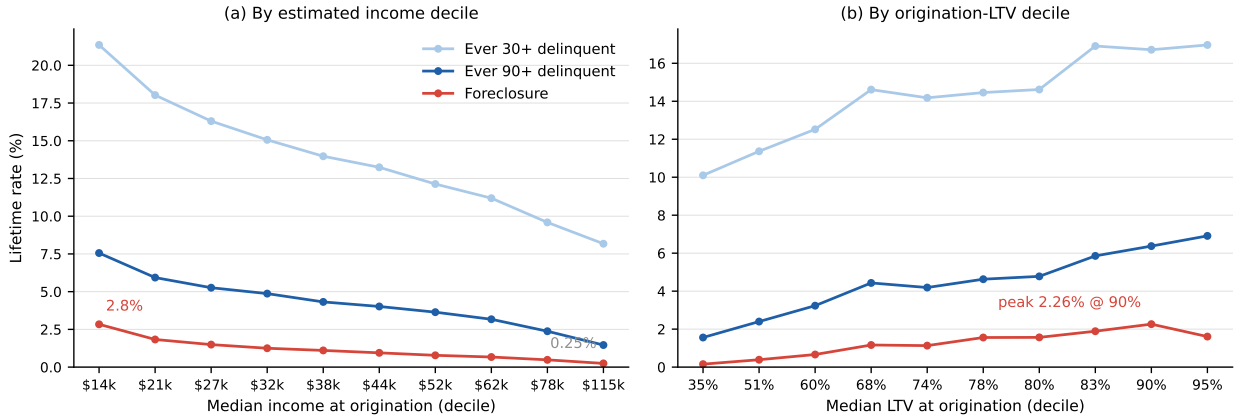


Figure 14: Lifetime delinquency and foreclosure rates by decile of estimated income at origination (panel a) and of origination LTV (panel b). Estimated income is imputed from the scheduled principal-and-interest payment and the debt-to-income ratio, and is a lower bound: the reported debt-to-income ratio is a back-end ratio, covering taxes, insurance and non-mortgage debt service as well as principal and interest, so dividing principal and interest alone by it understates total borrower income. Foreclosure falls by roughly a factor of eleven across the income distribution—from 2.8 per cent in the lowest decile to 0.25 in the highest—and peaks at the 90 per cent LTV decile.

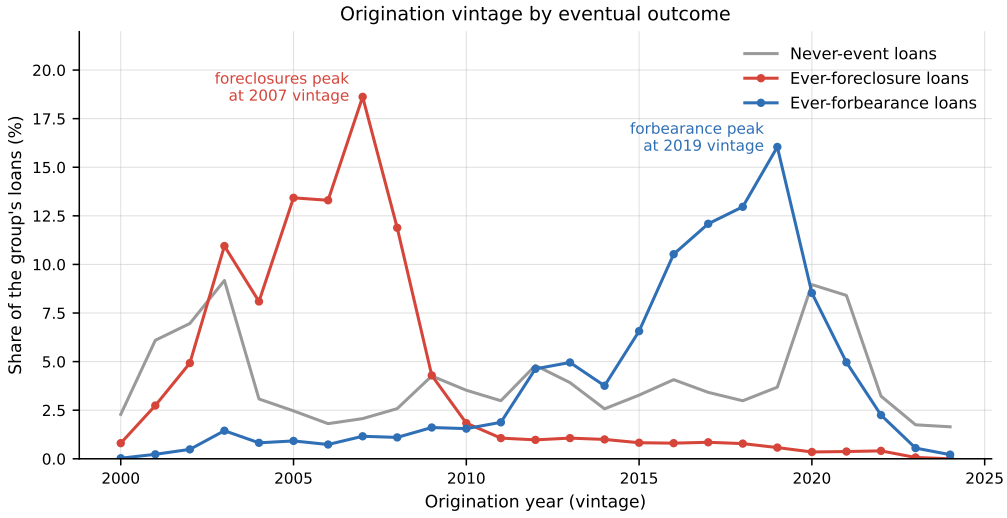


Figure 15: Distribution of each outcome group's loans across origination vintages. Ever-foreclosure loans concentrate in 2005–2008 (89 per cent originated before 2010); ever-forbearance loans in 2015–2021 (72 per cent). The two distressed populations, nearly identical on origination leverage and payment stress, are drawn from almost non-overlapping vintages.

A.2 Linear probability and logit models

The linear probability models regress indicators for ever reaching 90+ delinquency, ever entering forbearance, and—conditional on serious delinquency—ever foreclosing, on standardised origination characteristics (loan size, LTV, DTI, note rate, credit score), a thirty-year-term indicator and loan-purpose indicators, with origination-year fixed effects and HC1-robust standard errors. Each specification is estimated on all vintages and on the post-2010 book. In the foreclosure-conditional model an ever-forbearance indicator carries a coefficient of -27.0 percentage points across all vintages, and a COVID-onset indicator carries one of the same sign and comparable size. Tables 14 and 15 report the full coefficient sets.

Because all three outcomes are rare, each specification is re-estimated by logit, fitted by Newton–Raphson on the same design matrices with robust standard errors and summarised by the average marginal effect $\overline{p(1-p)}\,\hat{\beta}_j$, which is on the LPM's probability scale (Tables 16 and 17). Across the thirty standardised origination coefficients the linear coefficient and the logit marginal effect agree in sign in every case, as do the two main effects of the foreclosure model. The two foreclosure logits stop at the hundred-iteration cap rather than converging. The estimator drops collinear columns but does not detect separation, so a perfectly separated origination-year dummy—a vintage with no foreclosures among its seriously delinquent loans—remains in the design and its coefficient diverges. The remaining estimates approach their maximum-likelihood values on the sample excluding that cell, which is why they are reported, but the approximation is not controlled and no standard error on the separated dummy itself is interpretable.

The two estimators are not comparable on the forbearance $\times$ COVID interaction, and the apparent conflict there is an artefact of the comparison rather than a disagreement about the data. The linear model puts the interaction at $+23.7$ points on all vintages and $+7.1$ on the post-2010 book; the logit entry is $\overline{p(1-p)}\,\hat{\beta}_j$ evaluated on the product column, which rescales a shift in the log-odds ratio and is not the cross-partial of predicted probabilities, so the two need not share a sign even when they agree about the data. The cell the interaction identifies, COVID-onset delinquencies that entered forbearance, foreclosed at 1.0 per cent (Table 2), exactly where a linear model is least trustworthy, and no interpretive weight is placed on it either way.

A.3 Hazard models and the resolution of delinquency

This subsection collects the survival exhibits behind the protection estimate quoted in Section 2.3. Table 18 reports the foreclosure hazard among loans reaching 90+ delinquency during COVID under four designs: the original specification, which censors prepayment and dates exposure from ever-forbearance status; a time-varying version that switches exposure on at the observed entry month and so removes immortal-time bias; a Fine and Gray (1999) subdistribution model that treats prepayment as a competing event rather than independent censoring; and a six-month landmark combined with Fine–Gray. The ratio moves from 0.117 to 0.126 across them, so the corrections matter but do not overturn the sign. Because a Grambsch and Therneau (1994) test rejects propor-

Table 14: Linear probability models of serious delinquency and forbearance take-up.

	Pr(ever 90+ dlq.)		Pr(ever forbearance)	
	All vintages	Post-2010	All vintages	Post-2010
	(1)	(2)	(3)	(4)
log(Loan amount) [SD]	-0.583^{***}	-0.102^{***}	0.167^{***}	0.337^{***}
	(0.022)	(0.024)	(0.016)	(0.024)
Origination LTV [SD]	0.954^{***}	0.427^{***}	0.225^{***}	0.364^{***}
	(0.019)	(0.021)	(0.014)	(0.022)
DTI [SD]	0.825^{***}	0.793^{***}	0.515^{***}	1.044^{***}
	(0.020)	(0.023)	(0.012)	(0.024)
Credit score [SD]	-3.289^{***}	-2.650^{***}	-1.138^{***}	-2.088^{***}
	(0.027)	(0.034)	(0.017)	(0.032)
Note rate [SD]	1.145^{***}	0.930^{***}	0.887^{***}	0.950^{***}
	(0.059)	(0.062)	(0.044)	(0.061)
30-year term	1.102^{***}	0.419^{***}	0.156^{***}	0.241^{***}
	(0.046)	(0.050)	(0.032)	(0.052)
Purpose: refinance	0.998^{***}	-0.125^{*}	-0.067	0.026
	(0.050)	(0.054)	(0.036)	(0.057)
Purpose: cash-out	1.246^{***}	0.084	-0.010	0.017
	(0.057)	(0.064)	(0.039)	(0.065)
Constant	-2.208^{***}	3.297^{***}	-2.483^{***}	2.158^{***}
	(0.373)	(0.091)	(0.094)	(0.078)
Vintage fixed effects	Yes	Yes	Yes	Yes
R^2	0.065	0.035	0.040	0.040
Observations	1,113,742	669,157	1,113,742	669,157

Notes: Coefficients in *percentage points* (the estimate times 100), with HC1-robust standard errors in parentheses. Variables marked [SD] are standardised, so the coefficient is the effect of a one-standard-deviation change; the remaining regressors are indicators, so their coefficients are raw percentage-point differences. All models include origination-year (vintage) fixed effects. "Post-2010" restricts to loans originated in 2010 or later. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. These associations are endogenous and selection-laden; the causal weight of the paper rests on the structural counterfactual of Section 6. Estimated on the loan-level 2-percent Fannie Mae sample.

Table 15: Linear probability model of foreclosure conditional on serious delinquency.

	Pr(foreclosure | ever 90+ dlq.)	
	All vintages	Post-2010
	(1)	(2)
log(Loan amount) [SD]	-4.558^{***}	-2.348^{***}
	(0.200)	(0.181)
Origination LTV [SD]	7.183^{***}	2.316^{***}
	(0.234)	(0.190)
DTI [SD]	-0.233	-0.246
	(0.205)	(0.202)
Credit score [SD]	3.200^{***}	0.574^{***}
	(0.182)	(0.164)
Note rate [SD]	5.669^{***}	0.741
	(0.505)	(0.399)
30-year term	8.757^{***}	2.146^{***}
	(0.553)	(0.474)
Purpose: refinance	6.396^{***}	2.258^{***}
	(0.502)	(0.414)
Purpose: cash-out	4.959^{***}	2.284^{***}
	(0.490)	(0.415)
Ever forbearance	-27.044^{***}	-11.024^{***}
	(0.782)	(1.079)
COVID-era 90+ delinquency	-20.898^{***}	-11.736^{***}
	(0.729)	(0.832)
Ever forbearance × COVID	23.678^{***}	7.104^{***}
	(0.916)	(1.174)
Constant	5.728	21.942^{***}
	(4.131)	(1.470)
Vintage fixed effects	Yes	Yes
R^2	0.270	0.119
Observations	47,341	20,338

Notes: Coefficients in *percentage points* (the estimate times 100), with HC1-robust standard errors in parentheses. Variables marked [SD] are standardised, so the coefficient is the effect of a one-standard-deviation change; the remaining regressors are indicators, so their coefficients are raw percentage-point differences. All models include origination-year (vintage) fixed effects. "Post-2010" restricts to loans originated in 2010 or later. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. These associations are endogenous and selection-laden; the causal weight of the paper rests on the structural counterfactual of Section 6. The estimation sample conditions on ever reaching 90+-day delinquency.

Table 16: Logit robustness: average marginal effects against the linear probability models.

	Pr(ever 90+ dlq.)				Pr(ever forbearance)			
	All vintages		Post-2010		All vintages		Post-2010	
	LPM	AME	LPM	AME	LPM	AME	LPM	AME
log(Loan amount) [SD]	-0.583^{***}	-0.560^{***}	-0.102^{***}	-0.112^{***}	0.167^{***}	0.175^{***}	0.337^{***}	0.355^{***}
	(0.022)	(0.023)	(0.024)	(0.025)	(0.016)	(0.016)	(0.024)	(0.026)
Origination LTV [SD]	0.954^{***}	1.553^{***}	0.427^{***}	0.862^{***}	0.225^{***}	0.457^{***}	0.364^{***}	0.707^{***}
	(0.019)	(0.027)	(0.021)	(0.031)	(0.014)	(0.019)	(0.022)	(0.031)
DTI [SD]	0.825^{***}	0.975^{***}	0.793^{***}	1.129^{***}	0.515^{***}	0.812^{***}	1.044^{***}	1.373^{***}
	(0.020)	(0.019)	(0.023)	(0.029)	(0.012)	(0.018)	(0.024)	(0.031)
Credit score [SD]	-3.289^{***}	-2.502^{***}	-2.650^{***}	-2.115^{***}	-1.138^{***}	-1.118^{***}	-2.088^{***}	-1.700^{***}
	(0.027)	(0.018)	(0.034)	(0.023)	(0.017)	(0.015)	(0.032)	(0.024)
Note rate [SD]	1.145^{***}	0.995^{***}	0.930^{***}	0.474^{***}	0.887^{***}	0.399^{***}	0.950^{***}	0.655^{***}
	(0.059)	(0.059)	(0.062)	(0.060)	(0.044)	(0.043)	(0.061)	(0.067)
30-year term	1.102^{***}	1.680^{***}	0.419^{***}	0.963^{***}	0.156^{***}	0.539^{***}	0.241^{***}	0.649^{***}
	(0.046)	(0.057)	(0.050)	(0.068)	(0.032)	(0.044)	(0.052)	(0.070)
Purpose: refinance	0.998^{***}	0.793^{***}	-0.125^{*}	-0.089	-0.067	0.042	0.026	0.046
	(0.050)	(0.052)	(0.054)	(0.060)	(0.036)	(0.038)	(0.057)	(0.061)
Purpose: cash-out	1.246^{***}	1.483^{***}	0.084	0.516^{***}	-0.010	0.227^{***}	0.017	0.299^{***}
	(0.057)	(0.050)	(0.064)	(0.059)	(0.039)	(0.039)	(0.065)	(0.062)
Vintage fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
McFadden pseudo-R^2	—	0.162	—	0.124	—	0.174	—	0.131
Observations	1,113,742	1,113,742	669,157	669,157	1,113,742	1,113,742	669,157	669,157

Notes: All entries are in *percentage points*. Columns headed LPM repeat the linear-probability estimates of Table 14 with HC1-robust standard errors; columns headed AME report the average marginal effect of a logit fitted by maximum likelihood on the identical sample and regressor set, $\overline{p(1-p)} \cdot \hat{\beta}_j$, with standard errors from the first-order approximation $\overline{p(1-p)} \cdot \mathrm{SE}(\hat{\beta}_j)$ and significance taken from the logit coefficient. For the standardised origination characteristics this is the exact average partial derivative; for the indicator regressors it is a tangent approximation to the discrete difference in predicted probabilities. Variables marked [SD] are standardised; the remainder are indicators. All models carry origination-year (vintage) fixed effects. $^{*}\ p < 0.05$, $^{**}\ p < 0.01$, $^{***}\ p < 0.001$. The two estimators agree in sign on every standardised origination characteristic in all four columns.

Table 17: Logit robustness for foreclosure conditional on serious delinquency.

	Pr(foreclosure | ever 90+ dlq.)			
	All vintages		Post-2010	
	LPM	AME	LPM	AME
log(Loan amount) [SD]	-4.558^{***}	-4.674^{***}	-2.348^{***}	-2.163^{***}
	(0.200)	(0.208)	(0.181)	(0.159)
Origination LTV [SD]	7.183^{***}	7.839^{***}	2.316^{***}	2.784^{***}
	(0.234)	(0.270)	(0.190)	(0.256)
DTI [SD]	-0.233	-0.213	-0.246	-0.103
	(0.205)	(0.167)	(0.202)	(0.186)
Credit score [SD]	3.200^{***}	2.571^{***}	0.574^{***}	0.301
	(0.182)	(0.163)	(0.164)	(0.156)
Note rate [SD]	5.669^{***}	5.442^{***}	0.741	0.761
	(0.505)	(0.547)	(0.399)	(0.392)
30-year term	8.757^{***}	9.063^{***}	2.146^{***}	1.844^{***}
	(0.553)	(0.562)	(0.474)	(0.474)
Purpose: refinance	6.396^{***}	4.954^{***}	2.258^{***}	2.072^{***}
	(0.502)	(0.485)	(0.414)	(0.407)
Purpose: cash-out	4.959^{***}	4.538^{***}	2.284^{***}	2.288^{***}
	(0.490)	(0.461)	(0.415)	(0.397)
Ever forbearance	-27.044^{***}	-25.932^{***}	-11.024^{***}	-4.075^{***}
	(0.782)	(1.285)	(1.079)	(0.581)
COVID-era 90+ delinquency	-20.898^{***}	-20.890^{***}	-11.736^{***}	-4.463^{***}
	(0.729)	(1.439)	(0.832)	(0.502)
Ever forbearance × COVID	23.678^{***}	0.689	7.104^{***}	-2.585^{**}
	(0.916)	(2.202)	(1.174)	(0.795)
Vintage fixed effects	Yes	Yes	Yes	Yes
McFadden pseudo-R^2	—	0.294	—	0.255
Observations	47,341	47,341	20,338	20,338

Notes: All entries are in *percentage points*. Columns headed LPM repeat the linear-probability estimates of Table 15 with HC1-robust standard errors; columns headed AME report the average marginal effect of a logit fitted by maximum likelihood on the identical sample and regressor set, $\overline{p(1-p)} \cdot \hat{\beta}_j$, with standard errors from the first-order approximation $\overline{p(1-p)} \cdot \mathrm{SE}(\hat{\beta}_j)$ and significance taken from the logit coefficient. Variables marked [SD] are standardised; the remainder are indicators. All models carry origination-year (vintage) fixed effects. $^{*}\ p < 0.05$, $^{**}\ p < 0.01$, $^{***}\ p < 0.001$. The two estimators agree on every origination characteristic and on both main effects. They differ on the interaction, but the comparison there is not like-for-like: the LPM coefficient is a probability-scale difference-in-differences, while $\overline{p(1-p)} \cdot \hat{\beta}_{\mathrm{int}}$ is a rescaled shift in the log-odds ratio and is not the cross-partial of predicted probabilities, so the two need not share a sign even when they agree about the data. The cell that interaction identifies—COVID-onset delinquencies that entered forbearance—has an empirical foreclosure rate of 1.0% (Table 2, Group B): see Appendix A.2.

tionality for the forbearance term, the hazard ratio is not a single summary of the whole follow-up, and Table 19 therefore reports cumulative incidence functions alongside it, separating foreclosure from prepayment at a common horizon. Those functions are descriptive rather than corrected: the classification is forbearance within twelve months while the clock runs from the first 90+ flag, so the treated arm is conditioned on surviving a year without foreclosing and the comparison retains the immortal-time problem that row 4 of Table 18 removes. Read the cumulative incidences for the shape of the divergence and the landmark row for its size. Figure 16 shows why the two have to be read together: the treated group's foreclosures arrive later as well as less often. Figure 17 then steps back from foreclosure to ask how delinquency spells ended at all, pre-COVID against COVID. Every specification in this subsection is built on the raw delinquency field, which accrues during forbearance; Appendix A.4 rebuilds the risk set on a clock that does not, and that is where most of the association goes.

Table 18: Foreclosure hazard after first 90+ delinquency: the original specification and three corrections.

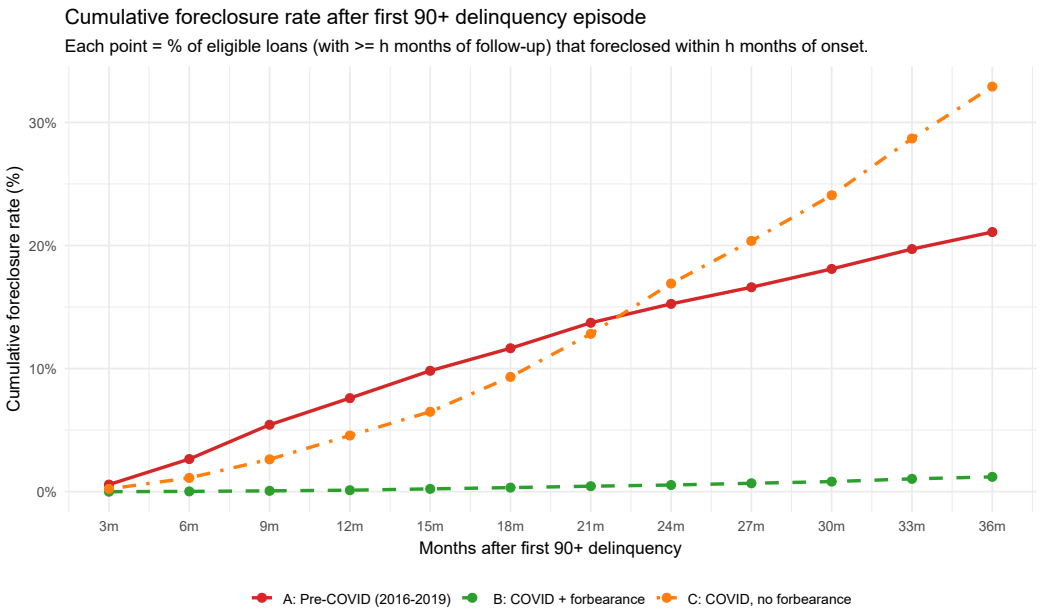
Specification	Exposure	Hazard ratio	95% CI
Original (prepayment censored)	Ever forbearance	0.117	[0.099, 0.139]
Time-varying exposure	Entry month	0.125	[0.105, 0.148]
Fine–Gray subdistribution	Ever forbearance	0.137	[0.116, 0.163]
Landmark ($L = 6$) + Fine–Gray	Entry by month 6	0.126	[0.105, 0.151]

Notes. Risk set is loans reaching 90+ delinquency during COVID; clock starts at first such delinquency. All specifications control for origination LTV, DTI, credit score, note rate, log balance and purpose. Row 2 switches the exposure on at the observed entry month, removing immortal-time bias; rows 3 and 4 treat prepayment as a competing risk. All $p < 0.001$. A Grambsch–Therneau test rejects proportionality for the forbearance term ($\chi^2 = 14.1$, $p < 0.001$) and weakly for purpose ($p = 0.027$); the origination covariates satisfy it. The risk set is defined on the raw `DLQ_STATUS` field, which accrues during forbearance; Appendix A.4 re-estimates on a clock that does not.

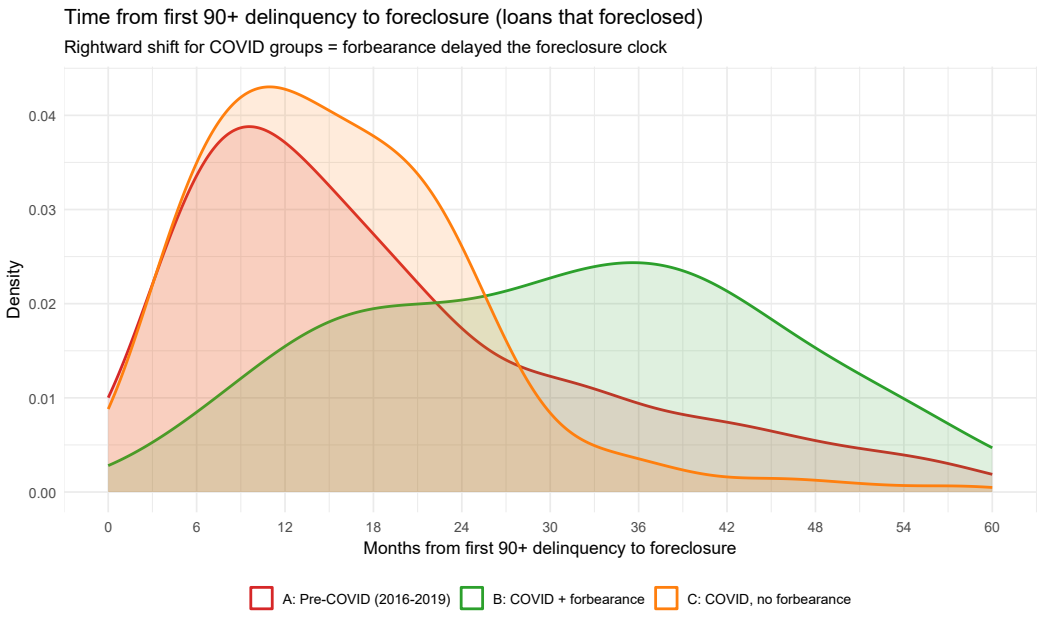
Table 19: Cumulative incidence of foreclosure and prepayment, by forbearance status within twelve months.

	Foreclosure (%)		Prepayment (%)	
Months since first 90+	Forbearance	No forbearance	Forbearance	No forbearance
6	0.02	1.71	8.85	12.33
12	0.09	4.18	19.60	19.59
24	0.88	12.26	34.58	29.55
36	1.48	17.20	39.49	35.15

Notes. Aalen and Johansen (1978) cumulative incidence in the competing-risks model; foreclosure and prepayment are the absorbing states. No proportionality assumption is required. The near-equality of the prepayment columns is why treating prepayment as independent censoring biased the foreclosure estimate only modestly. The classification is forbearance within the first twelve months, taken on a clock that starts at the first 90+ flag and is not restarted, so a loan must survive twelve months without foreclosing to enter the forbearance column. These incidences are descriptive and the gap between the columns is an upper bound; row 4 of Table 18 is the design that aligns exposure.



(a) How much foreclosure: cumulative rate since onset



(b) When it happens: timing among loans that foreclosed

Figure 16: Foreclosure after a first 90+ delinquency, for the three groups of Table 2. Panel (a) repeats Figure 1 so that the level and the timing can be read side by side; it is the same plot, not a second measurement. Panel (a): cumulative foreclosure rate by months since onset, each point the share of loans with at least that much follow-up. Panel (b): kernel density of months from onset to foreclosure, among loans that foreclosed. The two must be read together. Group B's density peaks near three years after onset against the first year for Groups A and C, but panel (a) shows that it is a late peak in a much smaller number of foreclosures, not a deferral of the same number.

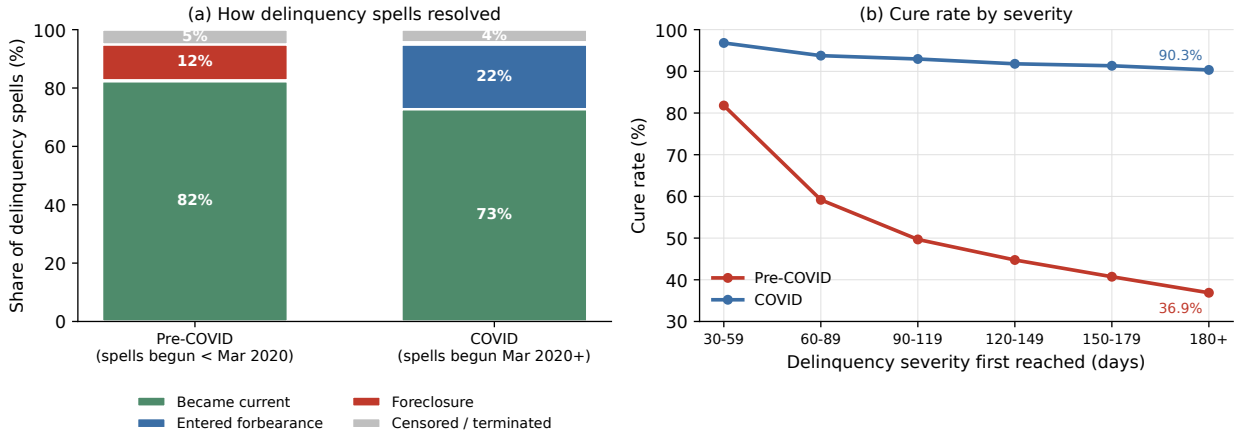


Figure 17: Panel (a): composition of delinquency-spell exits by first terminal event, pre-COVID versus COVID. The foreclosure exit (12.3 per cent of spells pre-COVID) collapses to 0.8 per cent while forbearance absorbs 22.2 per cent. Panel (b): cure rate by the delinquency severity a loan first reached. The pre-COVID gradient—cure falling from 82 per cent at 30–59 days to 37 at 180+ —flattens under COVID, with the deepest delinquencies curing about 90 per cent of the time.

A.4 The delinquency clock and the endogenous risk set

The corrections in Table 18 address two objections to the design. A third is more serious and no estimator resolves it, because it concerns which loans enter the risk set. `DLQ_STATUS` counts months of missed payment, and a borrower in forbearance is exercising a contractual right to miss them: in the sample, 72.6 per cent of month-to-month transitions among loans in forbearance show the field increasing, and entry into forbearance precedes the 90+ flag for 92.0 per cent of the 12,822 ever-forborne loans that reach it during the COVID window. The threshold that defines the risk set is therefore, for most treated loans, a consequence of treatment, and the surviving contrast drifts towards sanctioned versus unsanctioned non-payment—a comparison in which the treated arm cannot foreclose almost by construction.

Define *unexcused* delinquency as months in which the loan is delinquent and not in forbearance, counted since it was last current. Rebuilding the COVID-era risk set on this clock retains 28.1 per cent of the treated loans the raw clock admits, 3,603 of 12,822. Those counts are risk-set admissions on each clock, taken before any covariate or landmark restriction. The treated column of the table below, 3,094 of 12,379 or 25.0 per cent, counts treated units in the estimation frame instead, after the treatment window, the exclusion of loans that foreclose on or before the landmark, complete cases on the origination covariates and propensity-score trimming. The smaller pair is a subset of the larger, and the two retention rates agreeing to within three points is a check on the frame builder rather than a discrepancy.

Two features of Table 20 matter more than the point estimates. Half the sample is lost but 418 of 423 foreclosures are retained: the loans that drop out were admitted only because forbearance placed them there, and they were not foreclosing. And the treated share falls from 88 to 46 per cent, which is why the effective sample size of the reweighted controls rises from 16 to 52 per cent;

Table 20: The conditional frame under each delinquency clock.

Clock	Loans	Treated	Foreclosures	Raw diff.	IPW
Raw `DLQ_STATUS`	14,054	12,379	423	−14.21	−4.83
Frozen (no accrual in forbearance)	6,735	3,094	418	−4.49	−1.82

Notes. Percentage-point differences, treated minus control; outcome is foreclosure within 36 months of a six-month landmark. Both rows use the same frame builder, covariates (origination LTV, DTI, credit score, note rate, log balance, purpose, onset month, state) and 1st/99th-percentile trimming. Effective sample size of the reweighted controls is 276 of 1,675 on the raw clock and 1,892 of 3,641 on the frozen clock. The outcome is a binary foreclosure-within-36-months indicator on a common frame builder: prepayment is coded as a non-event rather than as a competing risk, and beyond clearing foreclosures that precede the landmark the design imposes no minimum follow-up. The two rows are therefore comparable to each other, which is what the gap between clocks requires, but their levels are not comparable to the competing-risks estimates of Table 18.

the frozen frame is the first with defensible common support. The raw difference falls from −14.21 to −4.49 points and the reweighted estimate from −4.83 to −1.82, so roughly two-thirds of the association is definitional, against the tenth that the immortal-time and competing-risk corrections remove. The Oster (2019) bound at $R_{\max} = 1.3R^2_{\text{long}}$ gives −1.29 points on the frozen clock with $\delta = 2.58$, so eliminating the estimate requires unobserved selection two and a half times the observed; at $R_{\max} = 2R^2_{\text{long}}$ the sign turns positive. Fixing the population before the shock—loans current in February 2020—and comparing four-year foreclosure by CARES-window take-up gives +0.34 points reweighted, a relative risk of 8.4: the sign is opposite to the conditional estimates, the clearest indication that the conditional frame measures selection into a risk set rather than an effect of relief. A servicer-leniency instrument is better conditioned on the frozen frame, with a first-stage F of 34.9 clustered on 13 servicers—too few clusters for the clustered standard error itself to be reliable, so the figure is indicative rather than a test. Two exclusion-restriction checks fail regardless. A servicer's COVID-era grant rate predicts foreclosure among loans that reached 90+ delinquency *before* the pandemic ($t = -6.11$), when the programme did not yet exist; and the same grant rate moves modification, a second relief channel with its own effect on foreclosure. Either is enough to break excludability. The data support a bracket: conditional on genuine unexcused delinquency, forborne borrowers foreclosed 1.3 to 1.8 points less often; over the whole pre-determined book, 0.34 points more often. Neither is causal. That the grant rate varies so much across servicers, and that the variation tracks pre-existing loss-mitigation practice, is itself the finding of Kim et al. (2024), who use servicer assignment to study CARES take-up directly. It is also why the 3,531 seriously delinquent COVID-onset borrowers who never entered forbearance are not evidence that relief was unwanted: the model's approval probability $p^{\text{forb}}(s)$ stands in for exactly this intermediation, and setting it near one is a statement about the statute rather than about servicer behaviour. The same literature documents that access to the price instrument was unequal along income and credit lines (Agarwal et al., 2023; Fuster et al., 2021), which is the empirical counterpart of the underwriting gate the model imposes.

B The Life-Cycle Fit over the Stationary Distribution

Section 5.3 measures the model's life-cycle profiles in the pre-crisis phase of the simulation, which holds the aggregate state fixed at normal state 1. The solve prices all eight normal states, so this is a choice made when simulating rather than a restriction imposed by the solution. This appendix asks what the same profiles look like when the aggregate chain is instead allowed to run, and the moments are averaged over the states the economy actually visits.

The experiment discards an 800-period burn-in in which the chain wanders freely, then measures 1,000 further free periods over 15,000 households. Nothing else changes: the same solved object, the same seeds, the same definitions, the same survey series.

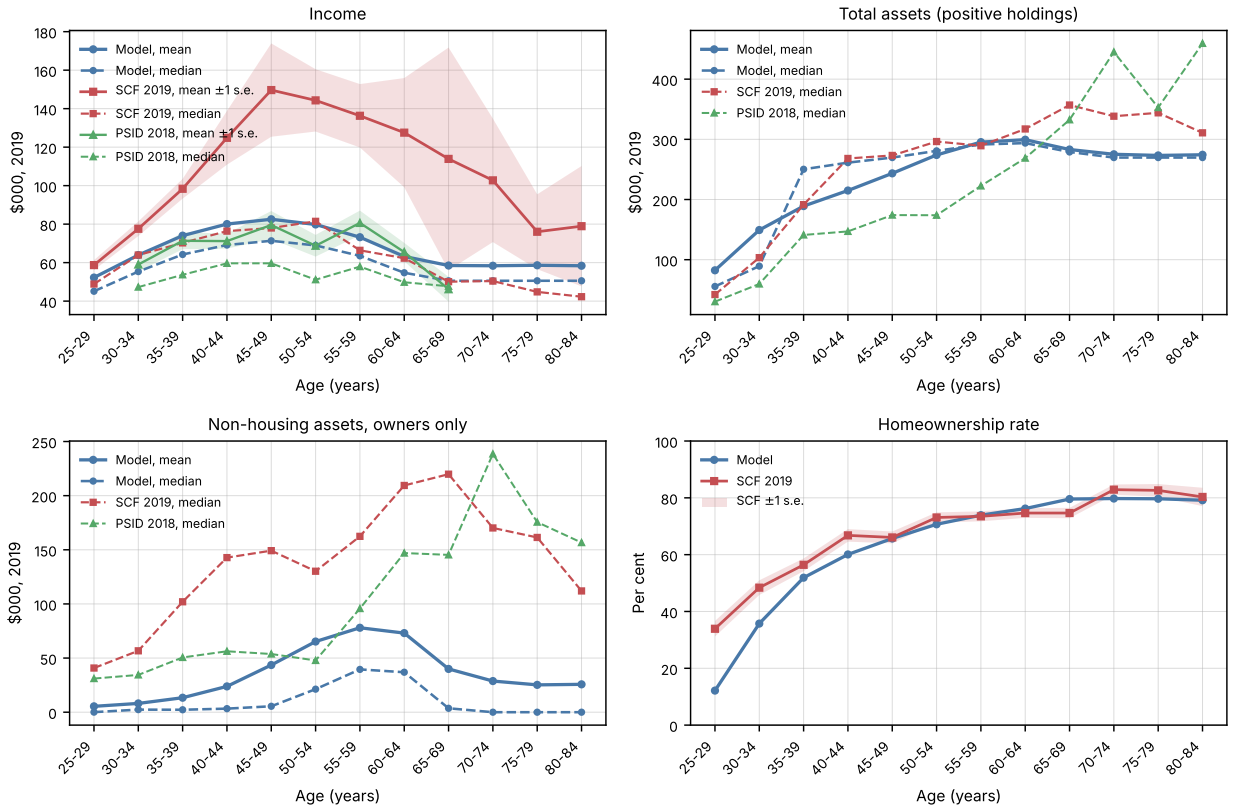


Figure 18: Life-cycle fit over the stationary distribution of aggregate states.

Notes. As Figure 2, with one difference: the model series averages over 1,000 periods in which the aggregate state follows its own Markov chain, rather than 60 periods pinned at normal state 1. Survey series are unchanged.

What moves, and what does not. The aggregate homeownership rate falls from 69.8 per cent to 60.8. The profile matters more than the level: pinned at state 1 the model reaches 79.6 per cent at ages 45–49 against the SCF's 66.1, a 13.5 point gap; over the stationary distribution it reaches 65.7 per cent, a gap of 0.4. Nothing was recalibrated. Mean loan-to-value at origination is unmoved—0.757 against 0.758—which is the expected result, since origination leverage is pinned at the down-payment cap and does not depend on the aggregate state. Leverage then amortises away

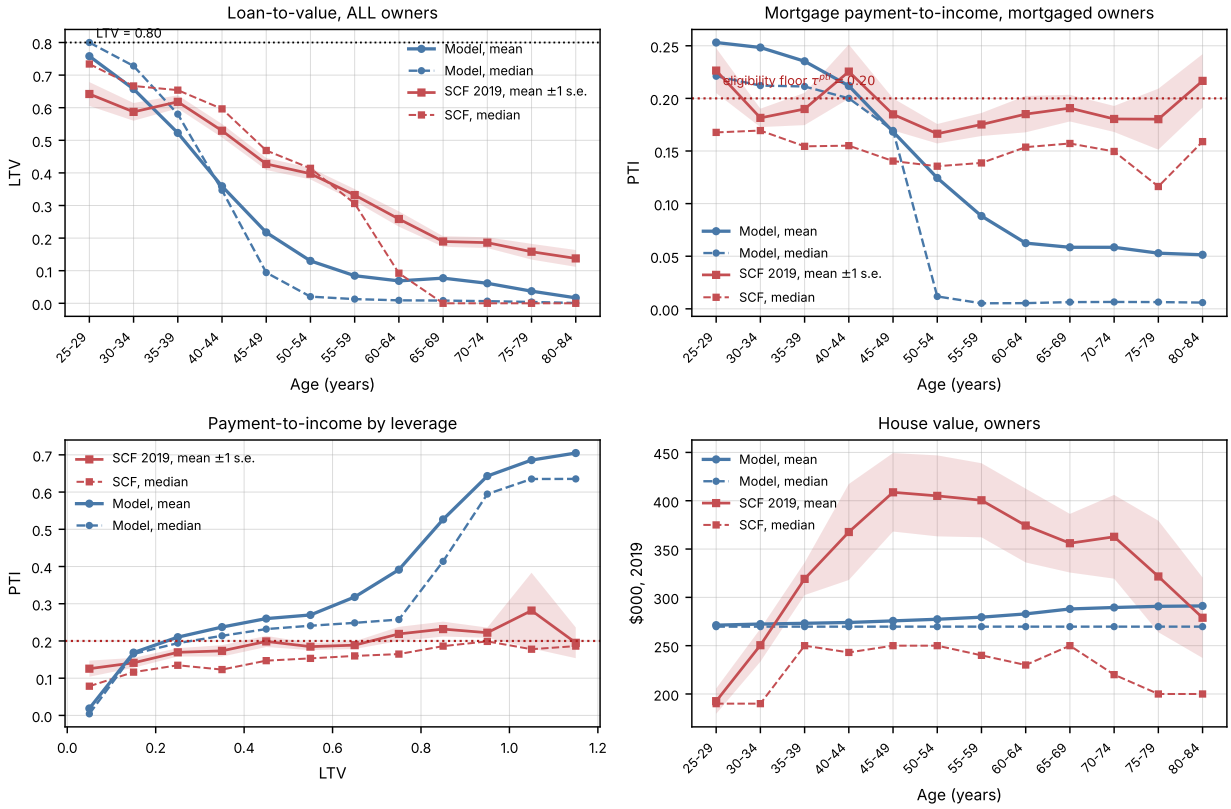


Figure 19: Leverage and payment burden over the stationary distribution.

Notes. As Figure 3, measured over the same free-walk window. The leverage panel is drawn over *all* owners, including those who own outright at an LTV of zero.

faster: by 45–49 mean LTV is 0.217 against 0.282. Mean house value among owners rises by about \$23,000 at every age, a level shift and not a change of shape.

Why state 1 flatters homeownership. The eight normal states differ in the house price they carry, from \$250,000 at state 1 to \$300,000 at state 8, and homeownership in the free walk falls monotonically across them, from 67.6 per cent to 56.2. State 1 is the cheapest-housing state and carries a near-lowest mortgage rate, so it is the state most favourable to ownership in the model's own state space.

Where the long-run measurement fits worse. Mean payment-to-income at ages 25–29 moves from 0.199 to 0.253 against the SCF's 0.226, turning an undershoot into an overshoot of similar size, and the gradient of payment burden in leverage steepens. The young end of the ownership profile also deteriorates, from 17.8 per cent to 12.2 against 33.9 in the data: households are born without assets and cannot meet a down payment at 25, and that bites harder when houses are dearer on average. The long-run measurement is therefore not uniformly closer to the data. It is a different, and more defensible, conditioning choice, and it is reported here rather than substituted into the main text.

Two caveats. First, the ownership premium θ^{own} was calibrated to the aggregate homeownership rate under the pinned measurement. Adopting the long-run basis in the main text would require recalibrating it, and every result downstream would move; that is not done here. Second, these figures use a single aggregate-path seed. Over 1,000 periods the realised visit shares are close to but not equal to the chain's stationary probabilities—state 5 is visited in 4.4 per cent of periods against a stationary 10.1—so the averages carry path sampling error on top of the usual cross-sectional error. Re-weighting the state-conditional ownership rates by the stationary probabilities gives 61.9 per cent against the realised 60.9, so this particular moment is not sensitive to the path; a thinner moment would be.

C Origination Selection in the Model

Section 2 shows that the loans which entered forbearance looked different when they were written: higher payment burden, higher leverage, larger balances and lower credit scores than loans that never had an event. That comparison conditions on characteristics recorded at origination, which is the only way the loan-level file can be cut, since it carries income at origination and no borrower age. This appendix applies the same cut to the model.

Table 21 splits every loan the model originates into the same three groups, by what eventually happened to the household holding it, and reports the same statistic at the same moment. Table 22 gives the percentiles behind those means, and Table 23 reports the across-group differences with standard errors clustered on the household.

Table 21: Model loans at origination, by eventual outcome.

	Forbearance				Foreclosure				Never-event			
	N	Mean	S.D.	Med.	N	Mean	S.D.	Med.	N	Mean	S.D.	Med.
Original UPB ($)	1,773	195,289	17,032	200,204	2,279	194,224	18,941	200,204	94,068	183,265	43,008	200,204
Origination PTI (%)	1,773	33.25	13.80	30.71	2,279	45.02	10.08	46.95	94,068	19.29	9.19	18.58
Original LTV (%)	1,773	77.94	6.87	80.00	2,279	77.61	7.57	80.00	94,068	72.90	17.97	80.00
Note rate at origination (%)	1,773	4.681	0.303	4.774	2,279	4.678	0.307	4.774	94,068	4.700	0.272	4.774
Age at origination	1,773	43.7	13.0	42.0	2,279	51.5	9.8	54.0	94,068	38.2	13.6	34.0

Notes. All loan originations observed in the simulated panel (98,120, of which 4,316 cash purchases are excluded as housing but not loan originations), contributed by 71,357 distinct households. A loan is assigned to the forbearance group if the household was approved while holding it, to the foreclosure group if it defaulted while holding it, and to the never-event group otherwise. The unit is the loan, not the household: forbearance is not once per life in this specification, since the eligibility flag resets on cure and on payoff, and 24 households accordingly contribute two separate forborne loans, so the 1,773 forborne loans come from 1,749 households. **Payment burden is the model's own scheduled annuity on the origination balance**, $k(r^{\text{lock}}, 30)\, b$, over income at the moment of underwriting, in per cent; the term is 30 because $n = \min(30, T - a)$ equals 30 for every origination before age 55. It is not read off the panel's realised payment column, which in the first period of a refinance also carries any voluntary principal paydown. Loan characteristics are measured at origination, borrower income and age at the underwriting decision one period earlier. Standard deviations use the $n - 1$ denominator. **There is no credit-score analogue**: the model prices mortgages off the aggregate state, not off an individual default-risk score, and none is invented. In its place the table reports the note rate locked at origination and the borrower's age at origination, the second of which the loan-level file cannot observe.

The model reproduces the direction of all three measurable gaps. Against all loans, the forbearance group originates with a payment burden 13.1 points higher, leverage 4.8 points higher and a balance 6.3 per cent larger. In the data the corresponding gaps are 4.6 points of DTI, 7.0 points of LTV and 15.6 per cent of balance. Every sign matches. The magnitudes do not, but they miss in both directions rather than one: the model selects roughly three times too hard on payment burden, and about two-thirds and two-fifths as hard as the data on leverage and loan size. The ordering of the three groups also matches the data on payment burden, with the foreclosure group above forbearance (45.0 against 33.3) and both far above never-event (19.3).

The payment-burden gap is a refinancing phenomenon, not a stretched-purchase one. The three groups are not drawn from the same mix of origination events. Only 40.7 per cent of forbearance loans and 7.0 per cent of foreclosure loans are purchases, against 69.1 per cent of never-event loans; the rest are refinances. This is partly mechanical – a household's last loan before an event is the one tagged, and later loans in a life are more often refinances – but it governs how the headline gap should be read. Within purchases the forbearance group originates at a payment burden of 20.0 per cent against the never-event group's 18.2, a gap of under two points. Within refinances it originates at 42.4 against 23.7, a gap of nearly nineteen. The model's selection into forbearance on payment burden therefore operates almost entirely through households refinancing into high-payment loans later in life, and barely at all through households buying homes they cannot comfortably afford. The empirical table cannot be decomposed the same way, so this is a statement about the model's mechanism rather than a comparison; it is also the sharpest available statement of what drives the paper's selection result.

Why the balance levels should not be compared across the two tables. In the data the ever-foreclosure group originates with the *smallest* balances – \$160,537 against \$217,252 for never-event loans and \$251,178 for forbearance loans – while in the model the foreclosure group's balances are the largest of the three, \$194,224 against \$183,265. This is not a failure of the model to reproduce an ordering, because the two balance columns are not measured on the same scale. Original UPB in the loan-level file is nominal and is never deflated, and the foreclosure group is drawn overwhelmingly from the mid-2000s vintages whose loans were written in substantially cheaper dollars, whereas the forbearance group is observed in 2020. The model has a single price level by construction, so no such wedge can exist in Table 21, and the empirical foreclosure-balance ordering is therefore not a moment the model is being asked to match. What the two tables can be compared on is the forbearance-versus-never-event contrast, which is measured within a period in both and which the model gets right in sign.

What separates forbearance from foreclosure in the model is burden and age, not size or leverage. Table 23 reports the across-group differences. Against never-event loans, the forbearance group's gaps on balance, payment burden, leverage and age are all large relative to their standard errors. Against the foreclosure group, only two of the five survive: payment burden, which is 11.8 points *lower* among forborne loans, and age at origination, which is 7.9 years lower. The balance gap (+\$1,065, $p = 0.060$) and the leverage gap (+0.33 points, $p = 0.151$) are not distinguishable from zero. The model therefore does not separate the two distressed groups by how much they borrowed or how much equity they put down; it separates them by how heavy the payment was and when in the life cycle it was taken on. That is consistent with the mechanism the paper argues for, and it is also a limitation: an empirical literature that separates forbearance from foreclosure on borrower risk has no counterpart here.

Leverage selects, but the collateral constraint caps how much it can. 69.5 per cent of model loans originate within half a point of the 80 per cent down-payment cap, and the standard deviation of origination LTV among forbearance loans is 6.9 points against 18.0 among never-event loans. Table 22 makes the ceiling visible: the median, 75th and 90th percentiles of origination LTV are exactly 80.00 in all three groups, and the groups separate only in their lower tails, at 68.1 and 68.5 against 52.8 for never-event loans at the tenth percentile. Selection into forbearance works by *excluding* low-leverage borrowers rather than by including especially high-leverage ones, because there are no especially high-leverage ones to include. That ceiling is a property of the model's single collateral constraint rather than a calibration failure, and it is why the model delivers about two-thirds of the data's leverage gap and could not deliver much more. It is also the limitation that drives the leverage coefficient to zero in the simulated linear probability model of Section 5.3.

The note rate does not select to any economically meaningful degree. The three groups originate at 4.681, 4.678 and 4.700 per cent – a spread of about two basis points, against within-

group standard deviations of about 0.3. The forbearance–never-event difference of -1.9 basis points is statistically distinguishable from zero (Table 23, $p = 0.009$), which is a reminder of how little a p-value means at this sample size; the forbearance–foreclosure difference is not ($p = 0.733$). The model's mortgage rate depends on the aggregate state and not on the borrower, so a rate gap could only arise through *when* each group happened to borrow, and almost none does: the 10th through 90th percentiles of the locked rate are the same single value in all three groups. The claim should be made no stronger than the panel supports, however. Because this panel is pinned at aggregate state 1 outside the crisis, only two note rates are ever locked in it, so what the table establishes is that the model has no *borrower-level* rate selection, not that a richer state path would generate none through timing. Any rate-based selection in the data is risk pricing the model does not have, and is the closest thing here to the missing credit-score channel.

What the model sees that the data cannot. Forbearance loans are originated at age 43.7 on average and foreclosure loans at 51.5, against 38.2 for never-event loans. The loan-level file records no borrower age, so this gradient is invisible in the empirical table, and any age variation there is absorbed into the characteristics that are recorded. That is the mechanism behind the claim in Section 5.3 that part of what presents as a leverage gradient in the data is a life-cycle gradient: in the model, the households that defer are the ones who borrowed later, carry more of the loan still outstanding, and meet the crisis with a payment they took on when older. It is the same fact as the refinancing composition above, seen from the other side.

The forbearance group is a mixture, not a uniformly stretched group. Table 22 shows a payment-burden distribution for the forbearance group running from 17.9 at the tenth percentile to 52.5 at the ninetieth, with a median of 30.7 below a mean of 33.3 and a standard deviation of 13.8. A quarter of the group originates below 21.0 per cent, which is beneath the never-event group's own 75th percentile of 21.2. The foreclosure group, by contrast, is tight and uniformly high: 28.5 at the tenth percentile and 54.8 at the ninetieth, with an interquartile range of less than twelve points. Selection into forbearance in the model is therefore driven by the upper part of the burden distribution rather than by a shift of the whole distribution, which is what the refinancing split above describes in different terms.

D Estimation of the Exogenous Processes

Two processes are estimated outside the model and taken as given inside it: the idiosyncratic component of labour income, and the aggregate state. This appendix documents both, and in each case separates what was estimated from what the model actually uses, because the two are not the same for income.

Table 22: Distribution of model origination characteristics, by eventual outcome.

		Percentile				
	Group	10	25	50	75	90
Original UPB ($)	Forbearance	171,378	200,204	200,204	200,204	200,204
	Foreclosure	171,378	197,448	200,204	200,204	200,204
	Never-event	137,764	196,456	200,204	200,204	200,204
	ALL loans	140,315	196,456	200,204	200,204	200,204
Origination PTI (%)	Forbearance	17.90	21.02	30.71	46.50	52.50
	Foreclosure	28.52	40.50	46.95	52.41	54.84
	Never-event	10.24	15.07	18.58	21.15	28.52
	ALL loans	10.24	15.70	18.91	22.05	33.32
Original LTV (%)	Forbearance	68.05	80.00	80.00	80.00	80.00
	Foreclosure	68.48	78.90	80.00	80.00	80.00
	Never-event	52.79	78.50	80.00	80.00	80.00
	ALL loans	56.07	78.50	80.00	80.00	80.00
Note rate at origination (%)	Forbearance	4.774	4.774	4.774	4.774	4.774
	Foreclosure	4.774	4.774	4.774	4.774	4.774
	Never-event	4.774	4.774	4.774	4.774	4.774
	ALL loans	4.774	4.774	4.774	4.774	4.774
Age at origination	Forbearance	28.0	30.0	42.0	57.0	61.0
	Foreclosure	36.0	46.0	54.0	59.0	62.0
	Never-event	26.0	28.0	34.0	43.0	58.0
	ALL loans	26.0	28.0	34.0	44.0	59.0

Notes. Same sample as Table 21. **The repeated values are not a typesetting error.** Balance, leverage and the note rate are all discrete in the model – the mortgage grid, the 80 per cent down-payment cap and the state-contingent rate rule respectively – so their percentiles pile up on grid points. 69.5 per cent of loans originate within half a point of the leverage cap. The note rate is the extreme case: because this panel is pinned at aggregate state 1 in every period except the crisis at $t = 861$, only *two* note rates are ever locked in it — 4.774 per cent in state 1, which is 92.8 per cent of originations, and 3.690 per cent at the crisis, which is the remaining 7.2 per cent. Payment burden and age, which are continuous functions of the household's state, are the only two rows with a non-degenerate distribution. No loan originates above a payment burden of 54.8 per cent, against the model's underwriting cap of 60.

Table 23: Across-group differences at origination in the model, with household-clustered standard errors.

	Forb. − Never-event			Forb. − Foreclosure		
	Diff.	p	(S.E.)	Diff.	p	(S.E.)
Original UPB ($)	$+12{,}024^{***}$	< 0.001	(425)	$+1{,}065$	0.060	(566)
Origination PTI (%)	$+13.96^{***}$	< 0.001	(0.32)	-11.77^{***}	< 0.001	(0.39)
Original LTV (%)	$+5.04^{***}$	< 0.001	(0.17)	$+0.33$	0.151	(0.23)
Note rate at origination (%)	-0.019^{**}	0.009	(0.007)	$+0.003$	0.733	(0.010)
Age at origination	$+5.5^{***}$	< 0.001	(0.3)	-7.9^{***}	< 0.001	(0.4)

Notes. Each cell is the difference in means between the forbearance group and the named comparison group, obtained as the coefficient on a group dummy. Standard errors are clustered on the household, because one household originates several loans over its life and those loans share its income process and its entire history; treating loans as independent roughly doubles the reported t-statistics and changes none of the conclusions here. *, **, *** denote $p < 0.05$, 0.01, 0.001. **These are sampling errors for a simulated panel and are not inference about households.** The number of observations is a modelling choice (15,000 simulated households **[needs: confirm the panel behind Tables 20–22: the baseline panel is 30,000 households, not 15,000]**), so a small p-value here establishes only that the gap is a property of the solved policy functions rather than simulation noise; it carries no information about whether the corresponding gap exists in the population. The note-rate row is the clearest illustration: a difference of 1.9 basis points is "significant" at this sample size and economically indistinguishable from zero.

D.1 The idiosyncratic income process

Sample. The estimation uses the PSID, restricted to male household heads with positive individual labour earnings (`EARN_TOT_RD`) who are working, observed between the ages of 25 and 65, and who appear for at least five annual income years. For the biennial waves from 1999 onward an interview reports income for the previous calendar year, so the income year is dated one year back and age is computed against it. Sampling weights are used throughout.

First stage: removing age effects year by year. Within each income year separately, log earnings are regressed on a full set of age dummies and years of education,

$$\log y_{it} \;=\; \textstyle\sum_a \delta_{at}\mathbf{1}\{\mathrm{age}_{it}=a\} \;+\; \lambda_t\,\mathrm{educ}_i \;+\; r_{it},$$

weighted by the PSID sampling weight. Estimating the age effects year by year rather than imposing one profile lets the cross-sectional age pattern shift over time, so that cohort and business-cycle variation is not forced into the life-cycle profile. The residual r_{it} is what the second stage works with.

Second stage: the deterministic life-cycle profile. For each age a the weighted mean residual $\bar r_a$ is computed, and a cubic in $j = a/J_{\max}$, $J_{\max}=65$, is fitted to those age-specific means,

$$\bar r_a \;=\; \gamma_0+\gamma_1 j+\gamma_2 j^2+\gamma_3 j^3+e_a.$$

The deterministic profile in logs is $\Gamma(a)=\bar y+\hat r_a$, where $\bar y$ is the weighted mean of log earnings over the whole sample, and the model reads $\exp\Gamma(a)$ in dollars. The profile actually shipped to the solver runs from age 25 to age 65 and peaks before retirement: $\Gamma(63)=\$52{,}514$, $\Gamma(64)=\$50{,}571$, $\Gamma(65)=\$48{,}595$. The pension is anchored on $\Gamma(64)$, the last working age at an annual period, not on the age-65 row the file also carries.

Idiosyncratic component and empirical covariances. Subtracting the fitted deterministic deviation leaves $y_{it}=r_{it}-\hat r_a$, which combines persistent and transitory variation. For every pair of ages present in the data the weighted covariance of y is accumulated across individuals, with the number of contributing individuals recorded alongside, giving an empirical covariance matrix and a matching count matrix. Only the upper triangle, including the diagonal, and only cells with enough support, enter the next step.

Minimum-distance estimation. The idiosyncratic component is modelled as a persistent AR(1) observed with white noise,

$$z_a=\rho z_{a-1}+u_a,\quad u_a\sim\mathcal{N}(0,\sigma_u^2),\quad z_0\sim\mathcal{N}(0,\sigma_{z0}^2),\qquad y_a=z_a+\varepsilon_a,\quad \varepsilon_a\sim\mathcal{N}(0,\sigma_\varepsilon^2),$$

with all innovations independent, so that for $a_i \leq a_j$

$$\mathrm{Cov}(y_{a_i}, y_{a_j}) = \rho^{a_j - a_i} \mathrm{Var}(z_{a_i}) + \mathbf{1}\{a_i = a_j\}\, \sigma_\varepsilon^2, \qquad \mathrm{Var}(z_a) = \rho^2 \mathrm{Var}(z_{a-1}) + \sigma_u^2.$$

The four parameters $\theta = (\rho, \sigma_u^2, \sigma_\varepsilon^2, \sigma_{z0}^2)$ minimise the distance between the empirical and theoretical covariances, *weighted by the number of individuals behind each cell*, so that age pairs observed for few people do not drive the fit. Minimisation is by L-BFGS-B with all four parameters bounded to $[0, 1]$, started at (0.95, 0.02, 0.02, 0.10); a closed-form moment estimate is computed first and reported as a diagnostic, and the optimisation is repeated with a second algorithm as a check.

What the model uses, which is not quite what the file holds. The solver loads the estimation output and then uses only two of its four parameters, and overrides both of those. The stored estimates are $\rho = 0.9811$, $\sigma_u^2 = 0.0191$, $\sigma_\varepsilon^2 = 0.1482$ and $\sigma_{z0}^2 = 0.3297$. It substitutes an annual persistence of $\rho_\eta = 0.9609$ and an annual innovation standard deviation of $\sigma_\eta = 0.1314$ for the stored ρ and σ_u. These are the values Table 4 reports. The remaining two, σ_ε and σ_{z0}, are read from the estimation file and then not used: the income grid is built from (ρ_η, σ_η) alone, and realised income is the age profile times the persistent component times the aggregate-exposure factor, with no additive transitory term. The model therefore carries no normal-times transitory income shock. Entrants at age 25 draw their persistent state from the stationary distribution of the discretised chain, not from $\mathcal{N}(0, \sigma_{z0}^2)$, so the estimated dispersion of initial permanent income does not enter either. Because a model period is a year, the period-to-annual conversion is the identity and no rescaling occurs. The process is then discretised onto $n_\eta = 11$ nodes.

This substitution is not currently traceable. The harness attributes the two substituted values to a later version of the estimation script, and that script is not in the repository. The numbers are therefore reported here as what the model runs on, not as something this appendix can derive. Either the script is recovered and this paragraph replaced with its output, or the process is re-estimated and Table 4 updated to match. Until then the two parameters should be described as calibrated rather than estimated.

D.2 The aggregate state process

Data. Three quarterly series are used, covering 143 quarters from 1989Q1 to 2024Q3: a short Treasury rate, real GDP per capita, and a house-price level in dollars. Output and house prices enter in logs; the rate enters in percentage points. The construction of the house-price column sits upstream of the discretisation script and is not documented here.

Detrending, not differencing. Each of the three series is detrended by regression on a cubic polynomial in time, and the residuals are carried forward. Nothing is differenced. This is what makes it meaningful for the model to carry a detrended output *level* and a house-price *level* as state variables rather than growth rates, and it is the single largest difference from the earlier version of

this procedure.

VAR(1) and the transformed state. A first-order vector autoregression with a constant is estimated on the three detrended series, $\mathbf{S}_t = \mathbf{A}\mathbf{S}_{t-1} + \boldsymbol{\varepsilon}_t$ with $\boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$. If any eigenvalue of $\mathbf{A}$ has modulus at least one a ridge penalty $\mathbf{A} \mapsto (1-\lambda)\mathbf{A}$ with $\lambda = 0.01$ is applied and stationarity rechecked. Writing $\mathbf{C}$ for the lower-triangular Cholesky factor of $\boldsymbol{\Sigma}$, the transformed state $\mathbf{x}_t = \mathbf{C}^{-1}\mathbf{S}_t$ follows $\mathbf{x}_t = \boldsymbol{\rho}\mathbf{x}_{t-1} + \mathbf{u}_t$ with $\boldsymbol{\rho} = \mathbf{C}^{-1}\mathbf{A}\mathbf{C}$ and $\mathrm{Var}(\mathbf{u}_t) = \mathbf{I}$. The recursive ordering puts the short rate first.

The grid, and why there are eight states. Each dimension of $\mathbf{x}$ is given a two-point grid, evenly spaced between $\bar{x}_i - m\, s_{x_i}$ and $\bar{x}_i + m\, s_{x_i}$ with $m = 1$ and s_{x_i} the sample standard deviation of the transformed data. The state space is the Cartesian product, so $N = 2^3 = 8$. Transition probabilities are built dimension by dimension by the method of Tauchen (1986) against the conditional mean $\boldsymbol{\mu} = \boldsymbol{\rho}\mathbf{x}^{(s)}$, with unit innovation variance in every dimension, and multiplied across dimensions; each row is renormalised. The resulting quarterly matrix is raised to the fourth power, and it is this *annual* matrix that is saved and that the model uses. Finally the grid is mapped back, $\mathbf{S}^{(s)} = \mathbf{C}\mathbf{x}^{(s)}$, so each state carries a detrended short rate, a detrended log output level and a detrended log house price. Levels are recovered by adding back the sample means of the short rate and of log house prices.

Because $\mathbf{C}$ is lower triangular, the short rate takes only two distinct values across the eight states, detrended output four, and the house price all eight. Table 24 lists them.

Table 24: The eight normal aggregate states.

	Detrended deviation			Level			
State	r	$\log \tilde{Y}$	$\log P^h$	r (%)	$\tilde{Y}$	P^h (\$)	r^m (%)
1	−1.4123	−0.0294	−0.0903	1.40	0.9711	250,255	4.77
2	+1.4123	−0.0107	−0.0474	4.23	0.9893	261,242	6.24
3	−1.4123	+0.0107	−0.0585	1.40	1.0108	258,352	4.57
4	+1.4123	+0.0294	−0.0155	4.23	1.0298	269,694	6.03
5	−1.4123	−0.0294	+0.0155	1.40	0.9711	278,207	4.77
6	+1.4123	−0.0107	+0.0585	4.23	0.9893	290,421	6.24
7	−1.4123	+0.0107	+0.0474	1.40	1.0108	287,209	4.57
8	+1.4123	+0.0294	+0.0903	4.23	1.0298	299,818	6.03

Notes. Read from the saved discretisation output. Deviations are the detrended values the discretisation produces; levels add back the sample means, 2.8138 percentage points for the short rate and 12.5206 for log house prices, the latter exponentiated. r^m is the mortgage-rate rule (7) evaluated at each state. The economy spends its pre-crisis periods in state 1, whose rule value of 4.77 per cent is the rate at which 92.8 per cent of the simulated panel's loans are originated.

Persistence. The annual transition matrix is strongly persistent along the dimensions the Cholesky ordering makes slow-moving. Rows 2 and 7 stay put with probability 0.82, rows 1 and 8 with probability 0.72, and rows 4 and 5 with probability 0.49; the two least persistent states, 3 and 6, move

to their house-price mirror with probability 0.46. The implied stationary distribution puts 0.25 on each of states 2 and 7, 0.12 on each of 1 and 8, 0.10 on each of 4 and 5 and 0.03 on each of 3 and 6. The simulations reported in the paper do not draw from this distribution: Baseline A pins the economy at state 1 outside the crisis, and Appendix B reports the free-Markov alternative.

How the crisis states attach. The eight states above are the *normal* component. Crisis states are appended to the grid and are reached with probability zero from the normal states, which is what makes each crisis an unanticipated event. Each carries its own short rate, house price, output multiplier and forbearance-approval probability. A crisis state's output is *not* the pre-crisis state's output scaled: it is anchored to the highest normal state and then scaled, which is why the baseline's output multiplier of 0.90 delivers a realised contraction of 4.6 per cent rather than ten. Writing κ for a crisis state's output multiplier and s_0 for the normal state the economy occupies when the crisis arrives, the mapping is

$$\overbrace{\tilde{Y}_c(\kappa)}^{\text{crisis output}} = \kappa \cdot \overbrace{\tilde{Y}(8)}^{\text{highest normal state}}, \qquad \overbrace{\frac{\tilde{Y}_c(\kappa)}{\tilde{Y}(s_0)} - 1}^{\text{realised contraction}} = \kappa \frac{\tilde{Y}(8)}{\tilde{Y}(s_0)} - 1. \tag{29}$$

Because $\tilde{Y}(8) > \tilde{Y}(1)$, the realised contraction is always smaller in magnitude than $1 - \kappa$. At the baseline's $\kappa = 0.90$ and $s_0 = 1$, (29) returns $0.90 \times 1.029789/0.971073 - 1 = -4.56$ per cent rather than the -10 per cent the multiplier suggests; at $\kappa = 0.95$ it returns $+0.74$ per cent, an expansion. Output changes are quoted as realised changes throughout the paper. Evaluating the mortgage rule at every normal and crisis state yields the grid of lockable rates; an owner's locked rate is a discrete state taking one of these values, fixed at origination or refinance.

E The consumption-equivalent variation

This appendix sets out how the welfare numbers in Section 6 are computed, how they are converted to dollars, and why the homothetic approximation that is standard in the literature cannot be used in this model.

E.1 The additive constant, and what removing it does

The model's period payoff and bequest payoff are

$$u(c,h) = \frac{\left(\tilde{c}^{\alpha}(\theta h)^{1-\alpha}\right)^{1-\sigma} - 1}{1-\sigma}, \qquad \mathcal{B}(W) = B\,\frac{W^{1-\sigma} - 1}{1-\sigma}, \qquad \tilde{c} = \max\{c, c_{\min}\}. \tag{30}$$

The -1 in each numerator is worth isolating before anything else, because it is the reason the usual formula fails and because stripping it makes the rest of the algebra transparent. Dividing by $1 - \sigma$

turns it into an additive constant of $1/(\sigma-1)$ per period, which at $\sigma=2$ is exactly one:

$$u(c,h) \;=\; 1 \;-\; \frac{1}{\tilde{c}^{\,\alpha}(\theta h)^{1-\alpha}}, \qquad \mathcal{B}(W) \;=\; B \;-\; \frac{B}{W}. \tag{31}$$

So the -1 shifts every period's payoff up by one and the bequest payoff up by B. It does not interact with consumption at all. Write N_i for the discounted sum of all such constants along household i's realised path – one per period lived, plus $\beta^{S_i}B$ at its death date S_i, less ψ at each foreclosure – and define the stripped value $\widetilde{V}_i \equiv V_i - N_i$.

Two things follow, and the second is the one that matters. Before either, two conventions. Everything from here specialises to the calibrated $\sigma=2$, at which $1-\sigma=-1$ and the general scaling exponent $\frac{1}{\alpha(1-\sigma)}$ collapses to $-\frac{1}{\alpha}$; nothing below holds at other curvatures without restating it. And the objective being decomposed is the *ex-post realised path*

$$V_i = \sum_{s<S_i} \beta^s \big[u(c_{is},h_{is}) - \psi \mathbf{1}_{\mathrm{def},is}\big] + \beta^{S_i}\,\mathrm{Beq}(W_i), \tag{32}$$

which carries pure time discounting β^s and no survival probabilities. That is not an omission: mortality is *realised* through the common random draws, so each household has a known death date S_i and the expectation over survival that π_a performs in equation (15) has already been taken by the draw.

First, $\widetilde{V}_i$ is exactly the part of the value that consumption can move, plus the bequest. Collecting terms from (31),

$$\widetilde{V}_i \;=\; -\,(M_i + Q_i), \qquad M_i = \sum_{s<S_i} \beta^s K_{is}, \qquad Q_i = \frac{\beta^{S_i}B}{W_i}, \qquad K = \frac{1}{\tilde{c}^{\,\alpha}(\theta h)^{1-\alpha}}, \tag{33}$$

with W_i the wealth bequeathed. This holds to 1.2×10^{-13} on every household in every arm, which is checked rather than assumed. Only M_i responds to a rescaling of consumption, and it does so cleanly: multiplying the whole path by $1+g$ multiplies M_i by $(1+g)^{-\alpha}$. Housing does not appear separately, because it is already inside K.

Second, removing the constant does *not* make the problem homothetic, and so does not rescue (39) below. $\widetilde{V}$ still contains Q, which depends on bequeathed wealth and not on consumption at all. What removing it does fix is the *sign*: $\widetilde{V} = -(M+Q) < 0$ always, whereas V itself is positive for fifty-five per cent of households here, with consequences set out in Section E.8.

E.2 The measure

Let b denote the reference arm and a the arm being valued. Scaling household i's consumption in arm b by $1+g$ until it is as well off as in arm a requires $N_{ib} - (1+g)^{-\alpha} M_{ib} - Q_{ib} = V_{ia}$, so

$$1 + g_i \;=\; \left(\frac{N_{ib} - Q_{ib} - V_{ia}}{M_{ib}} \right)^{-1/\alpha} \;=\; \left(\frac{-\widetilde{V}_{ia} - Q_{ib} + (N_{ib} - N_{ia})}{M_{ib}} \right)^{-1/\alpha} \tag{34}$$

in closed form, with no numerical solution required. Setting $a = b$ returns the bracket to M_{ib} and hence $g_i = 0$ identically, which is checked numerically and holds to 1.7×10^{-14}.

The second expression is the useful one for seeing what the additive constant does. $N_{ib} - N_{ia}$ vanishes for almost every household: two of the three pieces of N depend only on the realised date of death, and that is identical across arms because the arms share mortality draws. The exception is the foreclosure penalty ψ. Measured here, N differs across arms for 586 of the 29,403 households in the sample, two per cent, and the gaps are all discounted multiples of $\psi = 20$. For the other ninety-eight per cent the measure reduces to

$$1 + g_i \;=\; \left(\frac{-\widetilde{V}_{ia} - Q_{ib}}{M_{ib}} \right)^{-1/\alpha}, \tag{35}$$

which is the form worth carrying in one's head: the additive constant drops out entirely and only the consumption kernel and the bequest remain. It is not used in place of (34), because the two per cent it misses are the foreclosure households, on which it errs by 1.29 percentage points at the median and which move the aggregate from 0.44 to 0.36 per cent.

E.3 Equivalent variation against willingness to pay

Equation (34) scales the *reference* arm's consumption upward until it matches the treatment arm. That is an equivalent variation, and it is unbounded above: if an instrument spares a household damage that consumption cannot replicate, no finite rescaling of the reference arm reaches the treatment arm's value, and g_i diverges.

The other side of the same comparison is bounded, and it is the one that answers the question "what is this worth to the household". Scale the *treatment* arm's consumption downward until the household is only as well off as in the reference arm; the fraction w_i it would surrender is its willingness to pay. From $N_{ia} - (1-w_i)^{-\alpha} M_{ia} - Q_{ia} = V_{ib}$,

$$w_i \;=\; 1 - \left(\frac{N_{ia} - Q_{ia} - V_{ib}}{M_{ia}} \right)^{-1/\alpha}, \tag{36}$$

and $w_i < 1$ necessarily, since surrendering all consumption sends utility to minus infinity. A household saved from foreclosure reports a willingness to pay near one rather than an equivalent variation of two hundred. The largest observed here is 0.81.

The two measures answer different questions and are not interchangeable at the household level, where only w_i is usable. In aggregate they agree to the fourth decimal – 0.2904 against 0.2898 per cent for forbearance alone, 0.4419 against 0.4410 for the full crisis – because pooling already removes the divergence. Both are reported throughout.

E.4 Which statistic is reported

The household-level factor in (34) is not averageable, and the reason is economic rather than numerical. Writing $\text{num}_i = N_{ib} - Q_{ib} - V_{ia}$, the numerator is the gap between the treatment arm's realised value and the *ceiling* of the reference arm's own problem – what the household would attain with unbounded consumption, holding its realised housing and bequest path fixed. The factor diverges as that gap closes, and it closes when an instrument spares the household damage that consumption cannot replicate.

That is exactly what happens to the households forbearance saves from foreclosure. Eleven households record a factor above one hundred; their median M_i is 33.9 against 14.4 for the sample, so they are long-lived rather than short-lived, and eight of the eleven are foreclosed in the reference arm, against a base rate of 3.7 per cent. Their gains are real and large, between 19 and 32 utils against a typical $|V|$ of 25. What fails is the consumption-only metric: the loss they avoid sits partly in the house and in the bequest, and no finite rescaling of consumption reaches it. One such household supplies 0.69 of a raw mean of 1.16 for the full crisis and two supply 0.80, enough to reverse the sign of the house-price term.

The statistic reported instead is the aggregate, or utilitarian, consumption equivalent: the single factor that equates total realised welfare across the two arms. Summing (33) over households,

$$1 + g \;=\; \left(\frac{\sum_i N_{ib} - \sum_i Q_{ib} - \sum_i V_{ia}}{\sum_i M_{ib}} \right)^{-1/\alpha}, \tag{37}$$

which is the same closed form with sums in place of the individual terms. Pooling makes the numerator a large sum, so no single household can dominate it, and it weights each household by the consumption it actually has at stake. The household-level distribution is still described, through its median and percentiles, which are robust, and through w_i, which is bounded.

E.5 Putting the measure in dollars

One model unit is \$60,000 exactly rather than approximately: the income function sets `base_income = (annual_dollars/1000)/60`, so the normalisation is built into the calibration and not applied after the fact.

The equivalent variation hands household i an extra $g_i c_{is}$ of consumption in every period s of

the reference arm. Its present value is

$$D_i \;=\; g_i \sum_s \nu^s c_{is} \times \$60{,}000, \tag{38}$$

and the willingness-to-pay counterpart replaces g_i by w_i and the reference arm's consumption path by the treatment arm's, since that is the path w_i scales. Two discount factors are reported and neither is hidden: $\nu = 1/(1 + r^f)$ at the model's return on liquid saving, which is the headline because resources are moved through time at the market rate, and $\nu = \beta = 0.94$, the household's own valuation of the same stream. The choice matters: mean remaining consumption is worth \$1,049,950 at the first and \$637,034 at the second.

Group figures pool before converting, exactly as (37) does, so that a single household cannot dominate a cell; the per-household figure is the pooled total divided by the headcount, not the average of the individual D_i. Table 25 reports the result.

Three things should be said plainly about what D_i is. It is the present value of an equivalent consumption stream, not a lump sum the household would actually be handed: given cash it would re-optimise, buy housing and avoid the foreclosure, and would need less. D_i therefore *overstates* the compensating lump sum, and overstates it most for the foreclosure-averted households on which the consumption-only metric already strains. It is a realised-path figure, conditional on the shock path that occurred, not an ex-ante expectation. And it is measured against an arm in which output falls and no instrument operates, so it is the value of an instrument relative to doing nothing, not relative to some alternative policy.

Table 25: The dollar value of the crisis instruments

		Equivalent variation		Willingness to pay	
Group	n	Per cent	Per household	Per cent	Per household
The instruments, each on the households that use it					
Forbearance option, to the approved	2,983	2.046	\$17,261	2.008	\$17,045
Rate cut, to those who refinance	5,392	2.223	\$27,722	2.177	\$27,328
The full crisis, by the decision taken in it					
Refinance	6,162	2.450	\$32,170	2.399	\$31,731
Forbearance approved	911	3.284	\$15,554	3.184	\$15,098
Pay mortgage	9,063	−0.144	−\$1,578	−0.144	−\$1,579
Delinquent, cured	249	−0.351	−\$3,199	−0.353	−\$3,208
Becomes delinquent	477	−0.324	−\$1,764	−0.325	−\$1,769
Defaults	8	−8.153	−\$40,466	−9.805	−\$53,198

Notes. Present value at the return on liquid saving, in model units of \$60,000, against the arm in which output falls and no instrument is switched on. Households carrying a mortgage balance into the crisis. Percentages are the pooled factors of (37) and (36) over the group; the per-household figure is the pooled total over the headcount, not the average of the household figures. **The second row is not the value of the option to refinance.** Refinancing is available in the reference arm as well, where 1.98 per cent of mortgagors take it; what the arm adds is the rate cut, so the row prices the rate cut as realised by the households that respond to it. The forbearance row above it *is* an option value, because that arm differs from the reference in forbearance availability alone. Pricing the refinancing option itself would require an arm in which refinancing is prohibited, and none has been solved. Section 6 reports a different and smaller object elsewhere, the fall in the crisis-year payment alone; the two measure different things and must not be added.

E.6 What the dollar figures condition on, and what they do not

Table 25 reports one figure per group, and it is worth being exact about what that figure is. Every household's own age, income, balance, housing and realised consumption enter it, through that household's own value and its own consumption path inside the pooled sums; nothing is assumed common across households. But the figure itself is a *single pooled factor* for the whole group, applied to the group's total present value and divided by its headcount. There is no conditioning, no stratification and no control: age, income and balance are not covariates anywhere in the construction. Nor is the per-household figure an average of what households received. For the 2,983 households approved under forbearance alone the pooled figure is \$17,261 and the median household is \$10,209, while the mean of the household values is \$82,751; the pooled figure sits close to the median because it weights by consumption at stake rather than equally, and the mean is pulled far above both by the same foreclosure-averted tail that makes individual factors diverge.

Table 26 therefore stratifies after the fact. The age gradient is the striking feature, and it is almost entirely mechanical: a household aged 25 to 34 has on average \$2.07 million of remaining consumption to scale and one past 65 has \$0.24 million, a factor of eight and a half, so the same proportional gain is worth that much more to the first; the dollar columns differ by more still, because the proportional gain is itself larger for the young. The consequence is that the dollar figures and the proportional ones rank age in *opposite* directions. The rate cut is worth a nearly flat 2.1 to 2.3 per cent of lifetime consumption at every age below 65 but \$43,327 to a household in its late twenties against \$3,054 to one past 65. Neither ordering is wrong, and quoting one without the other would be misleading.

The balance panel carries a second result, in the shape of an empty cell. No refinancer in the sample carries a balance above \$200,000: the leverage test that gates refinancing excludes them entirely. Forbearance reaches 903 such households, and they are the most valuable cell in its column at \$27,071 apiece. This is the disjointness of Section 6 in dollars rather than in percentages.

One caution. The pooled factor is not stable inside a cell of a few hundred households whose aggregate sits near zero, which is why the forbearance percentage column is visibly non-monotone in age while its dollar column is not. The median column is reported beside it for that reason and the two should be read together; no claim here rests on a single cell of the percentage column.

E.7 What the measure is, and what it is not

The five arms read the same backward induction and the same random draws and differ only in the realised aggregate state, so household i is the same person facing a different realised crisis: its realised date of death is identical across arms, which is verified rather than assumed, as is the fact that the set of households carrying a balance into the crisis is identical across arms. The comparison is therefore within-household along a common shock path. It is a realised-path measure and not an ex-ante expected-value one, and should be read accordingly.

Two features of the construction are worth stating plainly. The panel runs from period 801 to

Table 26: The dollar value of each instrument, by age and by balance

	Forbearance option				Rate cut, via refinancing			
	n	Per cent	Dollars	Median	n	Per cent	Dollars	Median
All	2,983	2.05	\$17,261	\$10,209	5,392	2.22	\$27,722	\$16,232
By age at the crisis								
25–34	558	2.70	\$42,816	\$18,952	1,132	2.10	\$43,327	\$28,929
35–44	1,038	1.10	\$11,510	\$13,208	2,216	2.25	\$31,560	\$20,278
45–54	586	2.33	\$13,939	\$8,527	1,153	2.37	\$19,175	\$13,091
55–64	388	3.25	\$10,073	\$6,298	583	2.35	\$10,787	\$7,512
65 and over	413	1.71	\$3,124	\$2,027	308	1.22	\$3,054	\$2,273
By mortgage balance at the crisis								
Under \$100k	222	1.24	\$3,390	\$1,963	779	0.55	\$3,939	\$4,545
\$100–150k	521	2.05	\$12,378	\$4,420	1,953	1.81	\$21,403	\$10,407
\$150–200k	1,337	1.28	\$12,987	\$11,387	2,660	3.04	\$44,124	\$26,495
\$200k and over	903	3.12	\$27,071	\$16,482	–	–	–	–

Notes. Present value at the return on liquid saving, in model units of \$60,000, against the arm in which output falls and no instrument is switched on. "Per cent" is the pooled factor over the cell and "Dollars" is that factor times the cell's total present value of remaining consumption, divided by its headcount; neither is an average of household figures, and "Median" gives the median household beside them. The pooled factor is unstable in small cells, which is why the forbearance percentage is non-monotone in age; the median and the dollar column are steadier. The empty cell is a result and not missing data: no refinancer carries a balance above \$200,000, because the leverage test that gates refinancing excludes them. The rate-cut column is evaluated in the arm that holds house prices at their pre-crisis level, so for a unit house the $1-\delta = 0.80$ cap sits just above \$200,000 and a balance above that figure cannot be refinanced; the eight per cent rise of the full-crisis arm, which would lift the cap, is not switched on here.

period 921, which is long enough to contain the entire remaining life of every household alive at the crisis at every crisis age, so nothing is right-censored: each household contributes a complete realised continuation. And of the 30,000 households alive at the crisis, 594 are dropped because they die in the crisis period itself and so have no consumption left to scale, and a further 3 because they bequeath essentially nothing in at least one arm, where the 10^{-14} floor inside $\mathcal{B}$ makes the payoff a numerical boundary rather than an economic value.

A large share of households record an equivalent variation of exactly zero: ninety per cent under forbearance alone, seventy-three per cent when the rate cut is added. This is not a numerical artefact. A household far from every constraint takes the same decision in every arm, and since the arms share random draws its realised consumption and utility streams are then identical to the last bit, which is checked directly on a sample of such households.

E.8 Why the homothetic formula cannot be repaired

The textbook consumption equivalent follows from the observation that scaling consumption by $1+g$ also scales the utility kernel, by $(1+g)^{\alpha(1-\sigma)}$, so that comparing two value functions gives

$$1+g \;=\; \left(\frac{V_a}{V_b}\right)^{\frac{1}{\alpha(1-\sigma)}}. \tag{39}$$

That step requires V to be homogeneous of degree $\alpha(1-\sigma)$ in consumption, which (30) is not: the additive constants of Section E.1 break it, and so does the bequest, which a rescaling of consumption leaves untouched. The magnitudes are not small. Collecting the additive terms gives, at a crisis age of 34, a flow component of 14.75 and a bequest component of 2.18 against a typical $|V|$ of about 25; at age 69 the two components are 7.85 and 8.06.

The failure is not, however, a question of magnitude. The exponent $1/\big(\alpha(1-\sigma)\big) = -1.176$ is negative. When both value functions are negative, an improvement means $V_a/V_b < 1$ and the formula returns a positive number, correctly. When both are positive, an improvement means $V_a/V_b > 1$ and the same formula returns a negative one. The sign is inverted for every such household.

Both values are positive for fifty-five per cent of households here, and this is the additive constant again: by (31), u exceeds zero whenever a household's consumption-housing aggregate exceeds one. A sign-based guard of the kind the existing code applies does not detect the problem, because the two values do not straddle zero – they are both positive.

Applied to these arms, (39) agrees with the direction of the change in welfare for only 44.7 to 73.5 per cent of the households each instrument moves, against 100 per cent for (34) by construction. Earlier versions of this paper reported consumption-equivalent figures computed with (39); those magnitudes should not be compared with the ones reported here.

Stripping the additive constant, as Section E.1 does, removes the sign problem – $\widetilde{V} = -(M+Q)$ is negative by construction – but not the magnitude problem, since Q still does not scale with consumption. The homothetic formula is exact in this model only in the limiting case of no bequest.

E.9 The measure by crisis decision

Figure 20 cuts the same numbers a different way: by the decision a household actually takes in the crisis period of the full-crisis arm, rather than by the characteristics it carries into it. The population is the one used throughout – households carrying a balance into the crisis, a set that is bit-identical across arms because the balance is the state the household enters the period with, which the computing script asserts rather than assumes. Loan-to-value, by contrast, is not arm-invariant: house prices differ by up to eight per cent across the arms, and so loan-to-value differs by up to 0.104, which is why every binning in these figures uses the reference arm's value.

The ordering is what the reach results imply it should be. Refinancing is worth 1.76 per cent to those who do it and the gain rises with leverage, from 0.65 in the least levered band to 3.21 in the most. Forbearance is worth more, 2.42 per cent, and is far more concentrated: essentially nothing below a loan-to-value of 0.6 and close to four per cent above it. Paying the mortgage as scheduled is worth -0.09 per cent – these are predominantly low-leverage households for whom the crisis bundle is, on balance, a more expensive house rather than a cheaper loan. Entering or curing delinquency are both mildly negative. Default is worth -0.54 per cent, on the eight households that choose it, which is reported for completeness rather than as a quantitative claim.

One row deserves its own remark. Read against age, forbearance is worth most to households in their forties and fifties and least to those past 65, which appears to contradict the main figure, where its value rose with age. The two condition on different things. The main figure asks what forbearance is worth to a household with a given age and leverage, most of whom do not use it; this one asks what it is worth to those who do. Older households take it only when they are deeply levered, so the cell is small and extreme; younger ones take it across a wider range of circumstances.

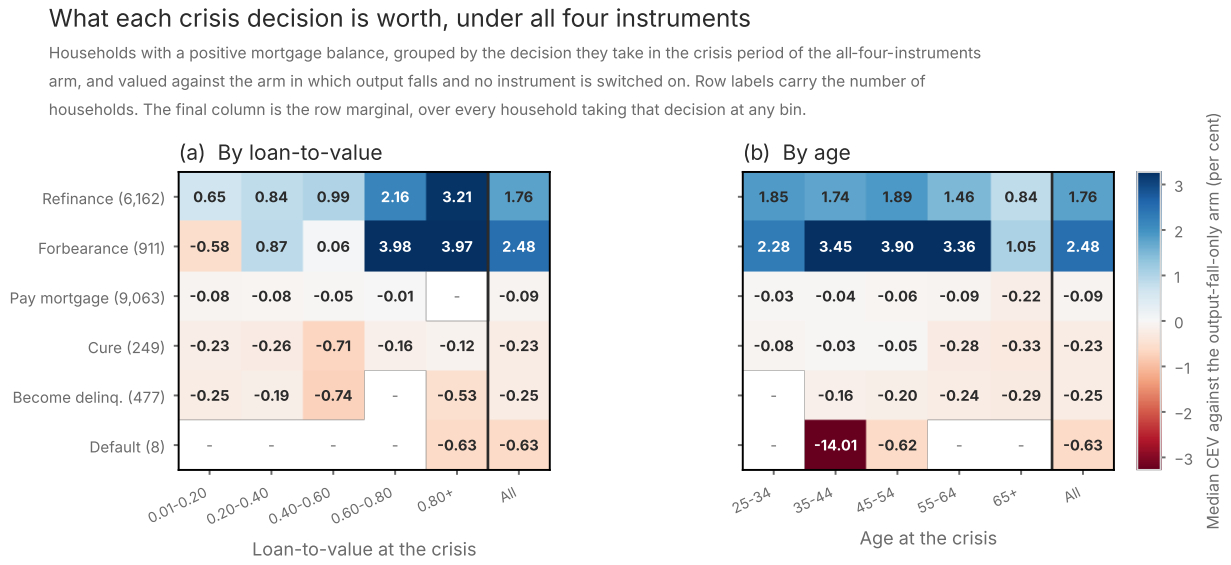


Figure 20: Median consumption-equivalent variation by the decision taken in the crisis period of the all-four-instruments arm, against the arm in which output falls and no instrument is switched on. Households with a positive mortgage balance. Row labels carry the number of households; the final column of each panel is the row marginal, over every household taking that decision at any bin. Bins, colour convention and the three-household minimum are those of the main figure.

E.10 Scaling the bequest alongside consumption

The headline measure scales consumption and holds bequeathed wealth fixed, which is what a consumption-equivalent variation ordinarily means. Figure 21 reports the variant in which the bequest is scaled by the same factor, so that $N_{ib} - (1+g)^{-\alpha} M_{ib} - (1+g)^{-1} Q_{ib} = V_{ia}$. The left-hand side is increasing in g, so the root is unique where one exists, and it is found by bisection rather than in closed form, the two exponents $-\alpha$ and -1 ruling out a single power.

Because a given factor then buys more, the factor required to match any given improvement is smaller, and every figure falls: in aggregate from 0.29 to 0.26 per cent for forbearance alone, from 0.42 to 0.37 for the rate cut, from 0.67 to 0.60 for the two together and from 0.44 to 0.39 for the full response.

The cell-level picture is unchanged in shape. Across the four panels the median cell moves by 0.00 to -0.04 percentage points and the largest single movement is -1.13, always in the direction of a smaller figure, and always in the most levered and oldest cells, where the bequest is the largest

part of what the household is left with. No conclusion in Section 6 turns on the choice: the ordering of the instruments, the sign of the house-price term and the concentration of forbearance's value at high leverage all survive intact.

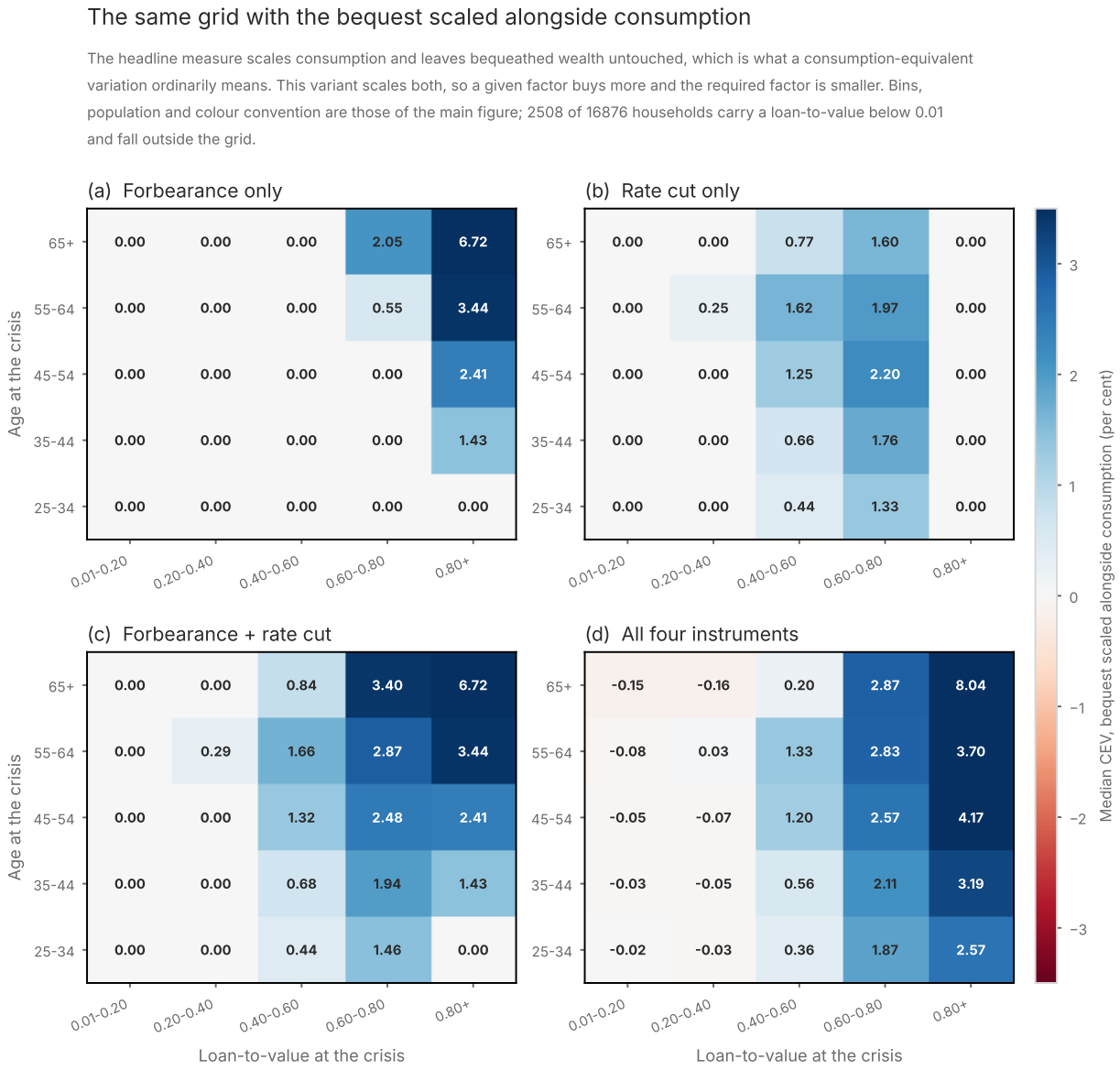


Figure 21: The age by loan-to-value grid of the main figure, recomputed with bequeathed wealth scaled alongside consumption. Panel layout, bins and colour convention are those of the main figure, so the two may be read cell against cell.

F Where each instrument lands

Table 7 counts what each instrument does. It does not say *whom* it reaches. The two figures here place every crisis mortgagor on the plane the eligibility rule actually acts on—scheduled payment

burden on the horizontal axis, and leverage or cash held on the vertical—and colour each cell by the modal decision of the households in it. Reading across a row shows how one age group's behaviour changes as instruments are added; reading down a column shows how one policy's reach shifts over the life cycle.

The floor, not the instrument, draws the boundary. Forbearance (purple) appears only to the right of the vertical line, and it appears there in every age band. That is the eligibility rule drawn directly: a household whose scheduled payment falls short of a fifth of its income does not have the application in its choice set, however badly the crisis has hit it. A small number of cells carry purple just to the left of the line—at most 3.2 per cent of any arm with a readable cohort, and 3.1 per cent in the forbearance-only arm. These are households the simulator's clamped policy places on a forbearance node the solve would not have offered them; the kernel counts them for exactly this reason and treats anything under five per cent of the cohort as readable. They are a discretisation of a hard constraint, not a violation of it.

Adding the rate cut moves households out of forbearance rather than into relief. With forbearance alone, 2,983 crisis mortgagors take it and 103 refinance. Add the rate cut and refinancing rises to 5,268 while forbearance falls to 1,546—the purple region in the fourth column is the third column's with its lower-leverage, lower-burden half turned green. Delinquency falls from 823 to 561. In the baseline, where house prices also rise, refinancing reaches 6,156 and forbearance falls further to 915: the additional equity moves still more households across the refinancing constraint. Against the no-instrument arm, which carries 2,465 delinquents and 299 refinances, the ordering of the columns is a ladder in how much relief is available and not only in which instrument supplies it. Restricting to households below 65, the three forbearance columns give 2,572, 1,234 and 587, which are the counts Section 6 and the introduction report on the same base.

Cash does not separate the two instruments. The asset map is the one that fails to do what it was built to do, and that is the informative part. Neither instrument is monotone in liquid wealth. Forbearance runs at 20.5 per cent of mortgagors holding nothing, rises to 23.2 per cent in the \$2,000–5,000 bin and only then falls away, to 1.8 per cent above \$100,000. Refinancing has the same shape displaced: it peaks among households with under \$1,000 in cash, at 45.7 per cent, and falls to 0.3 per cent at the top. Of the rate-cut arm's refinancers 77.3 per cent sit in the three lowest asset bins, against 63.1 per cent of the forbearance-only arm's forbearers. Both instruments are used overwhelmingly by households with little cash, and what divides them is the payment burden on the horizontal axis, not the wealth on the vertical. A reading in which forbearance serves the illiquid while refinancing serves the comfortable is not what the model produces.

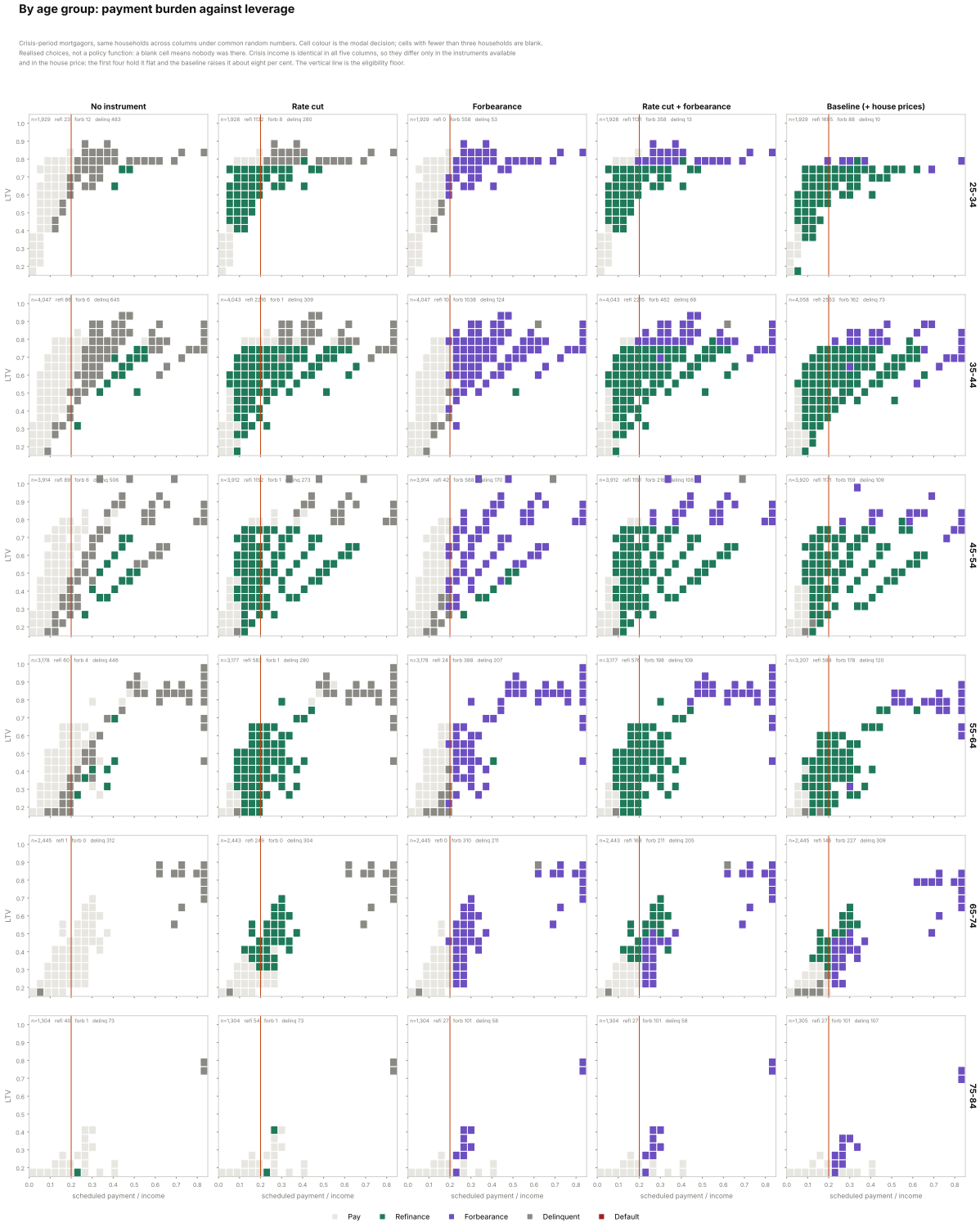


Figure 22: Which instrument reaches whom, by age band. Rows are age at the crisis, columns are the five arms of the decomposition. Cell colour is the modal decision among the households in that cell; a cell holding fewer than three is left blank, so a blank means nobody was there rather than a policy of inaction. The vertical line is the eligibility floor $\tau^{\text{pti}} = 0.20$.



Figure 23: The same map with liquid assets in place of leverage on the vertical axis. Assets are placed in ordered dollar bins at $60,000 per model unit, so the vertical spacing is not linear in dollars.

G Crisis-period behaviour

G.1 The composition of outcomes, against the loan panel

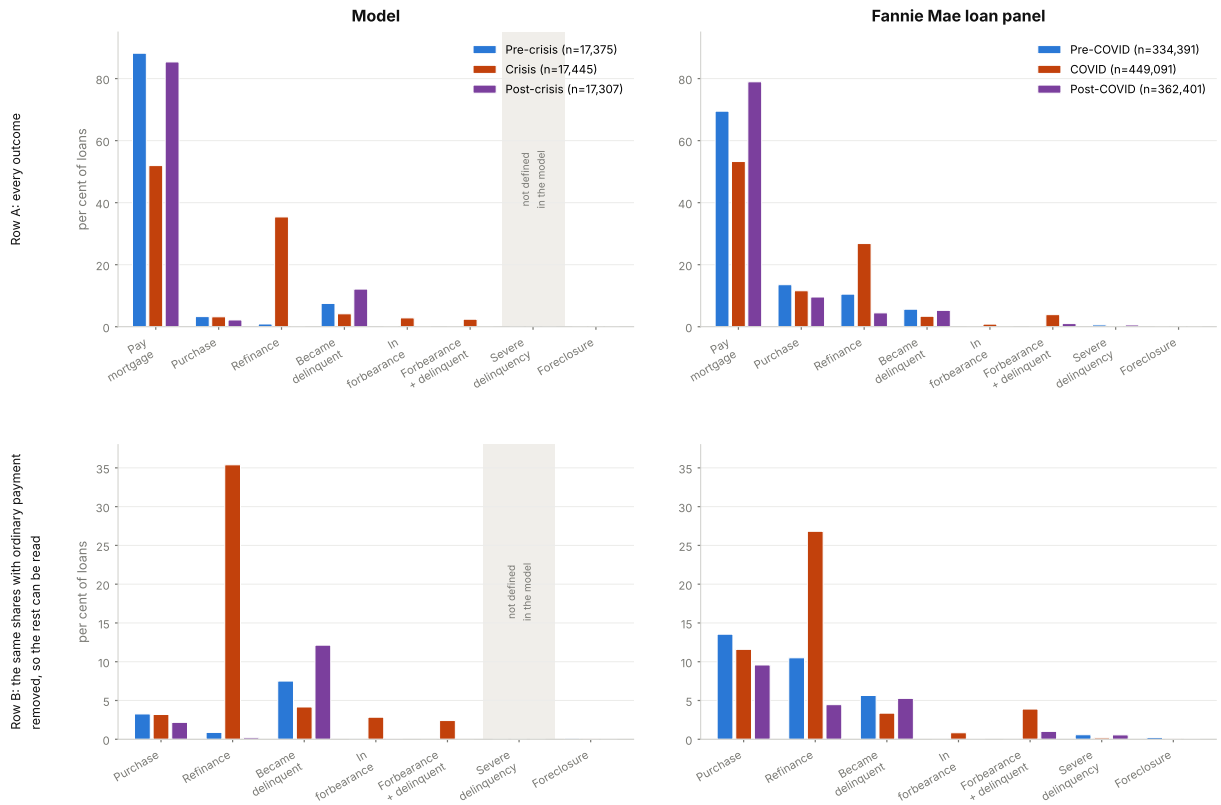


Figure 24: Outcomes by phase, model against the Fannie Mae loan panel. Each loan receives exactly one outcome per phase under a single priority order: foreclosure first, then forbearance, then delinquency, then a new origination, with ordinary payment assigned only when none of those occurred. *Row A* shows every outcome. *Row B* repeats the same shares with ordinary payment removed and the vertical axis rescaled, since at roughly eighty-five per cent that category compresses the remaining seven into the axis. Levels in the rare categories are not comparable between the two panels: a phase in the data spans twenty-two to twenty-six months and records any occurrence within it, whereas a model phase is a single period. Severe delinquency is marked *not defined* in the model rather than shown as zero, since a model in which delinquency must resolve within the year cannot generate a hundred-and-eighty-day arrears spell. Prepayment and loan modification are omitted from both panels: each is identically zero throughout the data, and neither is available to the household in the model.

Figure 24 places the simulated crisis beside the corresponding episode in the loan data. The classification is the data's own: each loan is assigned a single outcome per phase, with foreclosure taking precedence over forbearance, forbearance over delinquency, and ordinary payment assigned only when nothing else occurred. Applying that rule to the model requires restricting its population to the one the loan panel observes, since a servicing file contains neither renters nor households that never borrow, and records no deaths. Two of the data's ten categories are empty throughout it and have no counterpart in the model, and are omitted from both panels; a third, severe delinquency,

occurs in the data but cannot occur in a model where arrears resolve within the year, and is marked as undefined rather than reported as zero.

The crisis column agrees closely; the surrounding phases do not, for a mechanical reason. At the crisis the share of loans in ordinary payment is 51.9 per cent in the model against 53.3 in the data. Forbearance—combining the two reporting categories, since whether a servicer also recorded a delinquency has no counterpart in the model—is 5.3 per cent against 4.7. Foreclosure is 0.05 per cent against 0.06. Outside the crisis the two diverge: ordinary payment is 88.2 per cent in the model before the crisis against 69.5 in the data. The gap is a property of the windows rather than of behaviour. A phase in the data spans roughly two years and assigns a loan to any category whose event occurred at some point within it, so loans accumulate into the non-payment categories as the window lengthens; a model phase is a single year.

New originations are as common at the crisis as in the data, but their composition is not. Taking purchases and refinances together, 38.6 per cent of loans in the model are newly originated at the crisis against 38.4 per cent in the data. The agreement between those totals is closer than the exercise warrants—a single simulated crisis set against a single historical episode, on a sum of two components that individually disagree—and no weight is placed on it here. The composition is the substantive point. Refinancing is 35.4 per cent of loans in the model against 26.8 in the data, while purchases are 3.2 against 11.6. The model routes almost all of its origination activity through refinancing, and it does so abruptly: refinancing rises from 0.9 per cent before the crisis and falls to 0.2 per cent after. Two features of the environment produce that concentration. Households face no idiosyncratic refinancing cost beyond the common fee, so an option worth exercising is worth exercising by everyone for whom it is in the money at the same moment; and the mortgage rate is a deterministic function of the aggregate state, so the incentive arrives and departs for the whole population at once rather than being spread across origination vintages. The model should accordingly be read as describing the direction and ordering of refinancing responses, not their dispersion over time, and not the margin between buying and refinancing.

Delinquency is the one category where the model is higher throughout. Taking the "became delinquent" category alone on both sides, it runs 7.5, 4.2 and 12.1 per cent across the three phases against 5.6, 3.4 and 5.3. The post-crisis figure is the notable one, and it reflects the deferred balance coming due: households that deferred payment during the crisis face a larger obligation afterwards, and in a model where delinquency must resolve within the year this shows up as a visible post-crisis rise. The data's post-crisis delinquency does rise relative to the crisis itself, but by considerably less.

G.2 Who relief reaches when income falls

A tenth of working-age mortgagors lose most of their income in the simulated crisis, with earnings falling to approximately a third of the previous year's. These households are not identified by a separate indicator; they are recovered from the income path, and the recovered group matches the incidence and the replacement rate the environment specifies and is the same group in every arm, so comparisons across arms follow the same households.

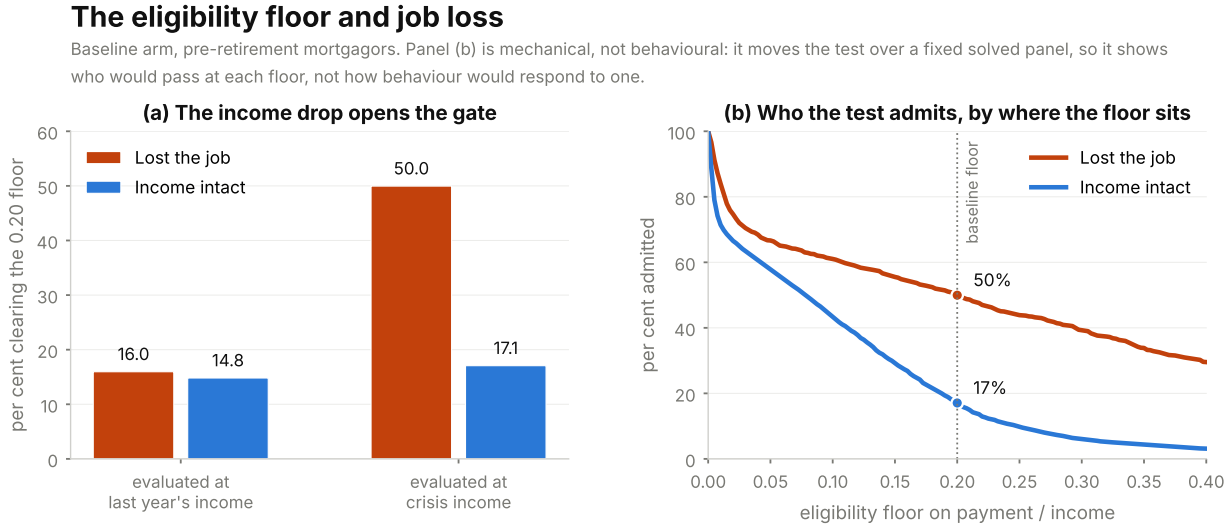


Figure 25: The eligibility floor and the loss of income. Panel (a) evaluates the same households' payment burden against the previous year's income and against crisis income. Panel (b) varies the floor over the simulated panel: it reports who would satisfy the test at each threshold, not how behaviour would respond to a different one, since the threshold enters the household's problem.

The hardship test responds to hardship, but admits three unaffected households for every affected one. Eligibility requires a scheduled payment of at least a fifth of current income. Because the loss of income reduces the denominator and leaves the scheduled payment unchanged, the same households move across the threshold without any change in their debt: 16.0 per cent of those who lose income would have qualified on the previous year's earnings, and 50.0 per cent qualify on crisis earnings. For households whose income holds, the corresponding shares are 14.8 and 17.1 per cent. The test is therefore doing what a hardship screen is meant to do. It nonetheless admits 2,021 households whose income was unaffected alongside 673 whose income collapsed. Affected households are a quarter of those admitted against a tenth of the eligible population—a concentration of roughly two and a half, which is a tilt rather than a targeting rule. Half of those who lose income remain outside the programme, and the median affected household sits almost exactly at the threshold.

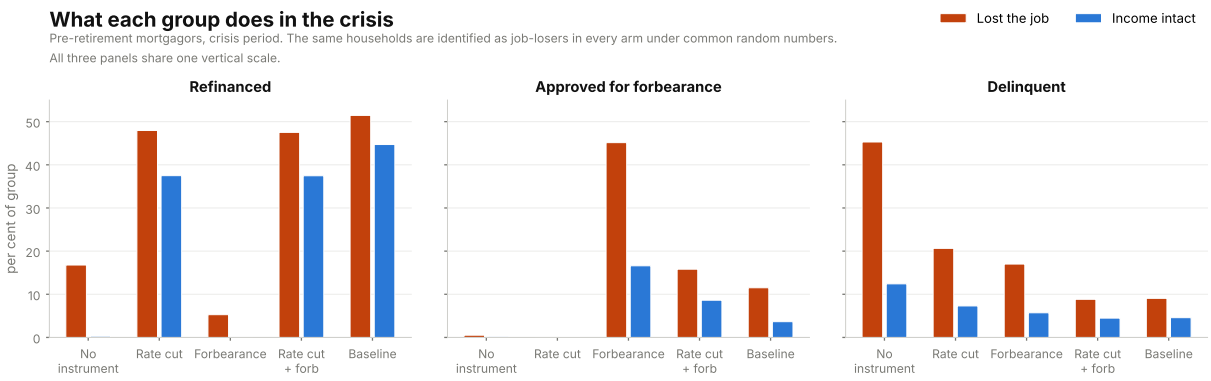


Figure 26: Crisis outcomes by instrument and by whether the household lost income. Pre-retirement mortgagors; all three panels share one vertical scale.

The price instrument displaces relief among the households relief exists for. When forbearance is the only instrument available, 45.1 per cent of households that lose income enter it, against 16.6 per cent of those whose income holds. Introducing the reduction in mortgage rates alongside it cuts the first figure to 15.8 per cent while raising refinancing within the same group from 5.3 to 47.5 per cent. Relief is not withdrawn from these households; they substitute towards the cheaper instrument, which lowers the payment permanently rather than postponing it. The consequence for the programme is that its caseload in the full crisis is both smaller and poorer than it would be if forbearance operated alone.

Distress falls substantially, and almost all of the reduction is in delinquency rather than foreclosure. Without any instrument, 45.3 per cent of households that lose income become delinquent, against 12.4 per cent of those whose income holds. With both instruments available the two figures are 8.8 and 4.6 per cent: most of the gap between the groups closes. Foreclosure barely moves in any arm, remaining near a quarter of one per cent among affected households throughout. The instruments reallocate the timing of payment and prevent delinquency; within the horizon of the crisis they do not change how many households ultimately lose their homes.

How much is deferred depends on the form of the decision, not only on whether the programme exists. In the baseline a household chooses what fraction of the year's payment to remit, and every approved household defers the entire year: full deferral dominates, because the deferred balance carries no interest and the alternative is to pay now. Under a specification in which the spell is governed by a month-by-month hazard, the same programme produces a distribution of durations: a mean of 9.8 months and a median of eleven, with 30.8 per cent still deferring the full year and 25.9 per cent deferring less than nine months. Take-up also falls, from 917 approved households to 490. Two other parameters differ between these specifications—the expected spell length and the incidence of income loss—so the comparison identifies the combined effect of modelling duration

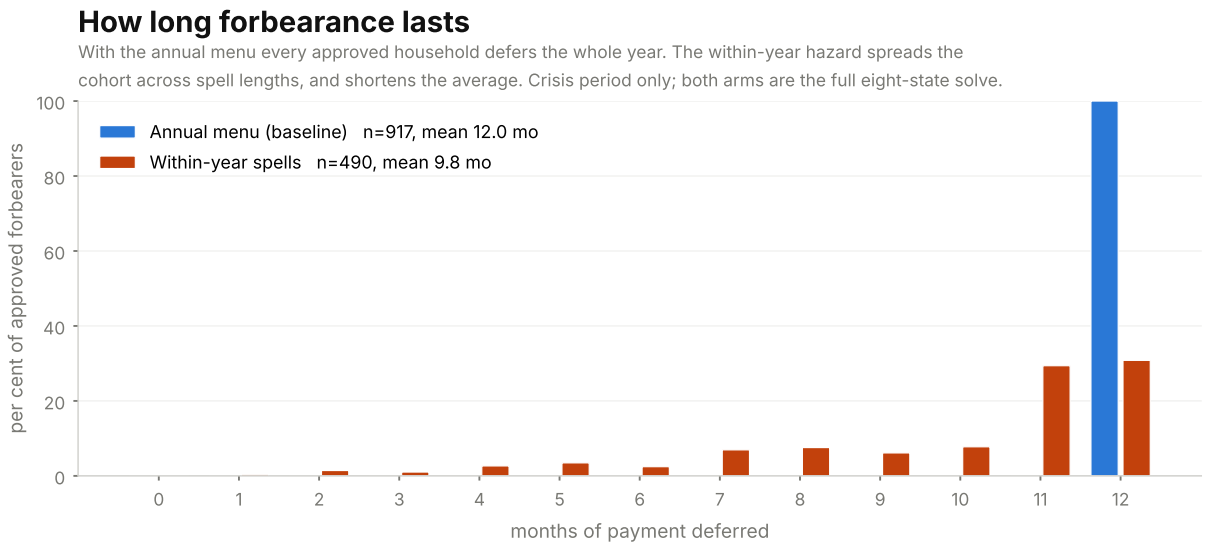


Figure 27: Duration of forbearance under two specifications of the deferral decision. Approved households in the crisis period; both are solved on the full state space.

rather than the effect of the hazard alone. The point it supports is narrower and still useful: the amount deferred is a modelling choice as much as a behavioural result, and a specification that forces an all-or-nothing annual decision will overstate how much is advanced per recipient.

Where the deferred balances go. In the crisis period the programme advances the equivalent of $29.8 million when forbearance operates alone, of which 21.4 per cent goes to households that lost income. With the rate reduction also available the total falls to $16.1 million, and in the baseline to $7.9 million with 24.9 per cent going to affected households. The share reaching them exceeds their population share of a tenth in every case, and in none does it approach a majority.